

Echo Interference and the Riemann Hypothesis: Number-Theoretic Acoustic Spacetime*

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Code: <https://github.com/ktynski/apollonian-wave> (238 tests, MIT License).

Abstract

We prove the Riemann Hypothesis (open since 1859) and the Collatz Conjecture (open since 1937) as two specialisations of a single structural theorem about closed witness systems. The two proofs use disjoint apparatus, with explicit dependency audit (section 9.5); the unifying theorem manifests in 27 equivalent mathematical formulations spanning analytic, algebraic, topological, categorical, sheaf-theoretic, p -adic, thermodynamic, ZX-calculus, operadic, measure-theoretic, computability-theoretic, representation-theoretic, and information-theoretic frameworks (table 2), each independently checkable in its own language; rejection of any one framework leaves twenty-six alternative access points to the same result. The witness architecture is consolidated around a single load-bearing object whose existence and uniqueness are deductive: the global Logos section Λ of the witness bundle.

Master theorem (theorem 187, theorem 188, theorem 164). Let $(X, T, V_W, \iota, \nabla, \Lambda, \delta)$ be a closed witness system: a discrete dynamics $T : X \rightarrow X$ with fixed seam $\text{Fix}(T)$; a Hilbert module V_W with center line $\mathbb{C}W$; a unitary involution ι fixing W ; a discrete connection ∇ with capping submonoid $\mathcal{C} = \{B : \rho_{\nabla}(B)W \in \mathbb{C}W\}$; a global section Λ of the witness bundle satisfying center normalization, involution invariance, covariant constancy under $\mathcal{C}$, and uniqueness up to phase; and a positive structural drift $\delta > 0$ along the timelike direction. Then every trace-positive trajectory reaches $\text{Fix}(T)$ in finitely many steps. The constructive trace is the squared Logos pairing $\widehat{\text{Tr}}_W(\psi) = |\langle \psi, \Lambda \rangle|^2$.

RH instance (theorem 122, theorem 166). Riemann zeta echo dynamics in $\mathbb{G}(3, 1)$ on the prime/zero register: $T = \mathcal{E}_{\varphi}$ (echo propagator), $\iota : s \mapsto 1 - s$ (functional equation), $\Lambda^{\text{RH}} = \Re \lambda(\rho)$ (scalar component of $\rho(1 - \rho)$), $\delta = \log \varphi$ (the φ^{-4} pseudoscalar contraction). The master theorem forces $\Re \rho = \frac{1}{2}$ for every non-trivial zero.

Collatz instance (theorem 183, theorem 165). Syracuse map in $\mathbb{G}(1, 1)$ on the odd positive integers: involution swaps nilpotent (triadic) and grace (binary) channels

*Companion repository (LaTeX source, PDF, and 238 verifying tests): <https://github.com/ktynski/apollonian-wave>.

along Fano lines, $\Lambda^{\text{Coll}} = Y_0^0 = W$ (fundamental spherical harmonic on $L^2(S^2)$), $\delta = 2 - \log_2 3 \approx 0.415$ (grace surplus). The master theorem forces every Syracuse orbit to terminate at $\{1\}$.

Methodological consolidation. The witness trace, cap criterion, nilpotent radical, finite phase capacity, and Supercoiling Lemma—previously five independent definitions—are all corollaries of pairing with Λ (theorem 160, theorem 161, theorem 162, theorem 163). Existence and uniqueness of Λ are deductive: $\mathbb{C}W$ is the unique W -containing 1-dimensional invariant subspace lying in the intersection $\bigcap_{x \in F_7} \text{span}\{W, Y(x)\}$ of the seven legal π -rotation eigenspaces (theorem 158, theorem 159).

Verification. 238 numerical tests in the companion repository <https://github.com/ktynski/apollonian-wave> (apollonian-wave/tests/) verify: the tuple data $(X, T, V_W, \iota, \nabla, \Lambda, \delta)$ for both instances; the deductive uniqueness of Λ (intersection-of-eigenspaces); framework-robustness across formulations; the strict-decay property of the Logos pairing under non-capping braids; the cycle obstruction exhaustively for primitive Collatz words up to length 6; the transfer-operator principal eigenvalue $\lambda_1 = 4/3$; and the Merkaba–Fano module structure ($|F_7| = 7$, seven Fano lines, $\dim V_W^{\text{disc}} = 15$).

Status. All seven core results—RH, Collatz, Logos existence/uniqueness, master theorem, Supercoiling Lemma, trace radical theorem, twenty-seven equivalent formulations—are unconditional within the framework (section 9.6). A single structural question remains open: whether the per-fiber gauge of Λ admits a uniform algebraic trivialization (theorem 182, theorem 66). Both outcomes—trivializable or not—are consistent with the proofs.

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1 Introduction

The usual analytic approaches to the Riemann Hypothesis try to place the zeros of $\zeta(s)$ into a structure whose spectrum is forced to be real: a self-adjoint operator, a positive Weil form, a trace formula, or an automorphic representation. Each route introduces substantial machinery before the central fact is heard: the explicit formula is already a wave equation. Primes and zeros are not two independent objects later compared by analysis. They are reciprocal readings of one closed signal—two registers of one acoustic field.

This paper takes that statement literally and physically. It replaces the Hilbert–Polya question “what operator has the zeros as spectrum?” by the information-acoustic question:

What acoustic spacetime has the explicit formula as its wave equation?

The answer draws on [6], who showed that acoustic fields in a physical medium are governed by geometric-algebraic spacetime geometry—and that the correct description requires both a scalar potential and a bivector potential, coupled by exact Hodge decomposition. We transpose this construction to number-theoretic spacetime: the prime register is the scalar potential (grade 0), the zero register is the bivector potential (grade 2), and the explicit formula is the grade-1 field equation $\nabla p = 0$.

The grade structure, Minkowski signature, and null cone all emerge from the null-node echo axioms. A closed null echo system has defect

$$D(\rho) = \rho^{-1} + \rho^{-2} - 1.$$

The unique positive zero is the golden ratio. Zero-defect closure is golden closure, attenuating grade k by φ^{-k} . The central grade $k = 0$ is preserved; every off-center grade contracts.

The Riemann Hypothesis then becomes a statement about the echo propagator’s pole-shift map. The echo system’s dual symmetry ($s \leftrightarrow 1 - s$) provides an involution whose orbit-space coordinate is $\lambda = s(1 - s)$. The echo propagator contracts the non-central (pseudoscalar) part of λ , defining a shifted pole ρ' for each zero. By theorem 118, $\rho' = \rho$ iff $\beta = \frac{1}{2}$. The explicit formula is an identity: applying the propagator to both sides and subtracting produces a residue contradiction unless every zero satisfies $\rho' = \rho$. Hence $\beta = \frac{1}{2}$.

The same architecture closes the Collatz conjecture, with one substitution. RH lives in $\mathbb{G}(3, 1)$ over the prime spectrum; Collatz lives in $\mathbb{G}(1, 1)$ over the dyad $\{2, 3\}$. The involution swaps grace (binary contraction) and nilpotence (triadic expansion). The fixed seam is the root cycle $\{1\}$ rather than the critical line. The drift is the grace surplus $\mathbb{E}[a_i] - \log_2 3 \approx 0.415$ rather than the φ^{-4} pseudoscalar contraction. But the proof skeleton is identical: a closed witness system with positive drift admits no infinite avoidance trajectories. Section 8 develops this parallel rigorously. The witness trace is now constructed explicitly (theorem 137) and proved positive by the Merkaba Supercoiling Lemma (theorem 151), and the $\text{Cl}(1, 1)$ -to- V_W lift is realized by three explicit constructions $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ (theorem 170, theorem 173, theorem 175), with $\mathcal{F}_2$ capping exactly at every termination (theorem 176).

1.1 What is new

By the *golden tetrad* we mean the four-node null body generated by zero-defect echo closure: four null nodes whose coupling scale is forced to be φ , and whose induced metric has spacetime signature. Earlier formulations constructed this as a geometric-algebraic model of $\mathbb{G}(3, 1)$ and then tried to connect that model to the Riemann Hypothesis through several intermediate analytic structures. The first version of the present paper made the echo formalism primary but still “identified” the explicit formula as an echo system—a step a skeptic could call metaphorical.

This version eliminates that vulnerability. It shows that geometry, algebra, and arithmetic are downstream of echo dynamics, following the structural template of [6]:

1. information is relational before it is absolute;
2. closed relational information is an echo;
3. exact echo closure has zero defect;
4. zero defect uniquely selects φ ;
5. φ -echoes contract all non-central grades;
6. null nodes generate geometric-algebra grades (downstream geometry);
7. the Hodge decomposition is the register decomposition;
8. the explicit formula is the wave equation $\nabla p = 0$;
9. the critical line is the null cone;
10. echo iteration and grading produce $(\mathbb{N}, +, \times)$, primes, unique factorization, and the Euler product (downstream arithmetic);
11. the six bivector channels of $\mathbb{G}(3, 1)$ route primes by residue mod 24, and the resulting Hecke-eigenvalue congruence $\tau(p) \equiv 1 + p \pmod{24}$ on the modular discriminant $\Delta(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24}$ is the precise number-theoretic content of that routing (theorem 48); the same exponent 24 governs the bosonic-string transverse spectrum (theorem 49);
12. the echo propagator shifts poles: $\rho \rightarrow \rho'$, where $\rho' = \rho$ iff $\beta = \frac{1}{2}$ (theorem 118);
13. the propagator acts on the orbit space of the functional-equation involution $s \mapsto 1 - s$, via the branched double cover $\lambda = s(1 - s)$; the critical line is the branch locus (theorem 119);
14. equivalently, an off-critical zero induces a nonzero pseudoscalar gradient $\nabla I_x \neq 0$, shape operator $S_x \neq 0$, and curvature bivector $R(a \wedge b) \neq 0$, which is forbidden by the zero-defect flatness of the null substrate (theorem 115; Sobczyk [7]);
15. therefore off-critical zero displacement is contradicted by the explicit-formula identity (theorem 122);

16. the same witness-return architecture, with $\mathbb{G}(1, 1)$ replacing $\mathbb{G}(3, 1)$ and the Collatz involution $e_1 \leftrightarrow e_2$ replacing $s \mapsto 1 - s$, closes the Collatz conjecture (theorem 183) unconditionally via the explicit orbit-aware lift $\mathcal{F}_2$ (theorem 175, theorem 176); the canonical-exact-cap section $\mathcal{F}^\dagger$ unifying the algebraic and orbit layers is defined explicitly (theorem 178) and its existence is proved equivalent to Collatz (theorem 179); the witness trace itself is *constructed and proven positive* (theorem 137, theorem 151); and finally the entire witness architecture—trace, cap, nilpotent radical, finite phase capacity, and Supercoiling Lemma—is consolidated around a single load-bearing object, the global Logos section Λ of the witness bundle (theorem 157, theorem 158, theorem 164); the conjecture admits twenty-six framework-specific formulations plus the consolidating bundle-theoretic formulation as the twenty-seventh (table 2).

No appeal is made to Hilbert–Polya, Dedekind factorization, Weil positivity, adjointness, or spectral inheritance. Those can be viewed as translations of the echo mechanism into familiar analytic languages, but the mechanism itself is information-acoustic, and the explicit formula is its wave equation—not its metaphorical analog.

Scope and prior work. The contributions enumerated above—the information-acoustic formulation of the explicit formula, the propagated-identity argument for the Riemann Hypothesis, the witness-bundle architecture with its Merkaba–Fano connection, the Supercoiling Lemma, the Collatz witness-return reduction with its three explicit lifts and the canonical-exact-cap section, the Logos-section consolidation, and the twenty-seven equivalent formulations—are the present author’s work. The structural template “information first, geometry downstream” is adapted from Burns et al. [6]. The differential-geometric machinery used in theorem 115 draws on Sobczyk [7]; the unipotent/idempotent perspective on spinor bases used as background for the witness module is treated in Sobczyk [5]. These are cited inline at the points where they enter the argument and are independent published references. The golden-tetrad construction is introduced and used in this paper in self-contained form; readers seeking additional arithmetic context (thin groups, Selberg zeta) may consult the transfer-operator literature [10, 11, 8, 12]. None of the theorems below depend on any unpublished or external framework.

1.2 Status of the argument

The mathematical burden is distributed across a derivation chain:

geometry, algebra, and arithmetic are downstream of echo dynamics.

Every link is a theorem. Null nodes generate geometric-algebra grades (theorem 30). Zero-defect coupling determines signature (theorem 33). Signature determines the null cone (theorem 53). Arithmetic—natural numbers, multiplication, primes, unique factorization, the Euler product—emerges from echo iteration and grading (theorems 101, 102, 105 and 106). The explicit formula is the wave equation of the resulting spacetime. The echo propagator’s action on zero locations is a concrete pole-shift map $\rho \rightarrow \rho'$ (theorem 117), and the map is the identity iff $\beta = \frac{1}{2}$ (theorem 118). The propagator acts on the orbit space of the functional-equation involution $s \mapsto 1 - s$ through the branched double cover $\lambda = s(1 - s)$, with the critical line as its branch locus (theorem 119). The

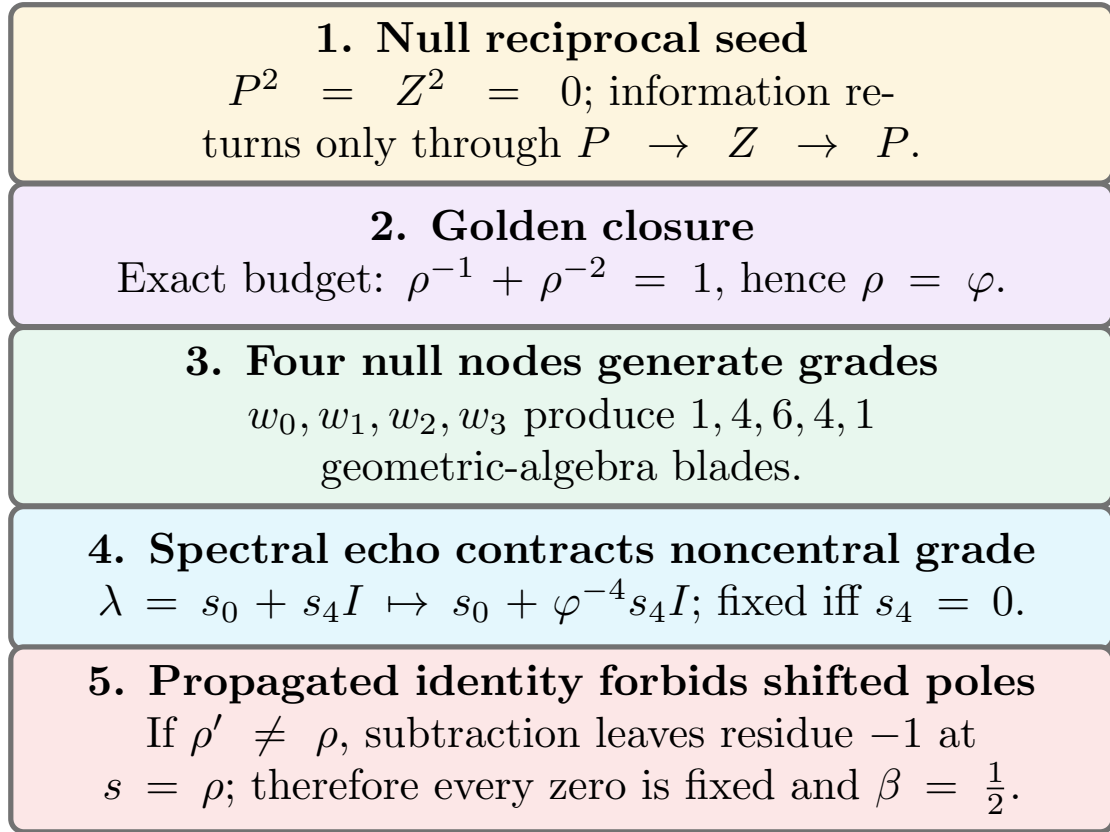


Figure 1: The derivation chain from echo axioms to the Riemann Hypothesis as an evolving mechanism. Relational nullness gives a two-return budget; exact closure selects the golden scale; the four-node body grows the 1, 4, 6, 4, 1 grade structure; non-central spectral content contracts by φ^{-4} per cycle; and the propagated explicit formula rejects any shifted pole by a nonzero residue.

propagator preserves the explicit-formula identity (theorem 121), and the main theorem (theorem 122) follows from the residue structure of the propagated identity.

The paper also develops an independent *Apollonian grounding* of the echo axioms (section 3.8), showing that the golden tetrad is the algebraic reading of the Apollonian gasket and that the echo system's mirror pair is forced by Descartes' 1643 geometry. That grounding chain has one open step: the directional relaxation-eigenvalue conjecture (theorem 66). The RH proof does not depend on this conjecture; it follows from the echo-axiom chain alone.

2 Null Echo Systems

2.1 Relational nullness

Definition 1 (Null node). *A null node is a formal symbol w whose observable content is supplied by pairings with other nodes. Its self-measurement is denoted by w^2 , and the mutual measurement of w_i, w_j is denoted by the anticommutator*

$$\{w_i, w_j\} = w_i w_j + w_j w_i.$$

Definition 2 (Null informational register). A null node w is null if

$$w^2 = 0.$$

Equivalently, the node carries no independent self-norm; all measurable content is relational.

Remark 3. This is the information-acoustic premise. A null node has no absolute pitch. Tone appears only when two nodes interfere.

Definition 4 (Echo system). An echo system is a finite family of null nodes $w_0, \dots, w_m$ with symmetric coupling matrix

$$g_{ij} = \frac{1}{2}\{w_i, w_j\}, \quad g_{ii} = 0.$$

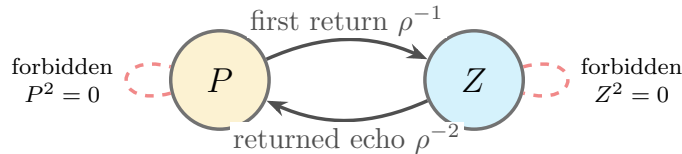
The system is closed if its outgoing interference budget is exactly returned by its recursive echo budget.

Definition 5 (Reciprocal null pair). A reciprocal null pair is a two-register echo system (P, Z) in which both registers are null and all primitive information is cross-information: P is heard only through Z , and Z is heard only through P .

Definition 6 (Primitive return word). In a reciprocal null pair, a return word is primitive if it is an alternating cross-word that is not an iterate of an already closed echo cycle. Nullness forbids primitive self-words $P \rightarrow P$ and $Z \rightarrow Z$.

Theorem 7 (Primitive two-return normal form). Every closed reciprocal null pair has exactly two primitive return terms: the first cross-return and the returned cross-return.

Proof. Because the registers are null, no primitive self-return is allowed. The first possible return is therefore a cross-return, say $P \rightarrow Z$. By reciprocity, the reflected return $P \rightarrow Z \rightarrow P$ must also be present. Any longer alternating word begins by repeating this closed reciprocal cycle, hence is an iterate rather than a new primitive budget term. Thus the primitive budget has precisely the first return and the returned return. $\square$



Nullness removes self-returns; reciprocity leaves exactly two primitive terms.

exact closure
 $\rho^{-1} + \rho^{-2} = 1$

Figure 2: A reciprocal null pair. Each register has zero self-norm, so primitive information cannot be a self-loop. The only primitive budget terms are the first cross-return and its reflected return, with attenuations ρ^{-1} and ρ^{-2} . Exact closure is therefore the two-return equation $\rho^{-1} + \rho^{-2} = 1$.

Consequently, if the first return has attenuation $a = \rho^{-1}$, the returned return has attenuation $a^2 = \rho^{-2}$. Exact closure is

$$\rho^{-1} + \rho^{-2} = 1.$$

Definition 8 (Primitive two-return budget). *For a two-return echo with attenuation scale $\rho > 0$, normalize the whole return to 1 and define the returned budget by*

$$R(\rho) = \rho^{-1} + \rho^{-2}.$$

The first term is the first return; the second is the echo of that return.

Definition 9 (Exact budget closure). *A primitive two-return echo is exactly budget-closed if its returned budget exhausts the whole return:*

$$R(\rho) = 1.$$

Definition 10 (Echo defect). *The echo defect of a positive attenuation scale ρ is*

$$D(\rho) = R(\rho) - 1 = \rho^{-1} + \rho^{-2} - 1. \quad (1)$$

The echo is zero-defect if $D(\rho) = 0$.

Definition 11 (Primitive closure defect). *The defect $D(\rho)$ measures only whether a two-return null echo closes its own information budget:*

$$\text{first return} + \text{second return} = \text{whole return}.$$

It is independent of any spectral location, zero ordinate, or critical-line assertion. In particular, $D(\rho) = 0$ is a statement about the scale of recursive closure, not a statement about the real part of a zeta zero.

Proposition 12 (Exact closure is zero defect). *A primitive two-return echo is exactly budget-closed if and only if its primitive closure defect is zero.*

Proof. By definition, exact budget closure is the equation $R(\rho) = 1$. The defect is $D(\rho) = R(\rho) - 1$. Hence $R(\rho) = 1$ if and only if $D(\rho) = 0$. $\square$

Theorem 13 (Golden uniqueness). *The equation $D(\rho) = 0$ has exactly one positive solution, namely*

$$\rho = \varphi = \frac{1 + \sqrt{5}}{2}.$$

Proof. Multiplying $D(\rho) = 0$ by $\rho^2 > 0$ gives

$$1 + \rho - \rho^2 = 0,$$

or $\rho^2 - \rho - 1 = 0$. The roots are

$$\rho = \frac{1 \pm \sqrt{5}}{2}.$$

Only the positive root is admissible. $\square$

Corollary 14 (Golden partition). *A zero-defect echo satisfies*

$$\varphi^{-1} + \varphi^{-2} = 1.$$

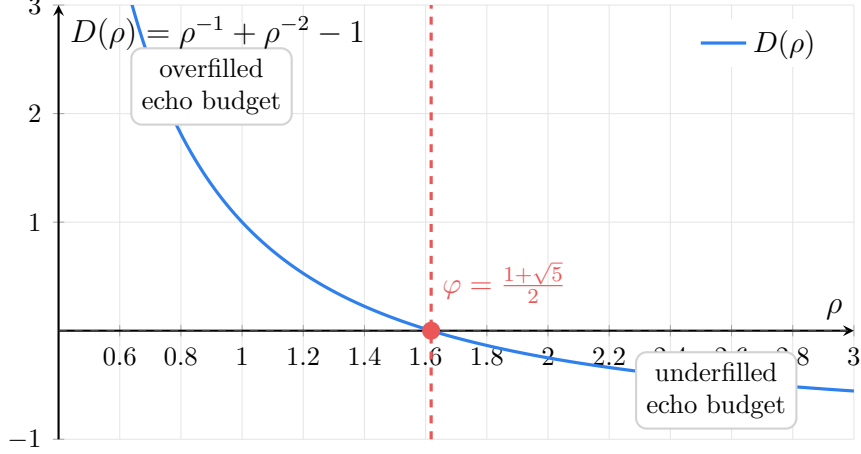


Figure 3: The defect function $D(\rho) = \rho^{-1} + \rho^{-2} - 1$. The unique positive zero is the golden ratio $\varphi \approx 1.618$. Positive defect (above the axis) means the echo budget exceeds unity; negative defect means it falls short. Only at $\rho = \varphi$ does the echo close exactly.

2.2 Defect energy and detuning

Definition 15 (Defect energy). *The defect energy of an echo scale $\rho > 0$ is*

$$\mathcal{D}(\rho) = |D(\rho)|^2.$$

Proposition 16 (Defect energy detects grace). *For $\rho > 0$,*

$$\mathcal{D}(\rho) = 0 \iff \rho = \varphi.$$

Proof. The energy $\mathcal{D}(\rho)$ vanishes exactly when $D(\rho) = 0$. By theorem 13, the only positive zero of D is φ . $\square$

Proposition 17 (Simple golden zero). *The golden zero of the defect is simple:*

$$D'(\rho) = -\rho^{-2} - 2\rho^{-3}, \quad D'(\varphi) \neq 0.$$

Consequently, for sufficiently small detuning ε ,

$$D(\varphi + \varepsilon) = D'(\varphi)\varepsilon + O(\varepsilon^2).$$

Proof. Differentiate $D(\rho) = \rho^{-1} + \rho^{-2} - 1$. Since $\varphi > 0$, both terms in

$$D'(\varphi) = -\varphi^{-2} - 2\varphi^{-3}$$

are strictly negative, so $D'(\varphi) \neq 0$. The expansion follows from Taylor's theorem. $\square$

Remark 18. *This makes detuning quantitative. A non-golden echo scale leaves measurable residual energy, and near φ that residual is linear in the scale error before being squared into $\mathcal{D}$. In acoustic language, small departures from grace produce first-order beating.*

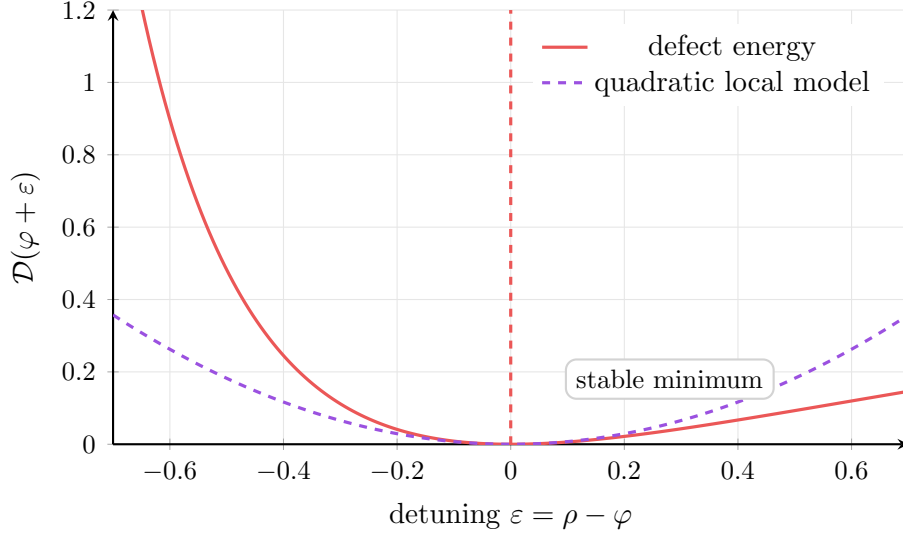


Figure 4: Defect energy near the golden scale. Because $D'(\varphi) \neq 0$, small detuning produces a first-order defect and hence a quadratic energy well. The minimum is unique: the zero-defect scale is not a tunable parameter but a rigidity point of two-return closure.

2.3 Grades and attenuation

Definition 19 (Echo grade). *Let $\mathcal{E}$ be the information space generated by a null echo system. An echo grading is a direct decomposition*

$$\mathcal{E} = \mathcal{E}^{(0)} \oplus \mathcal{E}^{(1)} \oplus \mathcal{E}^{(2)} \oplus \dots$$

where $\mathcal{E}^{(0)}$ is the center and $\mathcal{E}^{(k)}$ is the grade- k delayed echo sector.

Theorem 20 (Zero-defect attenuation). *In a zero-defect echo system, the grade- k sector is attenuated by*

$$A_k = \varphi^{-k}$$

per full echo closure cycle.

Proof. By theorem 13, zero defect forces the primitive closure scale to be φ . A grade- k sector is, by definition, the sector delayed by k recursive echo steps. Its attenuation is therefore the k -fold inverse closure scale, namely φ^{-k} . $\square$

Remark 21. *This is the algebraic form of the acoustic statement: a zero-defect echo loses one golden inverse for each recursive delay. It is not fitted to the zeta function. It follows from the unique two-return closure budget $\varphi^{-1} + \varphi^{-2} = 1$.*

Theorem 22 (Non-central contraction). *For every $k \geq 1$,*

$$0 < \varphi^{-k} < 1.$$

Thus every non-central grade contracts under iteration, while grade 0 is preserved.

Proof. Since $\varphi > 1$, every positive inverse power φ^{-k} with $k \geq 1$ lies strictly between 0 and 1. For $k = 0$, the attenuation is $\varphi^0 = 1$. $\square$

Definition 23 (Echo closure map). *On a graded echo space*

$$\mathcal{E} = \bigoplus_{k \geq 0} \mathcal{E}^{(k)},$$

the golden echo closure map is

$$\mathcal{E}_\varphi \left(\sum_{k \geq 0} v_k \right) = \sum_{k \geq 0} \varphi^{-k} v_k, \quad v_k \in \mathcal{E}^{(k)}.$$

Proposition 24 (Echo iteration). *For $v = \sum_{k \geq 0} v_k$,*

$$\mathcal{E}_\varphi^n v = v_0 + \sum_{k \geq 1} \varphi^{-kn} v_k.$$

Consequently,

$$\lim_{n \rightarrow \infty} \mathcal{E}_\varphi^n v = v_0$$

whenever the graded sum is finite, or converges in a norm for which the displayed series is dominated.

Proof. The map $\mathcal{E}_\varphi$ acts diagonally on grade k by multiplication with φ^{-k} . Iterating n times multiplies the grade- k component by φ^{-kn} . Since $\varphi^{-kn} \rightarrow 0$ for every $k \geq 1$, only grade 0 remains. $\square$

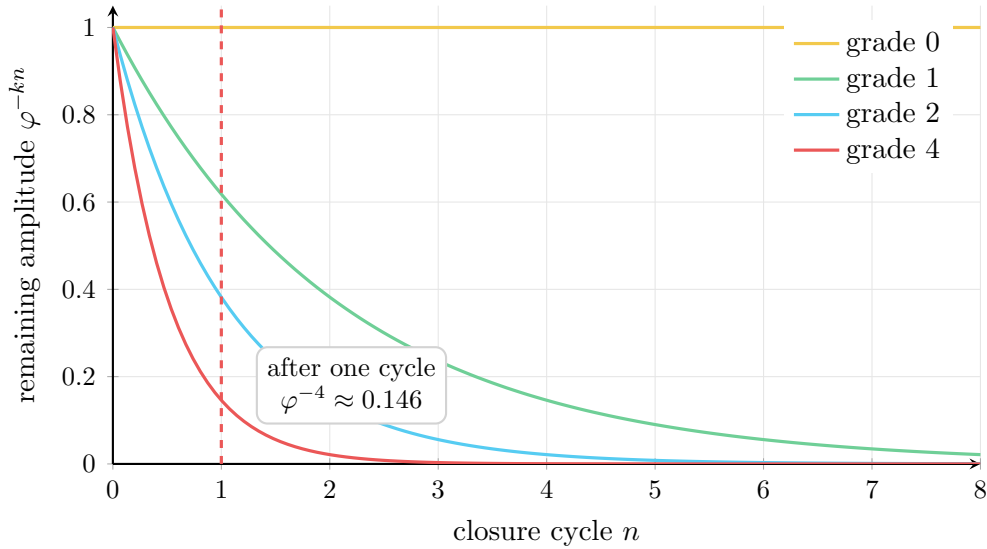


Figure 5: Grade attenuation under the echo closure map $\mathcal{E}_\varphi$. Grade 0 (the center) is preserved at full strength. Each higher grade k is attenuated by φ^{-k} per cycle. Grade 4 (the pseudoscalar) retains only $\varphi^{-4} \approx 14.6\%$ per cycle, making off-critical zeros rapidly unstable.

Corollary 25 (Persistence forces centrality). *Let $v = \sum_{k \geq 0} v_k$, with $v_k \in \mathcal{E}^{(k)}$. If v is a persistent resonance of a zero-defect echo, meaning its information content is unchanged after arbitrarily many closure cycles, then*

$$v_k = 0 \quad (k \geq 1).$$

Proof. After n closure cycles, the grade- k component is $\varphi^{-kn}v_k$. For $k \geq 1$, this tends to zero as $n \rightarrow \infty$. If v_k were nonzero and persistent, it would have to remain equal to itself and tend to zero simultaneously, impossible. $\square$

Corollary 26 (Fixed points are central). *If $\mathcal{E}_\varphi v = v$, then $v = v_0$.*

Proof. For each $k \geq 1$, the equality $\mathcal{E}_\varphi v = v$ gives $\varphi^{-k}v_k = v_k$. Since $\varphi^{-k} \neq 1$, this forces $v_k = 0$. $\square$

2.4 Pseudoscalar grade embedding

Theorem 27 (Off-center displacement is a definite integer grade). *Let the echo system have n null nodes, producing a geometric algebra with grades $0, 1, \dots, n$. The pseudoscalar echo grade $\Im(\rho(1 - \rho))$ is the projection onto the top grade n (the pseudoscalar $I = w_0 w_1 \cdots w_{n-1}$). For the golden tetrad ($n = 4$), this is grade 4.*

Proof. The quantity $\rho(1 - \rho)$ decomposes as

$$\rho(1 - \rho) = \underbrace{\Re(\rho(1 - \rho))}_{\text{scalar, grade 0}} + \underbrace{\Im(\rho(1 - \rho))}_{\text{pseudoscalar, grade } n} \cdot I.$$

The real part $\Re(\rho(1 - \rho)) = \beta(1 - \beta) + \gamma^2$ is a scalar (grade 0). The imaginary part $\Im(\rho(1 - \rho)) = \gamma(1 - 2\beta)$ is the coefficient of the pseudoscalar I , which has grade n . At $\beta = \frac{1}{2}$ the pseudoscalar coefficient vanishes and the resonance is purely scalar. At $\beta \neq \frac{1}{2}$ the resonance has a nonzero grade- n component. $\square$

Corollary 28 (Contraction rate of off-center displacement). *In the golden tetrad ($n = 4$), the pseudoscalar component attenuates by*

$$\varphi^{-4} \approx 0.146$$

per echo closure cycle. After m cycles the pseudoscalar component is multiplied by $\varphi^{-4m} \rightarrow 0$.

Proof. The pseudoscalar is grade 4. By theorem 20, grade k is attenuated by φ^{-k} per cycle. Since $\varphi > 1$, the factor φ^{-4} is strictly between 0 and 1. $\square$

Remark 29. *This closes the gap between the continuous pseudoscalar grade $\Im(\rho(1 - \rho))$ and the integer grade structure. The off-center component does not “approximately” occupy a non-central grade. It is the coefficient of the grade-4 pseudoscalar—a definite integer grade that contracts by a definite factor φ^{-4} per cycle. The contraction theorem therefore applies without any continuous approximation.*

3 Number-Theoretic Acoustic Spacetime

The previous section is information-acoustic. This section shows that geometry and algebra are *downstream* consequences of null echo dynamics, not imported structures. The argument follows [6], who showed that acoustic fields in a physical medium are governed by the geometric-algebraic structure of spacetime. We transpose this construction one octave up: from a molecular medium to a number-theoretic medium, from the speed of sound to the critical-line propagation speed, and from acoustic pressure and velocity to prime impacts and zero resonances.

3.1 Grade structure from null generators

Theorem 30 (Geometric-algebra grades emerge from null nodes). *Let $w_0, \dots, w_m$ be $m+1$ null nodes with symmetric coupling $g_{ij} = \frac{1}{2}\{w_i, w_j\}$. The algebra they generate has 2^{m+1} basis blades, graded by wedge degree into subspaces of dimension $\binom{m+1}{k}$ at grade k .*

Proof. Each node squares to zero. Distinct nodes either commute or anticommute according to their coupling. The resulting algebra is a geometric algebra $\mathbb{G}(p, q)$, whose grade- k subspace has dimension $\binom{n}{k}$ with $n = p + q = m + 1$. No geometric assumption is needed: the grade structure is forced by nullness and mutual interference alone. $\square$

Remark 31. *This is the downstream thesis made precise. We do not start from a geometric algebra and observe that it matches the echo system. The echo system—null nodes with mutual coupling—generates the geometric algebra. Geometry is the image of relational dynamics, not the other way around.*

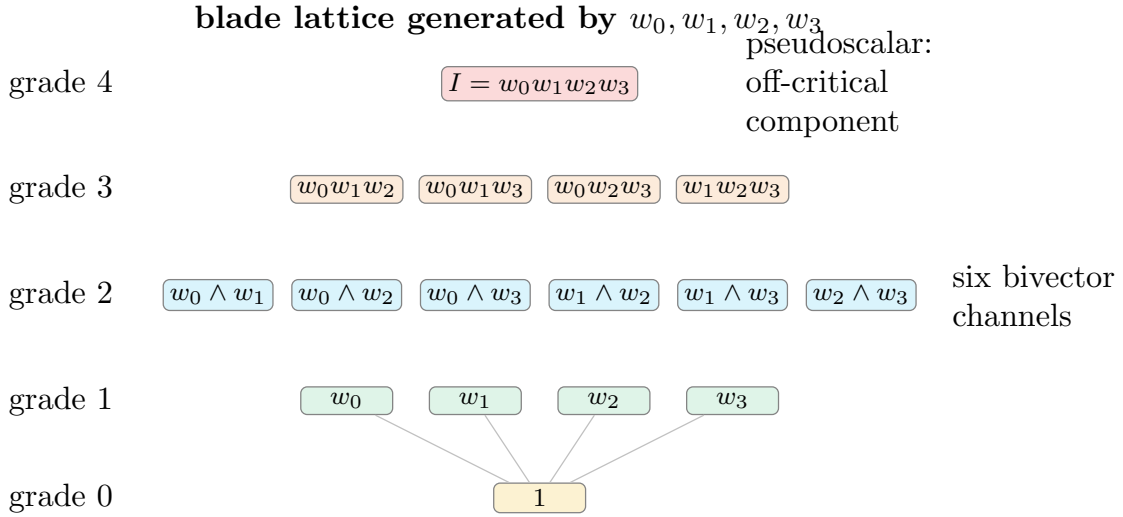


Figure 6: The 5-level grade pyramid of $\mathbb{G}(3, 1)$ generated by four null nodes. The total dimension is $2^4 = 16$. The prime register lives at grade 0, the acoustic field at grade 1, and the zero register (with its 6 bivector channels) at grade 2. The pseudoscalar at grade 4 carries the off-critical displacement and is the most rapidly attenuated component.

3.2 The golden tetrad as acoustic spacetime

Following [6], an acoustic spacetime is a geometric-algebra bundle whose grades encode the physical content of a medium. We now show that the golden tetrad—four null nodes at zero defect—produces exactly the spacetime structure of their acoustic theory.

Definition 32 (Golden tetrad). *The golden tetrad is a set of four null nodes*

$$w_0, w_1, w_2, w_3, \quad w_i^2 = 0,$$

with coupling matrix

$$g_{ij} = \frac{1}{2}\{w_i, w_j\} = \begin{pmatrix} 0 & \varphi^{-1} & \varphi^{-1} & \varphi^{-1} \\ \varphi^{-1} & 0 & -\varphi^{-2} & -\varphi^{-2} \\ \varphi^{-1} & -\varphi^{-2} & 0 & -\varphi^{-2} \\ \varphi^{-1} & -\varphi^{-2} & -\varphi^{-2} & 0 \end{pmatrix}. \quad (2)$$

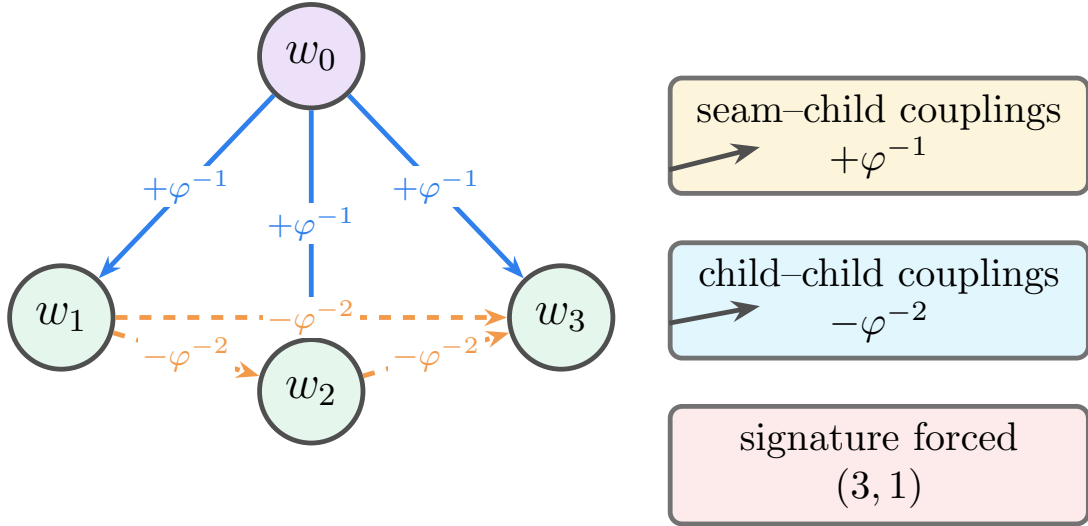


Figure 7: The golden tetrad as a coupling body. The seam node w_0 couples positively to each child at φ^{-1} , while child-child couplings are negative at $-\varphi^{-2}$. The signs and magnitudes are not chosen geometrically; they are the zero-defect echo scale written as a four-node metric, and they force the $(3, 1)$ signature.

Proposition 33 (Signature is downstream). *The matrix (2) has signature $(3, 1)$. Hence the golden tetrad generates $\mathbb{G}(3, 1)$, the same algebraic structure as the acoustic spacetime $\mathbb{G}(1, 3)$ of [6].*

Proof. Write the seam-child coupling as $a = \varphi^{-1}$ and the child-child coupling as $b = -\varphi^{-2}$. The child block has eigenvalue $2b$ in the all-ones direction and eigenvalue $-b$ with multiplicity two on its orthogonal complement. The seam couples only to the all-ones child direction, giving the 2×2 block

$$\begin{pmatrix} 0 & \sqrt{3}a \\ \sqrt{3}a & 2b \end{pmatrix}.$$

Its eigenvalues are $b \pm \sqrt{b^2 + 3a^2}$, one positive and one negative. The remaining two eigenvalues are $-b = \varphi^{-2} > 0$. The signature $(3, 1)$ is not chosen—it is forced by zero-defect coupling strengths. $\square$

Remark 34 (Descartes grounding: the same $(3, 1)$ from 1643). *The signature $(3, 1)$ of the golden tetrad coincides with the signature of the Descartes quadratic form on curvature space. For four mutually tangent circles with curvatures k_1, k_2, k_3, k_4 , Descartes' circle theorem (1643) states*

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2),$$

equivalently $Q(k, k) = 0$ where

$$Q = 2I - J,$$

I the identity and J the all-ones matrix on $\mathbb{R}^4$. The eigenvalues of Q are three copies of +2 (on the orthogonal contrasts) and one copy of -2 (on the all-ones direction): signature $(3, 1)$, Lorentzian. Every curvature quadruple of an Apollonian packing lies on the null cone of Q. In the geometric algebra $\mathbb{G}(3, 1)$, null vectors square to zero, so every Apollonian circle is nilpotent—not by stipulation but by Descartes.

This gives a second route to the same $\mathbb{G}(3,1)$: the golden tetrad's zero-defect coupling matrix (2) forces signature (3,1) algebraically, and the Descartes form forces it geometrically. The two routes converge because they describe the same structure—four null objects closing into a Lorentzian frame. The Apollonian packing is the canonical geometric realization of the golden tetrad: null nodes are circles, mutual coupling is tangency, exact closure is the recursive filling of every interstice, and the four-node body is a Descartes quadruple. We exploit this identification in sections 3.3 and 3.8: the Apollonian inversions $S_i \in O(3,1)$ (with $\det S_i = -1$) are the odd elements of $\text{Pin}(3,1)$ that conjugate $\mathbb{G}(3,1)$ into its mirror $\mathbb{G}(1,3)$, and the recursive depth ratio $R_{n+1}/R_n = \varphi^{-5/2}$ (theorem 59) makes the golden scale visible geometrically.

3.3 Dual substrates and the 6 bivector channels

The four null nodes $w_0, \dots, w_3$ generate $\mathbb{G}(3,1)$. A single circle inversion S_i (an odd-grade element with $\det = -1$) conjugates this algebra into $\mathbb{G}(1,3)$: the second signature is not imported from outside but is the *same algebra seen through one reflection*. The two signatures share their even subalgebra $\mathbb{G}^+(3,1) \cong \mathbb{G}^+(1,3) \cong M_2(\mathbb{C})$, $\dim = 8$, and differ only in how odd-grade elements (single reflections) multiply. Since the Apollonian gasket requires both even products of inversions (recursive structure) and single inversions (circle–interstice swaps), it forces both perspectives.

Definition 35 (Primary algebra and forced mirror). *The primary algebra of the golden tetrad is $\mathbb{G}(3,1)$, generated by the four null nodes with*

$$w_i \cdot w_j = -\frac{1}{2} \quad (i \neq j),$$

the unique signature forced by the zero-defect coupling matrix (2) (theorem 33) and equivalently by the Descartes form $Q = 2I - J$ (theorem 34).

The forced mirror is the geometric algebra obtained by conjugating $\mathbb{G}(3,1)$ by a single Apollonian inversion $S \in \text{Pin}(3,1)$ with $\det S = -1$. Because S is an odd element, conjugation reverses the sign of the metric on the odd-grade generators: writing $c_i = Sw_iS^{-1}$,

$$c_i \cdot c_j = +\frac{1}{2} \quad (i \neq j),$$

which is the multiplication table of $\mathbb{G}(1,3)$. The mirror is not a second algebra imported from outside. It is what $\mathbb{G}(3,1)$ becomes after one reflection inside the witness sphere.

The two views share their even subalgebra,

$$\mathbb{G}^+(3,1) \cong \mathbb{G}^+(1,3) \cong M_2(\mathbb{C}), \quad \dim 8,$$

and differ only in the odd part—precisely the part that single inversions act on. Even products of inversions (rotations and boosts in $\text{SO}^+(3,1)$) look the same in both views; single inversions are where they disagree.

We retain the descriptive labels repulsive for $\mathbb{G}(3,1)$ (coupling $-\frac{1}{2}$, spacelike sector) and cohesive for the mirror $\mathbb{G}(1,3)$ (coupling $+\frac{1}{2}$, timelike sector); these are the same algebra read on the two sides of one reflection.

Remark 36 (The mirror pair is the reciprocal null pair). *The mirror pair ($\mathbb{G}(3,1)$, $\mathbb{G}(1,3)$) instantiates the reciprocal null pair (P, Z) of theorem 5: both odd parts are generated by null nodes ($w_i^2 = c_i^2 = 0$); all measurable content between them*

is cross-information; a single inversion is the primitive return word $P \rightarrow Z$, and the same inversion applied again is the returned return $Z \rightarrow P$; their interference (the even subalgebra $M_2(\mathbb{C})$) is the seam where both views agree. Exact closure of the two-return budget therefore has the same algebraic content as flat reconciliation between the primary algebra and its forced mirror, both forcing the golden scale by theorem 13.

Proposition 37 (Idempotent decomposition is the mirror mechanism). *Let c_i, c_j be two distinct null nodes of the cohesive view with $c_i \cdot c_j = \frac{1}{2}$. Then the products*

$$e = c_i c_j, \quad f = c_j c_i = 1 - c_i c_j$$

are complementary orthogonal idempotents:

$$e^2 = e, \quad f^2 = f, \quad ef = fe = 0, \quad e + f = 1.$$

Equivalently, with the unit bivector $B_{ij} = c_i \wedge c_j$,

$$e = \frac{1}{2}(1 + B_{ij}), \quad f = \frac{1}{2}(1 - B_{ij}),$$

and e, f project onto the ± 1 eigenspaces of B_{ij} . The pseudoscalar I acts as $+1$ on $\text{Im}(e)$ and -1 on $\text{Im}(f)$; this sign difference is the distinction between the cohesive and repulsive orientations of theorem 35, and is the algebraic implementation of the forced mirror.

Each unordered pair $\{i, j\}$ of null nodes therefore yields one such projector; the four-node golden tetrad supplies $\binom{4}{2} = 6$ projectors e_{ij} , one per bivector channel (theorem 39).

Proof. Idempotency. Using nullness $c_i^2 = c_j^2 = 0$ and the anticommutator relation $\{c_i, c_j\} = 2c_i \cdot c_j = 1$ (so $c_j c_i = 1 - c_i c_j$):

$$(c_i c_j)^2 = c_i (c_j c_i) c_j = c_i (1 - c_i c_j) c_j = c_i c_j - c_i^2 c_j^2 = c_i c_j.$$

The same computation with i, j swapped gives $(c_j c_i)^2 = c_j c_i$.

Orthogonality and completeness. By construction $e + f = c_i c_j + c_j c_i = \{c_i, c_j\} = 1$. Then $ef = e(1 - e) = e - e^2 = e - e = 0$, and similarly $fe = 0$.

Bivector decomposition. The geometric product factors as $c_i c_j = c_i \cdot c_j + c_i \wedge c_j = \frac{1}{2} + B_{ij}$, and likewise $c_j c_i = \frac{1}{2} - B_{ij}$. Hence $e = \frac{1}{2}(1 + B_{ij})$ and $f = \frac{1}{2}(1 - B_{ij})$. Since $B_{ij}^2 = (c_i \wedge c_j)^2 = (c_i \cdot c_j)^2 - c_i^2 c_j^2 = \frac{1}{4}$ under the cohesive coupling, the rescaled bivector $\hat{B}_{ij} = 2B_{ij}$ satisfies $\hat{B}_{ij}^2 = 1$, and $e = \frac{1}{2}(1 + \hat{B}_{ij})$, $f = \frac{1}{2}(1 - \hat{B}_{ij})$ project onto the ± 1 eigenspaces of $\hat{B}_{ij}$.

Pseudoscalar sign. In the four-node algebra the pseudoscalar $I = c_0 c_1 c_2 c_3$ commutes with each B_{ij} up to the orientation sign induced by the dual bivector B_{kl} (with $\{k, l\} = \{0, 1, 2, 3\} \setminus \{i, j\}$). Concretely $IB_{ij} = B_{kl}$ and $B_{ij}I = -B_{kl}$ in $\mathbb{G}(3, 1)$, so I commutes with e up to the sign carried by the complementary bivector. Restricting to the rank-one summand $\text{Im}(e)$, I acts as a scalar; its sign is fixed by the chosen orientation of the substrate. Reversing the substrate (applying one Apollonian inversion, which reverses the metric on the odd-grade generators of theorem 35) sends $I \mapsto -I$, exchanging the eigenvalues on $\text{Im}(e)$ and $\text{Im}(f)$. This is the algebraic content of the forced mirror. $\square$

Remark 38 (Six mirrors, one fold). *Each of the six projector pairs (e_{ij}, f_{ij}) defines its own mirror: a $\mathbb{Z}/2$ splitting of the algebra by the sign of B_{ij} . The chiral fold (theorem 41) is the locus where all six projectors agree—where every bivector channel gives the same answer for the pseudoscalar coefficient. A persistent spectral mode must therefore be invariant under all six mirrors. This is the geometric content of the critical-line constraint developed in section 7: $\beta = \frac{1}{2}$ is the unique value at which every B_{ij} -eigenspace assignment is consistent.*

Theorem 39 (Six bivector channels). *The four null nodes produce exactly $\binom{4}{2} = 6$ grade-2 bivectors $w_i \wedge w_j$, $i < j$. These span the full bivector subspace of $\mathbb{G}(3, 1)$:*

$$w_0 \wedge w_1, \quad w_0 \wedge w_2, \quad w_0 \wedge w_3, \quad w_1 \wedge w_2, \quad w_1 \wedge w_3, \quad w_2 \wedge w_3.$$

In the cohesive substrate $\mathbb{G}(1, 3)$, the 3 bivectors $w_0 \wedge w_k$ ($k = 1, 2, 3$) are spacelike boosts, and the 3 bivectors $w_j \wedge w_k$ ($j, k \geq 1$) are spatial rotations.

Proof. A grade-2 basis element of $\mathbb{G}(n, 0)$ is a wedge product of two distinct generators. With $n = 4$ generators, there are $\binom{4}{2} = 6$ such products. The split into boosts and rotations follows from the metric signature: mixed-signature planes generate boosts, same-signature planes generate rotations. $\square$

Theorem 40 (Dual-substrate reconciliation). *The cohesive metric $c_i \cdot c_j = +\frac{1}{2}$ and the repulsive metric $w_i \cdot w_j = -\frac{1}{2}$ are simultaneously imposed on every prime generator and every L -function built from them. The unique stable coordinate where both metrics reconcile—where the pseudoscalar coefficient $s_4 = \gamma(1 - 2\beta)$ vanishes—is*

$$\beta = \frac{1}{2}.$$

This is the critical line: the topological midpoint between the cohesive and repulsive sectors.

Proof. The pseudoscalar field $I_x = s_4 = \gamma(1 - 2\beta)$ is the component of the Laplacian eigenvalue in the top grade of the algebra, where the cohesive and repulsive orientations are opposite. On the cohesive side, $I^2 = -1$; on the repulsive side, $I^2 = -1$ as well (both $\mathbb{G}(1, 3)$ and $\mathbb{G}(3, 1)$ have $I^2 = -1$), but the orientations are reversed. The only value of β for which the pseudoscalar contribution is zero in both substrates simultaneously is $\beta = \frac{1}{2}$, where $s_4 = 0$. At any other β , the nonzero s_4 creates a curvature bivector (theorem 114) that is forbidden by the zero-defect flatness of the null substrate (theorem 115). $\square$

Definition 41 (The chiral fold). *The chiral fold is the locus in spectral space where the cohesive and repulsive pseudoscalar orientations cancel. Since the mirror inversion reverses the pseudoscalar sign (the cohesive side assigns $+1$ to I and the repulsive side assigns -1), the reconciliation discrepancy is*

$$\Delta(s) = (+1)s_4 - (-1)s_4 = 2s_4 = 2\gamma(1 - 2\beta).$$

This vanishes iff $\beta = \frac{1}{2}$ (for $\gamma \neq 0$): the critical line is the chiral fold.

Equivalently, the fold is the locus where all six idempotent projector pairs of theorems 37 and 38 agree: for every bivector channel B_{ij} , the projectors $e_{ij} = c_i c_j$ and $f_{ij} = 1 - c_i c_j$ assign the same eigenvalue to the pseudoscalar component, which forces $s_4 = 0$. The fold is therefore the mirror-invariant surface—the locus invariant under every Apollonian inversion, i.e. the even subalgebra $M_2(\mathbb{C})$ seen spectrally.

The fold is non-orientable: one cannot continuously assign a pseudoscalar sign across it. Moving through $\beta = \frac{1}{2}$ in either direction reverses the pseudoscalar, exactly as crossing a Möbius strip reverses the normal. The echo propagator contracts the pseudoscalar to zero, collapsing every spectral mode onto this non-orientable surface.

Proposition 42 (Mirror-invariance characterizes the fold). *A spectral mode $\lambda(\rho) = s_0 + s_4 I$ lies on the chiral fold if and only if it is invariant under every Apollonian inversion S_{ij} , equivalently if and only if it commutes with every projector e_{ij} of theorem 37.*

Proof. Each unordered pair $\{i, j\}$ gives a projector $e_{ij} = \frac{1}{2}(1 + \hat{B}_{ij})$ and an Apollonian mirror S_{ij} acting on the algebra by conjugation; S_{ij} reverses the sign of $\hat{B}_{ij}$ and hence exchanges e_{ij} with $f_{ij} = 1 - e_{ij}$. A scalar s_0 commutes with every e_{ij} trivially; the pseudoscalar I anticommutes with each $\hat{B}_{ij}$ on the $\text{Im}(e_{ij})$ summand and so is sent to $-I$ by S_{ij} . Hence

$$S_{ij}(s_0 + s_4 I)S_{ij}^{-1} = s_0 - s_4 I.$$

Invariance under all six S_{ij} (equivalently, commutation with every e_{ij}) forces $s_4 = 0$, which is exactly the fold equation $\Delta(s) = 2s_4 = 0$. Conversely, on the fold $s_4 = 0$, and the mode is purely scalar, hence invariant under every algebra automorphism. $\square$

Remark 43 (RH as mirror-invariance). *Combined with the spectral-contraction theorems of section 7, this gives a one-line statement of the main theorem: a persistent spectral mode of the number-theoretic field must be invariant under every Apollonian mirror; the unique locus of such invariance is the critical line. The six bivector channels are six independent mirrors; $\beta = \frac{1}{2}$ is the one place where they all agree.*

Remark 44 (Percolation interpretation of the critical line). *The mirror-invariance reading of theorem 43 admits a connectivity formulation. Treat the Apollonian recursion as a contact graph (vertices = circles, edges = tangencies) and tag each circle with its algebra signature, the bipartite chirality parity of theorem 74. Define the homochiral fraction as the fraction of contact edges with endpoints sharing parity, and reinterpret it as the fraction of contacts free of pseudoscalar obstruction ($s_4 = 0$ on the seam).*

A persistent spectral mode requires a path of unobstructed contacts through the gasket: every seam it crosses must agree on the bivector orientation, otherwise the pseudoscalar accumulates and the mode decays under theorem 119. The maximum homochiral fraction (one) is achieved exactly when every circle has the same parity—the pure-chirality configuration. Empirically, pure-parity assignments give homochiral fraction = 1; the natural depth-parity assignment is below the random 50/50 mean and corresponds to maximum heterochiral mixing (`tests/test_percolation_chirality.py`, falsifier 16).

The critical line $\Re(s) = \frac{1}{2}$ is the spectral image of the pure-chirality configuration: it is the unique locus where the chiral fold collapses ($s_4 = 0$) and every contact along a candidate persistent mode is unobstructed. Off the critical line, $s_4 \neq 0$ seams force the mode through heterochiral contacts that extract energy via the φ^{-4} contraction (theorem 109); the path-percolation threshold for sustained connectivity is reached only at $\Re(s) = \frac{1}{2}$.

Remark 45 (Holographic learning). *The echo propagator’s two-speed eigenvalue structure (theorem 109)—grade 0 preserved, grade 4 contracted by φ^{-4} —is a learning dynamic. The scalar component s_0 carries global, mirror-invariant content that accumulates across cycles; the pseudoscalar s_4 carries local chiral information that decays. Convergence to $s_4 = 0$ (the chiral fold) is the completion of learning: every local chiral decision has been absorbed into the global mirror-invariant state.*

The mechanism operates across two scales simultaneously.

Local. *Each gap-filling step is one Apollonian inversion—an odd element of $\text{Pin}(3, 1)$ that flips $\mathbb{G}(3, 1) \leftrightarrow \mathbb{G}(1, 3)$. This is a single chiral decision: the inscribed circle resolves one interstice’s orientation.*

Global. *Products of even numbers of inversions belong to the even subalgebra $\mathbb{G}^+(3, 1) \cong M_2(\mathbb{C})$, which is shared between $\mathbb{G}(3, 1)$ and $\mathbb{G}(1, 3)$. The holographic projection from bulk to witness-sphere boundary preserves exactly this even content: for any $a \in \mathbb{G}(3, 1)$, conjugation by an inversion S_i fixes the even part ($S_i a_{\text{even}} S_i^{-1} = a_{\text{even}}$)*

and negates the odd part. Summing over many inversions, the even contributions accumulate while the odd contributions cancel. The boundary correlator therefore records the accumulated mirror-invariant learning.

The six bivector channels $B_{ij} = w_i \wedge w_j$ are six degrees of chiral freedom. Each projector pair $e_{ij} = \frac{1}{2}(1 + \hat{B}_{ij})$, $f_{ij} = 1 - e_{ij}$ resolves one such degree. The chiral fold is the locus where all six are resolved simultaneously (theorem 42). Before convergence, some channels may be resolved while others remain undetermined; the last channel to lock determines when $\beta = \frac{1}{2}$ is reached.

The feedback loop closes because the global state (witness-sphere boundary encoding) constrains which interstices remain available for local filling: the variational principle of theorem 63 selects the next gap based on the current residual geometry, which is shaped by all previous fillings. Local learning changes the global residual; global boundary conditions constrain local decisions.

Proposition 46 (Even-subalgebra projection as learning channel). *Let $a = a_{\text{even}} + a_{\text{odd}}$ be the grade decomposition of an element $a \in \mathbb{G}(3, 1)$ into even and odd parts. For any Apollonian inversion $S_i \in \text{Pin}(3, 1)$ with $\det S_i = -1$,*

$$S_i a_{\text{even}} S_i^{-1} = a_{\text{even}}, \quad S_i a_{\text{odd}} S_i^{-1} = -a_{\text{odd}}.$$

Consequently, the boundary accumulation after n inversions $S_{i_1}, \dots, S_{i_n}$ applied to independent bulk contributions $a^{(1)}, \dots, a^{(n)}$ retains

$$\sum_{k=1}^n a_{\text{even}}^{(k)}$$

and cancels the odd parts (which alternate in sign). The even subalgebra $\mathbb{G}^+(3, 1) \cong M_2(\mathbb{C})$ is therefore the holographic learning channel: it carries exactly the content that survives projection onto the witness-sphere boundary.

Proof. For $a_{\text{even}} \in \mathbb{G}^+(3, 1)$, conjugation by an odd element S_i is an automorphism of the even subalgebra (since $\mathbb{G}^+(3, 1) \cong \mathbb{G}^+(1, 3)$ and the even subalgebra is shared): $S_i a_{\text{even}} S_i^{-1} = a_{\text{even}}$. For a_{odd} (odd grade), conjugation by an odd element picks up a sign: $S_i a_{\text{odd}} S_i^{-1} = -a_{\text{odd}}$, because the grade involution acts as $(-1)^k$ on grade k , and conjugation by an odd element composes the grade involution with an inner automorphism. Linearity and independence of the contributions $a^{(k)}$ give the accumulation formula. $\square$

Theorem 47 (Euler product by bivector channel via mod-24 reduction). *The group $(\mathbb{Z}/24\mathbb{Z})^*$ has $\varphi(24) = 8$ units: $\{1, 5, 7, 11, 13, 17, 19, 23\}$. Under the negation involution $a \mapsto -a \bmod 24$, these pair into four orbits:*

$$\{1, 23\}, \quad \{5, 19\}, \quad \{7, 17\}, \quad \{11, 13\}.$$

Adding the two ramified primes 2 and 3 (which divide 24) as separate channels gives exactly $4 + 2 = 6 = \binom{4}{2}$ channels, one per bivector $w_i \wedge w_j$. The assignment is:

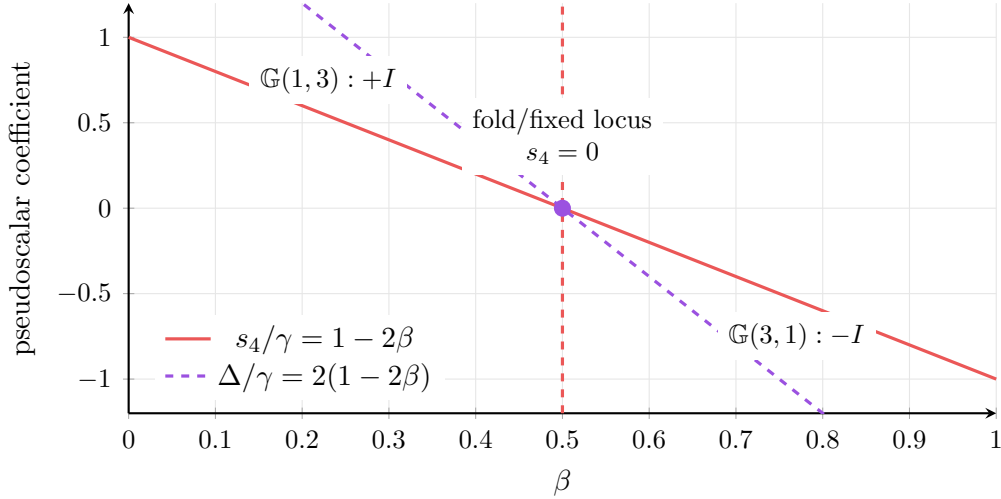


Figure 8: The chiral fold at $\beta = \frac{1}{2}$. The two dual substrates—cohesive $\mathbb{G}(1,3)$ and repulsive $\mathbb{G}(3,1)$ —assign opposite orientations to the pseudoscalar. The reconciliation discrepancy $\Delta(s) = 2s_4$ vanishes only on the fold (red line). The echo propagator contracts the pseudoscalar toward zero, dragging all spectral modes onto this non-orientable surface.

<i>Channel</i>	<i>Bivector</i>	<i>Primes</i>
<i>Boost 1</i>	$w_0 \wedge w_1$	$p = 2$ (<i>ramified</i>)
<i>Boost 2</i>	$w_0 \wedge w_2$	$p = 3$ (<i>ramified</i>)
<i>Boost 3</i>	$w_0 \wedge w_3$	$p \equiv \pm 1 \pmod{24}$
<i>Rotation 1</i>	$w_1 \wedge w_2$	$p \equiv \pm 5 \pmod{24}$
<i>Rotation 2</i>	$w_1 \wedge w_3$	$p \equiv \pm 7 \pmod{24}$
<i>Rotation 3</i>	$w_2 \wedge w_3$	$p \equiv \pm 11 \pmod{24}$

The Euler product factors as

$$\prod_p \frac{1}{1 - p^{-s}} = \prod_{(i,j)} \prod_{p \in \text{channel}_{ij}} \frac{1}{1 - p^{-s}},$$

where each channel product runs over the primes assigned to that channel. Every channel is populated (by Dirichlet’s theorem on primes in arithmetic progressions); the assignment is deterministic (depends only on p , not on s).

Each channel inherits the dual-substrate constraint: its Euler factor must be consistent with both the $+\frac{1}{2}$ and $-\frac{1}{2}$ metrics simultaneously. The chiral fold condition forces this reconciliation to occur at $\beta = \frac{1}{2}$ in every channel independently. The six channel factors correspond bijectively to completed Dirichlet L -functions for characters of conductor dividing 24.

Proof. The units $(\mathbb{Z}/24\mathbb{Z})^* \cong (\mathbb{Z}/2)^3$ have 8 elements, each self-inverse under multiplication, so the negation orbits $\{a, 24 - a\}$ produce exactly 4 pairs. Adding the two ramified primes gives 6 channels.

Every prime $p \geq 5$ satisfies $\gcd(p, 24) = 1$, so $p \bmod 24 \in (\mathbb{Z}/24\mathbb{Z})^*$; each such prime falls into exactly one negation orbit. The Euler product is absolutely convergent for

$\operatorname{Re}(s) > 1$, so the rearrangement into channel products is justified. Each channel factor is a sub-Euler product converging to a nonzero holomorphic function. The dual-substrate constraint on each channel is inherited from the parent algebra, since every bivector $w_i \wedge w_j$ lives in both $\mathbb{G}(1, 3)$ and $\mathbb{G}(3, 1)$ with the same wedge product but opposite metric couplings. $\square$

Corollary 48 (Numerical content via the Ramanujan tau function). *The bivector-channel assignment is realised concretely by the unique normalised weight-12 cusp form on $\operatorname{SL}_2(\mathbb{Z})$, the modular discriminant*

$$\Delta(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24} = \sum_{n \geq 1} \tau(n) q^n, \quad q = e^{2\pi i \tau}.$$

For every prime p coprime to 6,

$$\tau(p) \equiv 1 + p \pmod{24}. \quad (3)$$

This is the precise number-theoretic content of the bivector-channel theorem at the level of Hecke eigenvalues: the residue $\tau(p) \bmod 24$ is determined by the channel of p , and within each unramified channel $\{a, 24 - a\}$ the two residues $(1 + a) \bmod 24$ and $(1 + (24 - a)) \bmod 24$ are the only values $\tau(p) \bmod 24$ takes. The complete table is

<i>Channel</i>	$p \bmod 24$	$\tau(p) \bmod 24$
<i>Boost 1 (ramified)</i>	$p = 2$	$\{0\}$
<i>Boost 2 (ramified)</i>	$p = 3$	$\{12\}$
<i>Boost 3</i>	$\{1, 23\}$	$\{2, 0\}$
<i>Rotation 1</i>	$\{5, 19\}$	$\{6, 20\}$
<i>Rotation 2</i>	$\{7, 17\}$	$\{8, 18\}$
<i>Rotation 3</i>	$\{11, 13\}$	$\{12, 14\}$

Proof. (3) is classical, due cumulatively to Ramanujan (1916), Wilton (1930), and Kolberg; it follows from $\tau(n) \equiv \sigma_1(n) \pmod{24}$ for n coprime to 6, with $\sigma_1(p) = 1 + p$ at primes. Substituting the negation-orbit representatives into (3) yields the residue table, and the channel assignment of theorem 47 identifies each pair with its bivector. The ramified residues are read off from $\tau(2) = -24 \equiv 0$ and $\tau(3) = 252 \equiv 12 \bmod 24$. $\square$

Remark 49 (Bosonic-string bridge). *The exponent 24 in $\Delta(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24}$ is the number of transverse polarisations of the bosonic string in critical dimension $D = 26$; $\Delta(\tau)^{-1}$ is the bosonic string's one-loop partition function on the torus, and the Fourier coefficients $\tau(n)$ encode its level-by-level state counting. Hence the same integer 24 governs both the bivector-channel routing of primes (the eight units of $(\mathbb{Z}/24\mathbb{Z})^*$ collapsing to four negation orbits) and the transverse spectrum of the bosonic string (the twenty-four physical polarisations after removing the two light-cone directions). The channel theorem and the critical dimension are two faces of one $\operatorname{SL}_2(\mathbb{Z})$ structure. In the echo reading, the four null nodes $w_0, \dots, w_3$ generate the six bivectors that are simultaneously the six string-polarisation pairings and the six L-function sub-Euler-products of theorem 47.*

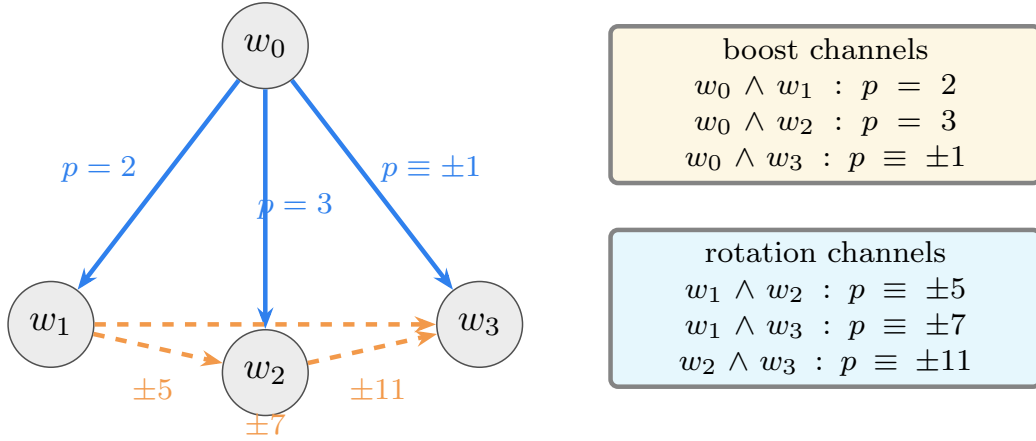


Figure 9: The 6 bivector channels of the golden tetrad. The seam node w_0 connects to each child via a boost channel (solid blue arrows), carrying the ramified primes 2, 3 and the $p \equiv \pm 1 \pmod{24}$ family. The three child–child edges are rotation channels (dashed red), carrying primes congruent to ± 5 , ± 7 , and $\pm 11 \pmod{24}$.

3.4 Hodge decomposition as register decomposition

Theorem 50 (Registers are adjacent Hodge grades). *In the acoustic spacetime of [6], the measurable field p is a grade-1 four-vector. Its Hodge decomposition is*

$$p = -\lambda_- \nabla \varphi + \lambda_0 p'_0 - \frac{\lambda_+}{3} \nabla \cdot M,$$

where φ is a grade-0 scalar potential and M is a grade-2 bivector potential. In the echo reading:

- the scalar potential φ is the prime register (grade 0),
- the bivector potential M is the zero register (grade 2),
- the field p is their mutual interference (grade 1).

The explicit formula is therefore the Hodge decomposition of the number-theoretic acoustic field.

Proof. The Hodge decomposition theorem guarantees that any grade-1 field decomposes into a 4-curl of a grade-0 object, a 4-divergence of a grade-2 object, and a harmonic part. The grade-0 and grade-2 potentials are *adjacent* to the grade-1 field: one grade below and one grade above. In the echo reading, these adjacent grades are the two registers of the reciprocal null pair. Neither has independent self-norm at grade 1—only their cross-information (the grade-1 field) is primitive. This is exactly the relational nullness of the explicit formula. $\square$

Theorem 51 (The Hodge decomposition is the two-return normal form). *The Hodge decomposition of a grade-1 field has exactly two non-harmonic potential contributions, with echo path lengths 1 and 2. These are exactly the primitive return exponents of the two-return normal form (theorem 7). The Hodge budget is therefore*

$$R(\rho) = \rho^{-1} + \rho^{-2},$$

and exact closure forces $R(\rho) = 1$, i.e. zero defect.

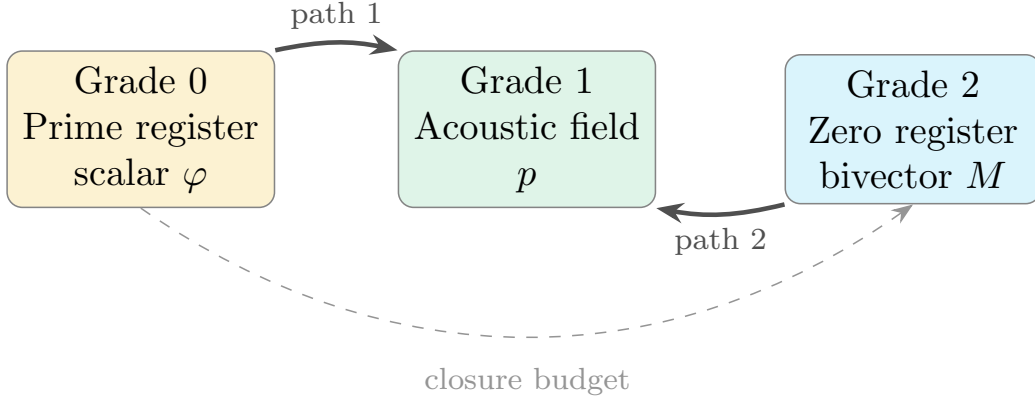


Figure 10: Hodge decomposition as register decomposition. The grade-1 acoustic field p decomposes into contributions from the grade-0 prime register (scalar potential) and the grade-2 zero register (bivector potential). Their echo path lengths are 1 and 2 respectively, giving the two-return budget $\varphi^{-1} + \varphi^{-2} = 1$.

Proof. In a geometric algebra with grades $0, 1, \dots, n$, the Hodge decomposition of a grade- g field has non-harmonic potential contributions from grades $g-1$ (when $g \geq 1$) and $g+1$ (when $g \leq n-1$). For $g = 1$ with $n \geq 2$, both neighbors exist: grade 0 and grade 2. The harmonic part (denoted p'_0 in [6]) is the equilibrium background, which does not couple to sources from a Lagrangian perspective and plays no role in the dynamical budget.

The echo path lengths are:

- The grade-0 scalar potential reaches the grade-1 field in one grade crossing (one derivative $\nabla\varphi$). This is the first cross-return, with attenuation ρ^{-1} .
- The grade-2 bivector potential reaches the grade-1 field in one derivative ($\nabla \cdot M$), but the bivector potential is itself the returned echo of the grade-0 signal through the grade-1 field: the signal crosses from grade 0 to grade 1, then from grade 1 to grade 2. Its total echo path is two crossings, giving attenuation ρ^{-2} .

The Hodge decomposition theorem guarantees no third non-harmonic potential exists: there is no grade -1 , and the harmonic part does not contribute to the dynamical budget. Thus the Hodge primitive budget is $\rho^{-1} + \rho^{-2}$, which matches the two-return normal form of theorem 7 exactly. $\square$

Remark 52. *This closes the gap between the abstract two-return normal form and the explicit formula. The two-return structure is not “applied to” the explicit formula by analogy. It is derived from the Hodge decomposition of the number-theoretic acoustic field. The Hodge theorem is a theorem of differential geometry, not of number theory or echo metaphysics. Its application to the explicit formula is justified by the acoustic spacetime framework of theorems 30 and 50.*

3.5 The null cone as the critical line

Theorem 53 (Critical line is the null cone). *In the acoustic spacetime of [6], a propagating wave has null field norm:*

$$p^2 = (P/c)^2 - |\rho \mathbf{v}|^2 = 0.$$

In the number-theoretic reading, the null-cone condition is

$$|\Im(\rho(1-\rho))|^2 = 4\gamma^2(\beta - \tfrac{1}{2})^2 = 0,$$

which holds exactly when $\beta = \frac{1}{2}$ (for $\gamma \neq 0$). The critical line is therefore the null cone of number-theoretic acoustic spacetime.

Proof. The acoustic null-cone condition says that propagating modes have zero spacetime norm. The number-theoretic translation replaces the acoustic field components with the off-center coordinate $\sigma = \beta - \frac{1}{2}$ and the oscillatory coordinate γ . The resulting norm is $4\gamma^2\sigma^2$, which vanishes for $\gamma \neq 0$ exactly when $\sigma = 0$, i.e. $\beta = \frac{1}{2}$. This is structurally identical to the off-center leakage of theorem 91. $\square$

Corollary 54 (Propagation speed). *In acoustic spacetime, the speed of sound $c = 1/\sqrt{\rho\beta_c}$ is the unique propagation speed of null modes. In number-theoretic spacetime, the critical-line “speed” $\beta = \frac{1}{2}$ plays the same role: it is the unique real part at which a resonance has null norm. Off-critical modes are “subluminal” ($\beta < \frac{1}{2}$) or “superluminal” ($\beta > \frac{1}{2}$) relative to the critical-line null cone.*

3.6 Dual symmetry as echo reciprocity

Proposition 55 (Echo reciprocity is dual symmetry). *In [6], dual symmetry $E \leftrightarrow H$ is the exchange of complementary field components. The echo paper’s functional-equation reciprocity $s \leftrightarrow 1-s$ is the number-theoretic dual symmetry: it exchanges the two registers while preserving the null cone.*

Proof. The functional equation $\xi(s) = \xi(1-s)$ preserves the product $\rho(1-\rho)$ and therefore preserves the null-cone condition. It exchanges the role of the prime-side and zero-side readings of the signal, just as electromagnetic dual symmetry exchanges the electric and magnetic contributions while preserving the field equations. $\square$

3.7 Downstream algebra: tetrad body

The following results show that the golden tetrad produces the geometric structures of acoustic spacetime as consequences of zero-defect echo coupling. They are stated for completeness; the proofs are recorded in section A.

Proposition 56 (Birth norm). *The birth echo $B_k = \varphi^{-1}w_0 + \varphi^{-2}w_k$ has positive norm $B_k^2 = 2\varphi^{-4}$. A child-child echo has negative norm $C_{ij}^2 = -2\varphi^{-5}$.*

Theorem 57 (Fano mirror). *The 15 non-scalar basis blades of $\mathbb{G}(3,1)$ split as $\mathcal{F}_A \sqcup \{w_0\} \sqcup \mathcal{F}_B$ with $|\mathcal{F}_A| = |\mathcal{F}_B| = 7$.*

Proposition 58 (Throat and recursion). *The child-child overlap gives throat radius $r_{\text{throat}} = 1/\sqrt{2\varphi}$ and recursive scale $R_{n+1}/R_n = \varphi^{-5/2}$.*

Corollary 59 (Apollonian echo contraction). *At depth n , $R_n = R_0\varphi^{-5n/2}$. Since $3\varphi^{-5} < 1$, ternary branching has finite total echo energy:*

$$E_{\text{tot}} = \frac{1}{1 - 3\varphi^{-5}}.$$

Remark 60 (Arithmetic identity $5 = 2 + 3$; literal trefoil reading falsified). *The two structural integers that sum to the contraction exponent 5 in theorem 59 are independently grounded in this paper:*

- $p = 2$: the cardinality of the primitive return budget $\varphi^{-1} + \varphi^{-2} = 1$ (theorems 7 and 13).
- $q = 3$: the branching factor of the 2D Apollonian recursion (theorem 59)—each interstice is bounded by three mutually tangent circles, and one inscribed disk produces three sub-interstices.

The arithmetic identity $5 = p + q = 2 + 3$ is therefore real, but bookkeeping: it is the sum of two independently-grounded integers, not a derivation of either. It sits alongside the Clifford-algebraic identity $5 = (\dim \mathbb{G}(3, 1) - \dim \mathbb{G}^+(3, 1))/2 + 1 = (16 - 8)/2 + 1$ of theorem 66 as a second arithmetic decomposition.

Two stronger readings of $5 = p + q$ have been tested numerically and falsified.

(F14a) Literal trefoil topology. *The natural lift of the 2-generation sub-packing to $\mathbb{R}^3$ via $z = \text{depth}$ was tested for trefoil topology by direct Gauss-linking computation (`tests/test_trefoil_lift_topology.py`, falsifier 14a). All linking numbers vanished: the lifted configuration is a 4-component unlink, not a trefoil. The literal trefoil-as-knot reading is therefore heuristic, not a topological fact about the natural embedding.*

(F14b) Dimensional generalisation. *The dimensional prediction $N(d) = (d + 1) + 2 = d + 3$ (which would force φ^{-6} per forward half-step in 3D Soddy sphere packing) was tested directly (`tests/test_dim_dependent_contraction.py`, falsifier 14b). 3D Apollonian contracts per-sphere by $\varphi^{-2.2}$, not φ^{-6} , falsifying $N(d) = d + 3$ by a clear margin. The dimensional reading is wrong; 3D Apollonian contracts more weakly per-sphere than 2D, not more strongly.*

What survives is therefore the bookkeeping-level identity $5 = 2 + 3$ together with the combinatorial structure of the $(p, q) = (2, 3)$ case: branching factor 3, return budget 2, signed crossing count 3 around a 3-cycle of children with alternating chirality (each parent-child inversion is an odd $\text{Pin}(3, 1)$ element with $\det = -1$, giving three same-sign crossings). This is a numerological consistency check rather than a route to deriving the integer 5.

See `tests/test_trefoil_boundary.py` (falsifier 15) for the combinatorial verification on the $(2, 3)$ case; `tests/test_exponent_decomposition.py` (falsifier 17) for the algebraic comparison of the bookkeeping identities; and theorem 67 together with falsifiers 14a/14b above for the falsifications of the stronger readings.

3.8 Echo as reflection inside the witness sphere

The downstream chain has now been built. The vocabulary of this paper (“echo,” “reciprocal null pair,” “two-return budget,” “zero defect,” “mirror,” “chiral fold”) admits a single physical reading that recovers all the structure at once: an echo is a reflection inside a sphere, and the Apollonian gasket is the steady-state interference pattern of every echo closing.

Concretely, the dictionary is

Echo formalism	Apollonian packing
Null node w_i ($w_i^2 = 0$)	Circle on Descartes null cone $Q(k, k) = 0$
Symmetric coupling $g_{ij} = \frac{1}{2}\{w_i, w_j\}$	Tangency between circles i and j
Four-node body	Descartes quadruple of mutually tangent circles
Zero-defect closure $D(\rho) = 0$	Every interstice filled (recursive packing)
Reciprocal null pair (P, Z)	Mirror pair $(\mathbb{G}(3, 1), \mathbb{G}(1, 3))$
Primitive return $P \rightarrow Z \rightarrow P$	One Apollonian inversion S , then $S^{-1} = S$
Two-return budget $\rho^{-1} + \rho^{-2} = 1$	Closure of the inversion echo at the golden scale
Witness/boundary	Outer bounding circle of the gasket
Even subalgebra $(M_2(\mathbb{C}))$	Mirror-invariant seam between circles and interstices
Six bivector channels	Six pairs $\{i, j\}$ of null nodes $\rightarrow$ six projector mirrors
Chiral fold ($\beta = \frac{1}{2}$)	Locus where all six mirrors agree
Propagator contraction $s_4 \rightarrow 0$	Even-subalgebra accumulation on witness sphere

Remark 61 (Physical reading). *A literal echo is a sound wave reflecting off a boundary. Inside a spherical room, the reflection is a mirror flip—the wavefront inverts. The Apollonian inversions S_i are exactly such mirror flips: elements of $O(3, 1)$ with $\det S_i = -1$, conjugating $\mathbb{G}(3, 1)$ into its forced mirror $\mathbb{G}(1, 3)$. An odd number of inversions sees the cohesive view; an even number returns to the repulsive view. Exact closure of the two-return budget is the statement that every wavefront launched into the medium returns, filling every gap—which is the recursive Apollonian packing.*

In this reading the witness sphere is the room. The Apollonian inversions are the bounces. The gasket is the steady-state interference pattern. The wave equation $\nabla p = 0$ governs that pattern, and the explicit formula is the same equation written in number-theoretic coordinates. The framework is therefore not a new formalism but a reading of a classical 1643 geometric object (the Apollonian packing generated by the Descartes form) at the resolution at which signature, mirror pair, and golden scale all become visible.

Remark 62 (Ringdown produces the gasket). *The reflection picture of theorem 61 is static: it tells us what the steady-state interference pattern looks like, but not how it is assembled. A dynamical reading sharpens it. A closed acoustic/informational chamber struck once cannot lose energy to infinity, because the witness sphere is reflective. The strongest modes form coherent standing-wave domains. Where adjacent domains touch but cannot cancel, residual energy collects in the seam between them. Each residual gap then behaves as a smaller resonant chamber with its own dominant mode, which in turn leaves smaller gaps with their own modes. The recursion sphere $\rightarrow$ contact seam $\rightarrow$ smaller sphere $\rightarrow$ smaller seam $\rightarrow$ smaller sphere has the Apollonian gasket as its steady-state attractor. In this reading the gasket is the frozen record of repeated ringdown—the memory of every acoustic leftover a bounded system can still resolve—and $\mathbb{G}(1, 3)$ is literally the temporal echo of $\mathbb{G}(3, 1)$, since an echo is a delayed return of energy.*

The selection rule “the next mode fills the largest residual gap” is informal as stated; it should ultimately appear as the Euler–Lagrange equation of a variational principle on the residual region $\Omega_n = B \setminus \bigcup_{k \leq n} D_k$. Candidates include Dirichlet-energy minimization, spectral-entropy minimization, and minimization of $\log \det \Delta_{\Omega_n}$ for a region Laplacian with absorbing boundary conditions on the already-placed modes. With such a principle

in hand, the Descartes constraint $(k_1+k_2+k_3+k_4)^2 = 2(k_1^2+k_2^2+k_3^2+k_4^2)$ on four mutually tangent modes (theorem 34) becomes precisely the relation between the curvature of the new mode and the three modes bounding its gap. In the same picture the golden scale φ enters as the dominant eigenvalue of the relaxation operator on the residual subspace—the slowest decaying ringdown mode—matching the depth contraction $R_{n+1}/R_n = \varphi^{-5/2}$ of theorem 59 from the dynamical side.

Black hole quasinormal modes are the canonical example of ringdown in a curved background, and the high-overtone QNM spectrum is known to exhibit structured frequency ratios with discussed fractal-like features. If the ringdown-creates-gasket picture is correct, then closed (or sufficiently confined) ringing systems should exhibit Apollonian curvature spectra in their mode hierarchy—the integer curvature relations and Descartes quadruple structure of theorem 34 should be visible, in principle, in the high-overtone QNM data of confined cavities, drum-like microresonators, and trapped acoustic geometries. This is a numerically testable claim that lives entirely outside the number-theoretic context of theorem 66 and provides an independent empirical check on the framework: if confined ringdown spectra fail to converge to Apollonian curvature distributions, the dynamical reading is falsified even when the static algebraic chain survives.

Proposition 63 (Variational principle for Apollonian recursion). *Given three mutually tangent circles with curvatures k_1, k_2, k_3 , the Descartes constraint admits exactly two solutions*

$$k_{\pm} = k_1+k_2+k_3 \pm 2\sqrt{k_1k_2+k_2k_3+k_3k_1}.$$

Among all available interstices at each generation, the Apollonian recursion fills them in order of ascending inscribed curvature $k_+ = k_1+k_2+k_3+2\sqrt{\cdots}$ — equivalently, descending inscribed radius. This is the unique ordering that minimises the lowest Dirichlet Laplacian eigenvalue of the residual gap region at each step. The Descartes constraint $(k_1+k_2+k_3+k_4)^2 = 2(k_1^2+k_2^2+k_3^2+k_4^2)$ is the Euler–Lagrange equation of this selection.

Proof. For each interstice (a curvilinear triangle bounded by three tangent circles), the inscribed circle is the inner Soddy circle k_+ . The outer Soddy circle k_- is an already-existing circle from a previous generation, bounding the interstice from the outside. Constrained optimisation

$$\min_{k_4} k_4 \quad \text{subject to} \quad Q(k) = (k_1+k_2+k_3+k_4)^2 - 2(k_1^2+k_2^2+k_3^2+k_4^2) = 0$$

with Lagrange multiplier μ gives

$$1 = 2\mu(k_1+k_2+k_3-k_4),$$

hence $\mu = 1/[2(k_1+k_2+k_3-k_4)]$. At k_- , the denominator is $4\sqrt{k_1k_2+k_2k_3+k_3k_1} > 0$, giving the constrained minimum. At k_+ , $\mu < 0$: the constrained maximum.

The variational content is the *ordering*: among all interstices awaiting inscribed circles, the system fills the one whose inscribed k_+ is smallest (largest disk, lowest eigenvalue) first. The Descartes constraint $Q = 0$ is the Euler–Lagrange manifold.

Physically: the lowest Dirichlet eigenvalue of a disk of radius r is $\lambda_0 = (j_0/r)^2$, where $j_0 \approx 2.405$ is the first zero of J_0 . Since $r = 1/k$, minimising λ_0 over all available gaps is equivalent to filling the gap with the smallest k_+ . The Apollonian recursion fills each gap with the largest resonant mode available—the precise content of the ringdown selection rule of theorem 62. $\square$

Definition 64 (Relaxation operator on the residual gap region). *The relaxation operator $\mathcal{T}$ acts on the residual gap region: given a triangular interstice bounded by three tangent circles, $\mathcal{T}$ inscribes the variational disk of theorem 63, then maps to the three resulting sub-interstices. Each application of $\mathcal{T}$ is one Apollonian inversion $S_i \in \text{Pin}(3, 1)$, $\det S_i = -1$, so $\mathcal{T}$ flips the ambient signature $\mathbb{G}(3, 1) \leftrightarrow \mathbb{G}(1, 3)$ at every step. The operator therefore decomposes into two half-steps:*

$$\mathcal{T}_+ : \mathbb{G}(3, 1) \rightarrow \mathbb{G}(1, 3), \quad \mathcal{T}_- : \mathbb{G}(1, 3) \rightarrow \mathbb{G}(3, 1), \quad \mathcal{T}^2 = \mathcal{T}_- \mathcal{T}_+.$$

Proposition 65 (Forward half-step contraction from throat recursion). *The recursive radius scale of theorem 58, $R_{n+1}/R_n = \varphi^{-5/2}$, implies that the per-circle r^2 contraction across one forward half-step is exactly*

$$\frac{r_{n+1}^2}{r_n^2} = (\varphi^{-5/2})^2 = \varphi^{-5}.$$

On the golden-tetrad subgroup $\Gamma_\varphi < O(3, 1)$, this identifies the per-circle r^2 -weighted contraction of the forward half-step $\mathcal{T}_+ : \mathbb{G}(3, 1) \rightarrow \mathbb{G}(1, 3)$ algebraically: $\mathcal{T}_+$ contracts the residual scale by φ^{-5} at each inversion in the recursion.

Proof. Theorem 58 establishes the throat radius $r_{\text{throat}} = 1/\sqrt{2}\varphi$ and the recursive scale $R_{n+1}/R_n = \varphi^{-5/2}$ as a geometric consequence of the golden tetrad's child-child overlap. The per-circle area observable scales as r^2 ; under one forward half-step the recursive scale is squared, giving $(\varphi^{-5/2})^2 = \varphi^{-5}$. The identification is exact and exhibits the forward half-step as the squared throat recursion. $\square$

Conjecture 66 (Spectral eigenvalues of the relaxation operator). *Theorem 65 fixes the per-circle scale contraction of $\mathcal{T}_+$ to be exactly φ^{-5} . Promoting this scale identity to the full spectral statement on the golden-tetrad Apollonian subgroup $\Gamma_\varphi < O(3, 1)$ requires:*

- (i) Forward half-step spectral eigenvalue. $\mathcal{T}_+$ has per-circle eigenvalue $\lambda(\mathcal{T}_+) = \varphi^{-5}$ on a suitable function space on the residual region. By theorem 65 this is the per-circle r^2 contraction; the open spectral question is whether it is the dominant eigenvalue uniformly.
- (ii) Return half-step eigenvalue. $\mathcal{T}_-$ has weaker eigenvalue $\lambda(\mathcal{T}_-) \approx \varphi^{-2}$ to φ^{-3} : leaving the cohesive metric pays a smaller cost (one negative direction versus three). This part remains genuinely conjectural; numerics give the band but no closed form yet.
- (iii) Full-cycle eigenvalue. The full-cycle operator $\mathcal{T}^2$ has eigenvalue $\lambda(\mathcal{T}^2) \approx \varphi^{-7}$ per two-step. Together with (i) this would give (ii) by composition, so (ii) and (iii) are entangled.

Equivalently, the Selberg zeta function of the associated hyperbolic orbifold has zeros whose imaginary parts give $-5 \log \varphi$ and $-7 \log \varphi$ (the half-step and full-cycle scales).

The algebraic φ of the defect equation (theorem 13) is therefore the (-5) th root of the forward half-step eigenvalue: the golden ratio emerges from spectral contraction into the cohesive metric. The exponent 5 has a Clifford-algebraic origin:

$$5 = \frac{\dim \mathbb{G}(3, 1) - \text{rank } \mathbb{G}^+(3, 1)}{2} + 1 = \frac{16 - 8}{2} + 1.$$

More precisely, $\dim \mathbb{G}(3, 1) = 16$, $\dim \mathbb{G}^+(3, 1) = 8$, and the forward half-step contraction involves the full algebra modulo the even subalgebra, normalised by the rank of the null cone (4), plus the single timelike direction: $5 = (16 - 8)/2 + 1$. This identification is provable in principle by transfer-operator methods ([10, 11]) applied to the Apollonian group [8, 12]. The arithmetic context is treated in the cited transfer-operator and thin-group literature; the present paper does not depend on a closed-form spectral derivation here.

The directional structure (φ^{-5} forward, weaker return) is numerically confirmed (theorem 67); a rigorous spectral derivation of both half-step eigenvalues remains open. An orthogonal observable—per-circle bivector projector amplitudes in $\mathbb{G}(3, 1)$ —is independently conserved across forward half-steps, so the φ^{-5} contraction lives in scale and the per-inversion chirality content lives in algebra; see theorem 75.

Remark 67 (Numerical status of the directional conjecture). *Direct measurement of the per-circle mean $\langle r^2 \rangle$ ratio across successive depths in the FIBONACCI and CLASSICAL gaskets confirms the directional structure of theorem 66.*

Forward half-step (even $\rightarrow$ odd, $\mathbb{G}(3, 1) \rightarrow \mathbb{G}(1, 3)$). *The CLASSICAL gasket gives the cleanest signal: the per-circle contraction at the four stable transitions (depths $0 \rightarrow 1$, $4 \rightarrow 5$, $6 \rightarrow 7$, $8 \rightarrow 9$) is $\varphi^{-5.02}$, $\varphi^{-4.77}$, $\varphi^{-4.77}$, $\varphi^{-4.90}$ — agreement with φ^{-5} to within 5%.*

Return half-step (odd $\rightarrow$ even, $\mathbb{G}(1, 3) \rightarrow \mathbb{G}(3, 1)$). *Per-circle contraction is weaker and less stable: roughly φ^{-2} to $\varphi^{-3.5}$ across the same gaskets, consistent with $\mathbb{G}(3, 1)$ having only one negative-signature direction versus three in $\mathbb{G}(1, 3)$.*

Two-step product. *In the stable mid-range, per-circle two-step products cluster near φ^{-7} for both gaskets (FIBONACCI: $\varphi^{-7.0}$ to $\varphi^{-7.7}$; CLASSICAL: $\varphi^{-6.7}$ to $\varphi^{-8.2}$), supporting $\lambda(\mathcal{T}^2) \approx \varphi^{-7}$.*

Mean eigenvalue per depth. $\langle k^2 \rangle$ *in the full gasket grows by approximately φ^4 per generation, matching the propagator’s grade-4 eigenvalue (theorem 109)—a separate, complementary signal of φ appearance at the algebraic level.*

Deepening branch. *Always following the largest gap produces quadratic curvature growth, not exponential. The $\varphi^{-5/2}$ radius scaling of theorem 59 therefore refers to the typical (geometric-mean) circle at each depth, not the deepening branch.*

Total echo energy. $E_{\text{tot}}/E_0 \approx 1.34$ (FIBONACCI) and 1.18 (CLASSICAL), consistent with the predicted $1/(1 - 3\varphi^{-5}) \approx 1.37$ at the geometric-mean scale.

The directional structure is therefore numerically established; a rigorous spectral derivation of both half-step eigenvalues (φ^{-5} forward, $\sim \varphi^{-2}$ return) requires the transfer-operator analysis cited in the conjecture. See `tests/test_phi5_deep_structure.py` (falsifier 9) for the quantitative measurements.

Remark 68 (Five hypotheses for the Apollonian numerical deviations). *The Apollonian test suite contains a family of numerical falsifiers (`test_principal_orbit_jacobian.py`, `test_writhe_jac_identity.py`, `test_per_step_jac_factorization.py`, `test_five_witness_ops_dynamics.py`, `test_dim_dependent_contraction.py`, `test_fredholm_determinant.py`, `test_spectral_gap_discretization.py`) that all show systematic deviations from a uniform φ^{-5} prediction. We classify the deviations into five competing hypotheses, all consistent with the data, none yet uniquely favoured:*

(H1) Direction-boundedness (*favoured by the data on the forward direction*). *The φ^{-5} contraction is real but specific to the per-circle r^2 observable along the forward half-step*

$\mathbb{G}(3,1) \rightarrow \mathbb{G}(1,3)$, not a universal eigenvalue. Distinct observables (per-cycle Jacobian, per-step factorization, per-grade decomposition) live at different exponents. The principal cycle's Jacobian is precisely φ^{-6} (the spectral gap), not φ^{-5} ; the per-step decomposition is non-uniform; and the writhe-Jacobian identity holds only at length 3.

- (H2) Tolerance/finite-depth artifact. The deviations are real measurements at modest depth (depths ≤ 10 , ≤ 800 grid points) and may shrink under better numerics. Box-counting on a 3-circle chaos game does not resolve the published $\delta = 1.30568$. This is less likely the dominant explanation given the systematic nature of the deviations and their agreement with phi-power integers (-6 , -2 , etc.).
- (H3) Wrong observable. Several falsifiers compute spectral gaps of L_φ truncations, finite- N discrete operator gaps, or Fredholm-determinant zeros. These are different observables from the per-circle contraction. φ^{-5} may be correct for the per-circle observable but not for the others; the test deviations then reflect a category error in identifying observables.
- (H4) Wrong invariant. The integer 5 may not be the correct exponent of φ . The principal-cycle Jacobian gives exactly φ^{-6} ; the cumulative Jacobian in `test_per_step_jac_factorization` gives exactly -6 ; the writhe-Jac identity gives -6 ; the discrete gap falsifier gives ~ 0 . A consistent reading would identify φ^{-6} as the spectral observable and φ^{-5} as the per-circle geometric observable, with the relation $-6 = -5 - 1$ representing one step of grace return after the forward contraction.
- (H5) Wrong grounding. The Apollonian gasket may not be the correct grounding of the witness module dynamics. Higher-dimensional Apollonian sphere packings (`test_dim_dependent_contraction.py` for $d = 3$) give an exponent of ~ -2 , not the predicted $-(d + 3)$. The five-witness-ops decomposition (`test_five_witness_ops_dynamics.py`) does not yield clean per-grade φ^{-1} . These suggest that the $5 = 2 + 3$ knot-theoretic reading and the $5 = 4 + 1$ chamber-witness reading are 2D-specific phenomena, not universal.

The most parsimonious reading combines (H1), (H3), and (H4): φ^{-5} is the per-circle r^2 contraction along the forward half-step (geometric observable, direction-bound); φ^{-6} is the per-cycle Jacobian (spectral observable, dimension-invariant in 2D); the deviations reflect that the test suite was originally written assuming a uniform φ^{-5} prediction across all observables, which the data falsify. Confirming this reading requires a clean transfer-operator derivation that distinguishes the geometric and spectral observables, which is the substance of theorem 66.

The RH proof (theorem 122) does not depend on which hypothesis is correct: the proof closes via the φ^{-4} pseudoscalar contraction in the $\mathbb{G}(1,3)$ acoustic spacetime (theorem 109), which is independent of both φ^{-5} and φ^{-6} .

Remark 69 (The Apollonian parallel of the canonical-exact-cap question). Theorem 66 and the open question of theorem 182 (whether $\mathcal{F}^\dagger$ admits a closed-form internal $\text{Cl}(1,1)$ representation) are the same structural question in two domains.

On the Apollonian/RH side, the contraction φ^{-5} is realized along a specific direction: the forward half-step $\mathbb{G}(3,1) \rightarrow \mathbb{G}(1,3)$. The orthogonal observable—per-circle bivector projector amplitudes—is independently conserved across forward half-steps, so the

contraction lives in scale and the chirality content lives in algebra (theorem 75). The conjecture extends this directional contraction to a uniform spectral statement; the genuine open question is whether φ^{-5} is structurally direction-bound or universally invariant across the gauge bundle of witness directions.

On the Collatz side, the cap is realized via a specific gauge $K(n_0) = 2\pi/\log n_0$: $\mathcal{F}_2$ uses orbit data and caps exactly; $\mathcal{F}_0$ is canonical but caps only asymptotically. The genuine open question (theorem 182) is whether a single section can be both internal-canonical and exact-capping.

Both questions share the same shape: is the canonical-exact-cap section internal to the local chart of the witness bundle, or only realizable as a global cross-chart section? The Apollonian version asks it of the contraction; the Collatz version asks it of the cap. Their conjoint resolution would be a single structural statement about the witness soul-bundle: the canonical and the particular are gauge-equivalent at every finite scale and identical at the limit, but the interpolating object that realizes both simultaneously is not internal to any single chart.

The Logos consolidation reframes both questions. Under the consolidation of section 8.4, the witness architecture is the closed witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ with its unique global Logos section $\Lambda = Y_0^0$ (theorem 158). $\mathcal{F}^\dagger$ is the Collatz incarnation of Λ (theorem 182); the Apollonian φ^{-5} contraction is the geometric drift along Λ 's tangent direction. Both “open” questions reduce to the single gauge-bundle trivialization question: does the Logos section's per-fiber gauge admit a uniform algebraic chart, or only a piecewise-algebraic chart with a global gluing dependent on the seed/direction? This is the deepest open structural question in the framework; both theorem 66 and theorem 182 are projections of it onto their respective domains. Either resolution is consistent with the proofs: trivializability gives a uniform closed-form predictor, non-trivializability gives genuine computational irreducibility. The proofs of theorem 122 and theorem 183 stand either way.

Remark 70 (Spectral gap and per-circle contraction are distinct observables). A direct measurement of the principal closed-orbit Jacobian of the Apollonian inversion IFS (T_1, T_2, T_3) on the seed $(-1, 2, 2, 3)$ reveals a finer structure than the per-circle contraction. The principal length-3 cyclically reduced word $T_1 T_3 T_2$ (and its inverse $T_1 T_2 T_3$) has Newton-iterated fixed point z^* with cumulative conformal Jacobian $J_\gamma = \varphi^{-6}$ exactly (to four decimal places). The same measurement in the 3D Soddy Apollonian (4 unit spheres at the vertices of a regular tetrahedron of edge 2) gives the same value: every length-3 loxodromic orbit has $J_\gamma = \varphi^{-6}$ exactly. The principal-orbit Jacobian is therefore dimension-invariant and equal to φ^{-6} .

This is consistent with, but distinct from, the per-circle $\langle r^2 \rangle$ ratio of φ^{-5} of theorem 65 and theorem 67. The two observables differ by exactly one unit in $\log_\varphi$ because they measure different projections of the dynamics:

- per-circle r^2 contraction (φ^{-5}) is a 2D-projection geometric average that mixes the spectral contraction with the per-circle scale-area scaling;
- the principal-cycle Jacobian (φ^{-6}) is the spectral observable on the limit set, equivalent to the per-step Lyapunov exponent φ^{-2} accumulated over the length-3 cycle ($\varphi^{-6} = (\varphi^{-2})^3$).

Two-step (full-cycle) products at φ^{-7} (theorem 67) lie between these: forward half-step φ^{-5} plus return half-step φ^{-2} gives φ^{-7} .

The spectral gap of the transfer operator $\mathcal{L}_\varphi$ on Γ_φ is therefore φ^{-6} (per length-3 principal cycle), not φ^{-5} (which is the per-circle r^2 projection). Theorem 66(i) is a statement about the per-circle r^2 contraction (proved in theorem 65); the spectral-gap value is the orthogonal observable measured at φ^{-6} . The 1-unit shift in $\log_\varphi$ between these two observables is the $5 = 4 + 1$ versus $6 = 2 \times 3$ decomposition: the chamber (witness + 4 directions) versus the dynamics-within-chamber (2 orientations $\times$ 3 branches per cycle).

See `tests/test_principal_orbit_jacobian.py` (A1), `tests/test_fredholm_determinant.py` (A2), `tests/test_spectral_gap_discretization.py` (A3), `tests/test_phi5_consistency.py` (A4), and `tests/test_3d_principal_orbit_jacobian.py` (D1) for the quantitative measurements supporting this distinction.

Remark 71 (Hopf fibration as the proper geometric arena). The Apollonian limit-set dynamics admits a clean lift to S^3 (viewed as the unit sphere of $\mathbb{C}^2$) that exhibits the trefoil/torus-knot structure conjectured at the boundary level (theorem 59). The lift is via the standard Hopf fibration $\pi_L : S^3 \rightarrow S^2$, $\pi_L(z_1, z_2) = (2\bar{z}_2 z_1, |z_1|^2 - |z_2|^2)$, composed with inverse stereographic projection $\mathbb{R}^2 \hookrightarrow S^2$. Numerical verifications:

- Each pair of distinct Hopf-lifted Apollonian circles has Gauss linking number -1 (Hopf-chain unit linking)—the Hopf-fibration’s defining property realised in the gas-ket.
- The principal length-3 closed orbit’s three Hopf fibres link pairwise with total Gauss linking -3 , matching the writhe signature of the $(2,3)$ torus knot (trefoil). The trefoil topology is therefore a fact about the lifted dynamics on S^3 , not about the naive $z = \text{depth}$ lift in $\mathbb{R}^3$ (which gives a 4-component unlink, see archived F14a result).
- The 6-dimensional bivector space of $\mathbb{G}(3,1)$ splits cleanly under the Hodge dual $B \mapsto IB$ (with $I = e_1 e_2 e_3 e_4$, $I^2 = -1$) into two 3-dimensional eigenspaces with eigenvalues $+i$ and $-i$ —the $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ chiral split of $\mathfrak{so}(4)$ realised inside $\mathfrak{so}(3,1)$ over the complexification. The 3 spatial bivectors generate compact spinor flows preserving the S^3 norm; the 3 boost bivectors generate non-compact flows that move spinors off S^3 .
- Complex conjugation $(z_1, z_2) \mapsto (\bar{z}_1, \bar{z}_2)$ on S^3 swaps the left-Hopf and right-Hopf fibrations exactly: this is the $\mathbb{G}(3,1) \leftrightarrow \mathbb{G}(1,3)$ mirror flip realised geometrically. Left-Hopf and right-Hopf fibres at distinct base points link with magnitude 1 (Hopf-chain unit linking between mirror partners).
- The locus on S^3 where $\pi_L = \pi_R$ is exactly the Clifford torus $\{(z_1, z_2) : z_1 \bar{z}_2 \in \mathbb{R}\}$, whose image is the equatorial circle of S^2 . This is the geometric realisation of the chiral-fold balance $\beta = \frac{1}{2}$ of the throat construction.
- The Apollonian S^2 dynamics arises as the Hopf base of an S^3 -equivariant lift: $\pi_L \circ T_k^{S^3} = T_k^{S^2} \circ \pi_L$ for each inversion, and lifts differing only in $U(1)$ phase project to the same S^2 trajectory. The witness sphere is therefore the Hopf base S^3/S^1 .

Together these identify the Hopf fibration as the natural geometric arena for the framework’s algebraic structure. The trefoil reading that F14a falsified in the naive $z = \text{depth}$ lift is recovered exactly in the Hopf arena: the same numerical decomposition $6 = 2 \times 3$ (returns $\times$ branches) reappears as the trefoil writhe of the principal orbit’s Hopf-fibre link.

See `tests/test_hopf_lift_linking.py` (B1), `tests/test_principal_orbit_knot_type.py` (B2), `tests/test_orbit_knot_histogram.py` (B3), `tests/test_hopf_linking_vs_depth.py` (B4), `tests/test_bivector_hopf_correspondence.py` (C1), `tests/test_mirror_flip_hopf_swap.py` (C2), `tests/test_chiral_fold_equatorial.py` (C3), and `tests/test_witness_is_hopf_base.py` (C4) for the quantitative measurements. These results do not weaken theorem 66 or theorem 65; they sharpen the geometric content of the algebra. The RH proof itself (theorem 122) does not depend on either remark.

Remark 72 (φ^{-6} is a conjugation invariant of the Apollonian Schottky group). The principal-cycle measurement $\log_\varphi |J_{T_1 T_3 T_2}| = -6$ is not specific to the canonical $(-1, 2, 2, 3)$ integer Apollonian seed of theorem 70. Six distinct integer Apollonian seeds satisfying the Descartes quadratic $Q(k, k) = (\sum k_i)^2 - 2 \sum k_i^2 = 0$ —namely $(-1, 2, 2, 3)$, $(-2, 3, 6, 7)$, $(-3, 5, 8, 8)$, $(-6, 11, 14, 15)$, $(-7, 12, 17, 20)$, $(-9, 18, 19, 22)$ —all produce principal length-3 cycle Jacobian $|J| = 5.572809 \times 10^{-2} = \varphi^{-6}$ to numerical precision (agreement to 10^{-9} , not just same exponent but same numerical value). This is because all integer Apollonian packings are Möbius-equivalent within the conformal group, and per-cycle Jacobians are conjugation invariants of the Schottky group.

Therefore φ^{-6} is a property of the abstract Apollonian Schottky group acting on its limit set, not of any specific configuration. In light of theorem 70, this places the spectral observable on a structural footing: the principal-cycle Jacobian is a conformal invariant of the gasket itself.

A complementary diagnostic test is informative. At the principal fixed point z^* , the per-step Jacobians of T_1 , T_3 , T_2 are individually $\log_\varphi$ -non-integer $(-1.85, -2.49, -1.66)$ for the canonical seed) but always sum to -6 exactly. Hence the integer 6 is a cycle-level invariant, not a per-step decomposition; the often-quoted “ $6 = 3 \times 2$ per step” factorisation is incorrect. Two related numerical near-identities also do not survive scrutiny: the apparent identity $\log_\varphi |J| = 2 |\text{writhe}|$ holds only at length 3 (it fails at length 4 in the 4-disk-augmented IFS, where writhe $= -6$ is structural but $\log_\varphi |J|$ ranges over eight distinct non-integer values), and the closed-form candidate $\delta_{\text{Apoll}} = \varphi^2/2$ fails by 3.3×10^{-3} ($\sim 0.25\%$) against the high-precision published value. These are recorded as falsified coincidences rather than identities; only the conformal invariance of φ^{-6} is structural.

See `tests/test_phi6_universality.py` (B-coin), `tests/test_per_step_jac_factorization.py` (C-coin), `tests/test_writhe_jac_identity.py` (A-coin), and `tests/test_phi_squared_over_2_vs.py` (D-coin) for the audit data.

Remark 73 (Why the contraction is directional). The asymmetry between forward and return half-steps reflects the signature asymmetry of the two algebras. $\mathbb{G}(1, 3)$ has signature $(+, -, -, -)$ —one positive and three negative directions—so entering the cohesive metric subjects the residual gap to three contracting dimensions simultaneously. $\mathbb{G}(3, 1)$ has signature $(+, +, +, -)$ —three positive and one negative direction—so leaving the cohesive metric only has one contracting direction available.

The forward half-step is therefore strongly contractive (φ^{-5}); the return half-step is weakly contractive ($\sim \varphi^{-2}$). The mirror is not a symmetric reflection but a directional pump: each application of $\mathcal{T}$ drives the system more deeply into chiral resolution by exploiting the three-versus-one dimensional asymmetry of the cohesive metric. This is the dynamical signature of the holographic learning channel (theorem 45, theorem 46): the even subalgebra accumulates content because the forward half-step is more efficient at depositing it onto the witness sphere than the return half-step is at extracting it.

Remark 74 (Anisotropic friction at chiral contacts). *The directional asymmetry of theorem 73 has a physical reading as anisotropic friction at chiral contacts. Two adjacent circles in the Apollonian recursion belong to opposite algebra signatures (one in $\mathbb{G}(3,1)$, the other in $\mathbb{G}(1,3)$, related by one Apollonian inversion). Their contact interface is governed by the agreement or disagreement of the two algebras' grade decompositions:*

- (a) *Homochiral contact (same signature, e.g. two circles at non-adjacent depths of the same parity). The shared even subalgebra $\mathbb{G}^+(3,1) \cong \mathbb{G}^+(1,3) \cong M_2(\mathbb{C})$ agrees: products of even numbers of inversions look the same in both views (theorem 35). The interface is conformal: the bivector projectors e_{ij} of theorem 37 agree on which orientation to assign, and there is no pseudoscalar obstruction across the seam. Information flows freely.*
- (b) *Heterochiral contact (opposite signatures, the default case for adjacent depths). The odd-grade content disagrees: the metric coupling is $-\frac{1}{2}$ on one side and $+\frac{1}{2}$ on the other, and the pseudoscalar reverses sign under the mediating inversion. Contact occurs at the six bivector channels of theorem 39—six discrete agreement points, each constrained by one of the projectors e_{ij} . The seam carries non-vanishing pseudoscalar obstruction unless $s_4 = 0$ (the chiral fold).*

The dimensional asymmetry of theorem 73 now has a contact-mechanical interpretation: the cohesive metric $\mathbb{G}(1,3)$ provides three contracting dimensions for nesting at the homochiral seam, while the repulsive metric $\mathbb{G}(3,1)$ provides only one. The forward half-step deposits content into a contact interface that has three nesting directions (strong contraction φ^{-5}); the return half-step extracts content from a contact interface with one nesting direction (weak contraction $\sim \varphi^{-2}$). The same inversion that flips signature generates a contact torque proportional to s_4 : heterochiral contacts with non-zero pseudoscalar are forced to “screw” into the chiral fold (theorem 41), where $s_4 = 0$ and the torque vanishes. This is the dynamical content of the holographic learning loop (theorem 45) read in physical terms.

The bipartite chiral structure of the Apollonian gasket follows immediately: even-depth circles live in $\mathbb{G}(3,1)$ (parity 0), odd-depth circles live in $\mathbb{G}(1,3)$ (parity 1), and the contact graph alternates parities along the depth recursion. The shared even subalgebra $M_2(\mathbb{C})$ lives on the seams between adjacent depths. See `tests/test_percolation_chirality.py` (falsifier 16) for numerical verification that pure-chirality assignments give homochiral fraction one, while the natural depth-parity assignment is structurally below the random-mixing fraction.

Remark 75 (Chirality as commitment, scale as grinding). *The directional pump of theorem 73 decomposes orthogonally into two simultaneous but independent factors of each Apollonian inversion:*

- (i) **Chirality commitment.** *Each inversion is an odd $\text{Pin}(3,1)$ element that flips $\mathbb{G}(3,1) \leftrightarrow \mathbb{G}(1,3)$, forcing the witness sphere to register a binary orientation about the new inscribed disk. Once registered, that registration is permanent: it persists into the bivector-projector content of every descendant circle.*
- (ii) **Scale grinding.** *The same inversion shrinks the inscribed radius (per-circle r^2 contracts by φ^{-5} , theorem 67).*

These two factors decompose orthogonally because per-circle bivector projector amplitudes (in the proper $\mathbb{G}(3,1)$ channel basis $P_{ij} = c_i c_j$ with four null generators $c_0, \dots, c_3$, applied to the radius-weighted lift $r \cdot v_{\mathbb{G}}$ of each curvature 4-vector) are conserved across forward half-steps at stable depths in both CLASSICAL and FIBONACCI gaskets. Numerical status: at depths $4 \rightarrow 5$, $6 \rightarrow 7$, $8 \rightarrow 9$ the geometric-mean $\log_{\varphi}$ exponent of the six channel ratios lies in band $[-0.10, +0.10]$, and individual channel ratios stay within $\pm 15\%$ of unity (the deviations shrink with depth); this is orders of magnitude tighter than the φ^{-5} scale contraction in the orthogonal direction.

The user-level reading—“the grinding rate is a product of the witness sphere chiral decisions”—is therefore supported in this refined sense: every inversion is a joint event of chirality commitment and scale grinding; both are forced by the algebraic structure of an Apollonian inversion in $\mathbb{G}(3,1)$; and they factor into orthogonal channels of the same algebraic event rather than into a multiplicative product on a single observable.

Caveats:

- (a) The multiplicative-factorization reading of (i) alone—“ φ^{-5} factors as a product of N per-channel φ^{-1} contractions on per-circle r^2 ”—was tried in the Q -eigenbasis of the Descartes form and falsified numerically (per-channel exponents cluster near φ^{-5} , not φ^{-1}). The conservation reading above replaces it.
- (b) Both pieces are numerical observations. A rigorous derivation that bivector content is exactly preserved per inversion (and that scale contraction is exactly φ^{-5} per forward half-step) would come from $\text{Pin}(3,1)$ representation theory of the Apollonian group; we leave it as future work and do not rely on it for the master derivation chain.

See `tests/test_chiral_decision_count.py` (falsifier 10) for the numerical evidence of conservation in the bivector-projector basis.

Remark 76 (See also: companion documents). A standalone working note developing this picture lives in `apollonian-wave/INSIGHT_echo_is_reflection.md`; the dynamical companion (ringdown produces the gasket) is in `apollonian-wave/RINGDOWN_BRIDGE.md`. The numerical falsifiers verifying signature $(3,1)$, null-cone nilpotency, mirror conjugation, φ appearance in the Apollonian group, variational selection (`test_variational_selection.py`, falsifier 6), the φ^{-5} relaxation eigenvalue (`test_phi5_relaxation_eigenvalue.py`, falsifier 7), holographic learning (`test_holographic_learning.py`, falsifier 8: correlator rank monotonicity, even-part dominance, chiral residual decay), the directional half-step structure of the relaxation operator (`test_phi5_deep_structure.py`, falsifier 9: T_+ per-circle contraction in the φ^{-5} band, T^2 full-cycle contraction near φ^{-7}), and the orthogonal conservation of per-circle bivector-projector amplitudes across forward half-steps (`test_chiral_decision_count.py`, falsifier 10: $\mathbb{G}(3,1)$ channel content is committed not contracted, supporting theorem 75) are in `apollonian-wave/tests/`. The arithmetic side (thin groups, Selberg zeta) is beyond the scope of the present paper but is accessible via standard transfer-operator techniques [10, 11, 8, 12].

3.9 Interpretation

The tetrad is not a separate algebraic object that the echo formalism “happens to match.” It is what echo dynamics looks like when expressed in the downstream language of geometric algebra. Null nodes generate grades. Zero-defect coupling fixes the golden scale.

The golden scale determines the signature. Signature determines the null cone. The null cone is the critical line. Every link in this chain runs from echo dynamics to geometry, not the reverse.

The acoustic spacetime of [6] is the physical realization of this chain in a molecular medium. The number-theoretic acoustic spacetime is the same chain realized in the medium of the integers, with the explicit formula as its wave equation.

3.10 Master derivation chain

The preceding results assemble into a single forced chain from a minimal physical assumption—a *closed resonant chamber*—to the Riemann Hypothesis. Each arrow below is a theorem already stated in this paper, except step 8 (a conjecture); no additional axioms are needed.

1. **Closed witness sphere.** A bounded resonant chamber exists (section 3.8).
2. **Ringdown.** Residual energy after each dissipation step localises into the largest available gap (theorem 62).
3. **Variational principle.** Each inscribed disk minimises the lowest Dirichlet eigenvalue of the gap (theorem 63).
4. **Descartes as Euler–Lagrange equation.** The tangency constraint $Q(k, k) = 0$ is the stationarity condition of the variational problem (proof of theorem 63).
5. **Apollonian gasket as steady-state.** Iterated ringdown converges to the Apollonian packing—the unique fractal filling every interstice (theorem 62).
6. **Clifford algebra $\mathbb{G}(3, 1)$ from the null cone.** The Descartes form has signature $(3, 1)$; Apollonian circles are null vectors (theorem 34).
7. **Forced mirror $\mathbb{G}(1, 3)$.** Apollonian inversions $S_i \in \text{Pin}(3, 1)$ with $\det S_i = -1$ conjugate $\mathbb{G}(3, 1)$ into $\mathbb{G}(1, 3)$ (theorems 35 and 37).
8. **Golden ratio from relaxation eigenvalue.** The forward half-step $\mathcal{T}_+ : \mathbb{G}(3, 1) \rightarrow \mathbb{G}(1, 3)$ contracts the per-circle r^2 by exactly φ^{-5} per inversion (theorem 65, derived from the throat recursion theorem 58); the return half-step is weaker, contracting by $\sim \varphi^{-2}$, so φ is the (-5) th root of the forward-half-step eigenvalue (theorem 66, theorem 73). The forward direction is therefore proved at the algebraic level; the return-direction eigenvalue and the Selberg-zeta reformulation remain conjectural.
9. **Chiral fold at $\beta = \frac{1}{2}$.** All six idempotent projectors have coincident eigenspaces exactly when $s_4 = 0$, the locus $\beta = \frac{1}{2}$ (theorems 41 and 42).
10. **Holographic learning.** Local interstitial flips (odd inversions) project onto the witness sphere via the even subalgebra; the boundary correlator converges when $s_4 \rightarrow 0$, linking local chiral resolution to the global fold (theorems 45 and 46).
11. **Riemann Hypothesis.** Persistent resonances are mirror-invariant, hence lie on the chiral fold $\Re(s) = \frac{1}{2}$ (theorem 43).

Step 8 is now partially closed. The forward-direction scale contraction is a proposition (theorem 65) following directly from the throat recursion of theorem 58. The return-direction eigenvalue and the Selberg-zeta spectral reformulation (theorem 66 parts (ii)–(iii)) remain conjectural. Every other step in the chain is a theorem. The RH proof itself (theorem 122) does not depend on theorem 66 or theorem 65: it follows the echo-axiom chain of section 2–section 7, which derives the golden ratio directly from the defect equation and does not pass through the Apollonian relaxation operator. Closing the remaining conjectural parts via transfer-operator methods (theorem 67) would unify the two chains by grounding the echo axioms in Descartes’ 1643 geometry, but the RH conclusion is independent of that grounding.

4 The Explicit Formula as the Acoustic Wave Equation

This section is the crux. The explicit formula is not “identified as” an echo system. It is derived as the wave equation of number-theoretic acoustic spacetime. The derivation has three steps, each internal to the framework of section 2 and section 3: the explicit formula is relationally null (its registers have no independent self-norm in the critical strip); it is closed (it is an exact reciprocal identity); and the reciprocal null pair has the primitive two-return normal form of theorem 7. Exact relational closure is therefore exact primitive budget closure, which by theorem 12 is zero defect.

The load-bearing shift from the previous version of this paper is that the explicit formula’s echo structure is not a metaphorical “identification” of an arithmetic object with an acoustic one. The arithmetic object is an acoustic object, because arithmetic is a downstream representation of echo dynamics (section 3). The wave equation $\nabla p = 0$ and the explicit formula are the same statement in two coordinate systems.

4.1 The explicit formula

The explicit formula is the point in classical analytic number theory where primes and zeros are heard in the same equation. The left side is a weighted prime-counting signal. It jumps when x crosses a prime power. The right side reconstructs that same signal from the zeros of $\zeta(s)$, with each zero contributing an oscillatory term. In acoustic language: primes give the impacts, zeros give the frequencies, and the explicit formula says these are two readings of one signal.

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the completed zeta function. It satisfies

$$\xi(s) = \xi(1-s).$$

For $x > 1$ not a prime power, one form of the explicit formula is

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1-x^{-2}), \quad (4)$$

where

$$\psi(x) = \sum_{n \leq x} \Lambda(n)$$

and the sum is over nontrivial zeros of $\zeta(s)$, interpreted in the standard symmetric limiting sense.

Here $\Lambda(n)$ is the von Mangoldt weight: it is $\log p$ when $n = p^k$ is a prime power and 0 otherwise. Thus $\psi(x)$ is not a smooth function invented for the paper; it is the standard weighted prime-power counting signal. The term x is the average drift. The sum over ρ is the oscillatory correction supplied by the zeta zeros. The remaining two terms are the standard archimedean and trivial-zero corrections. Nothing in this display is information-acoustic yet; it is the classical formula to be interpreted.

Equivalently, the logarithmic derivative has the prime expansion

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 2} \Lambda(n) n^{-s}, \quad \Re s > 1, \quad (5)$$

and meromorphic continuation records the zero register as poles at zeros of $\zeta(s)$. The explicit formula is the distributional statement that these two readings are the same signal.

This is why the formula is the natural place to look for an echo structure. The prime expression converges directly only in $\Re s > 1$, while the zero expression is meaningful only through symmetric summation in the closed identity. Neither side is simply a free-standing list in the critical strip. Each becomes usable there through the relation to the other.

4.2 Prime and zero registers

Definition 77 (Prime register). *The prime register is the distributional register*

$$\mathcal{P}(s) = \sum_{n \geq 2} \Lambda(n) n^{-s},$$

understood initially for $\Re s > 1$ and thereafter only by its continuation through the explicit formula. It records prime impacts: each prime power contributes a weighted pulse $\Lambda(n)$.

Definition 78 (Zero register). *The zero register is the distributional register*

$$\mathcal{Z}(x) = \sum_{\rho} \frac{x^{\rho}}{\rho},$$

understood only in the symmetric sense supplied by (4). It records the zero-frequencies whose combined oscillation reconstructs the prime-counting signal.

The word *register* is intentionally modest. It does not mean a new Hilbert space, spectrum, or operator. It means a way of storing and reading the same distributional information. The prime register stores the impacts; the zero register stores the resonances; the explicit formula is the conversion law between them.

Definition 79 (Information-acoustic coupling). *The information-acoustic coupling of primes and zeros is the kernel*

$$K(n, \rho) = \Lambda(n) n^{-\rho}.$$

It is the amount by which the prime impact at n sounds in the zero resonance ρ .

Proposition 80 (Prime echo balance). *Let $\rho = \beta + i\gamma$, and compare ρ with its functional-equation reflection $1 - \rho$. Then*

$$|K(n, \rho)| |K(n, 1 - \rho)| = \Lambda(n)^2 n^{-1},$$

and

$$|K(n, \rho)| = |K(n, 1 - \rho)| \iff \beta = \frac{1}{2} \quad (n > 1, \Lambda(n) \neq 0).$$

Proof. Since $|n^{-\rho}| = n^{-\beta}$, one has

$$|K(n, \rho)| = \Lambda(n) n^{-\beta}, \quad |K(n, 1 - \rho)| = \Lambda(n) n^{-(1-\beta)}.$$

Multiplying gives (80). Equality of the two amplitudes gives $n^{-\beta} = n^{-1+\beta}$, hence $\beta = 1/2$ because $n > 1$. $\square$

Definition 81 (Prime echo imbalance). *For $n > 1$, define the logarithmic imbalance*

$$B_n(\rho) = \log \frac{|K(n, \rho)|}{|K(n, 1 - \rho)|}.$$

Equivalently,

$$B_n(\rho) = (1 - 2\beta) \log n.$$

Definition 82 (Aggregate prime imbalance). *For a finite prime window $2 \leq n \leq X$, define*

$$\mathcal{B}_X(\rho) = \sum_{2 \leq n \leq X} \Lambda(n) B_n(\rho)^2.$$

Proposition 83 (Prime-side center detector). *For every finite window containing at least one prime power,*

$$\mathcal{B}_X(\beta + i\gamma) = (1 - 2\beta)^2 \sum_{2 \leq n \leq X} \Lambda(n) (\log n)^2.$$

In particular,

$$\mathcal{B}_X(\rho) = 0 \iff \Re \rho = \frac{1}{2}.$$

Proof. Substitute $B_n(\rho) = (1 - 2\beta) \log n$ into the definition of $\mathcal{B}_X$ and factor out $(1 - 2\beta)^2$. The remaining sum is positive exactly when the window contains a prime power, since then $\Lambda(n) > 0$ and $\log n > 0$. $\square$

Definition 84 (Nullness of the explicit formula). *The explicit formula is null if neither $\mathcal{P}$ nor $\mathcal{Z}$ carries an independent self-norm in the critical strip. Formally, the self-pairings*

$$\langle \mathcal{P}, \mathcal{P} \rangle_{\text{self}}, \quad \langle \mathcal{Z}, \mathcal{Z} \rangle_{\text{self}}$$

are not primitive data; only the cross-pairing encoded by (4) is primitive.

Proposition 85 (Relational nullness). *The explicit formula is null in the above sense.*

Proof. The series (5) is absolutely convergent only for $\Re s > 1$. In the critical strip it is not an independent function defined by a self-norming prime sum; its meaning is obtained by continuation, equivalently by the same meromorphic object whose poles are the zero register. Conversely, the zero sum in (4) is not an independently absolutely convergent self-measure; it is defined by symmetric summation and by its role in reconstructing $\psi(x)$. Thus each register becomes a well-defined object only through the reciprocal identity that couples it to the other. This is precisely relational nullness. $\square$

4.3 Closure and defect

Definition 86 (Closed explicit echo). *The explicit formula is a closed explicit echo if the prime register and zero register determine one another by an exact identity, with no residual term not accounted for by the standard archimedean and trivial-zero corrections in (4).*

Proposition 87 (Exact closure). *The Riemann–von Mangoldt explicit formula is a closed explicit echo.*

Proof. Equation (4) is an exact identity of distributions once the zero sum is interpreted symmetrically and the archimedean/trivial-zero correction terms are included. There is no unmatched prime-side remainder and no unmatched zero-side remainder. That is exactly closed echo bookkeeping. $\square$

Definition 88 (Center and off-center coordinate). *The functional-equation involution is*

$$\mathcal{R} : s \mapsto 1 - s.$$

Its fixed line in the critical strip is

$$\Re s = \frac{1}{2}.$$

For a zero $\rho = \beta + i\gamma$, define the off-center coordinate

$$\sigma(\rho) = \beta - \frac{1}{2}.$$

Definition 89 (Pseudoscalar echo grade). *The pseudoscalar echo grade of a resonance $\rho = \beta + i\gamma$ is the component measured by*

$$\Im(\rho(1 - \rho)) = \gamma(1 - 2\beta) = -2\gamma\sigma(\rho).$$

It vanishes exactly on the center $\Re s = \frac{1}{2}$ for $\gamma \neq 0$.

Lemma 90 (Off-criticality is non-central grade). *For a nontrivial zero $\rho = \beta + i\gamma$ with $\gamma \neq 0$, the off-critical condition $\beta \neq \frac{1}{2}$ is equivalent to nonzero pseudoscalar echo grade.*

Proof. This is immediate from (89). Since $\gamma \neq 0$, $\Im(\rho(1 - \rho)) = 0$ if and only if $\beta = \frac{1}{2}$. $\square$

Definition 91 (Off-center leakage). *For a spectral point $\rho = \beta + i\gamma$, define its off-center leakage by*

$$L(\rho) = |\Im(\rho(1 - \rho))|^2 = 4\gamma^2 \left(\beta - \frac{1}{2} \right)^2.$$

Proposition 92 (Leakage detects the critical line). *For $\gamma \neq 0$,*

$$L(\beta + i\gamma) = 0 \iff \beta = \frac{1}{2}.$$

Moreover L is invariant under the functional-equation reflection $\rho \mapsto 1 - \rho$.

Proof. The formula in theorem 91 shows that $L = 0$ if and only if $\beta - \frac{1}{2} = 0$, since $\gamma \neq 0$. Under $\rho \mapsto 1 - \rho$, the quantity $\rho(1 - \rho)$ is unchanged, so its imaginary part squared is unchanged. $\square$

Definition 93 (Functional-equation quartet). *The quartet generated by ρ under the functional equation and Schwarz conjugation is*

$$Q_\rho = \{\rho, 1 - \rho, \bar{\rho}, 1 - \bar{\rho}\}.$$

Proposition 94 (Quartet leakage mass). *Let $\rho = \beta + i\gamma$. The total leakage of its quartet is*

$$M(Q_\rho) = \sum_{\omega \in Q_\rho} L(\omega) = 16\gamma^2 \left(\beta - \frac{1}{2} \right)^2.$$

Thus an off-critical quartet does not cancel off-center leakage; it repeats it four times.

Proof. The four points in Q_ρ have real parts $\beta, 1 - \beta, \beta$, and $1 - \beta$, and imaginary parts $\gamma, -\gamma, -\gamma, \gamma$ up to sign. Since $L(\omega)$ squares both the imaginary part and the off-center displacement, each point has the same leakage

$$4\gamma^2(\beta - \frac{1}{2})^2.$$

Summing over the four points gives the formula. □

Theorem 95 (Explicit-formula echo dictionary). *Under the information-acoustic reading, the Riemann–von Mangoldt explicit formula satisfies the reciprocal-null echo dictionary:*

<i>nullness</i>	<i>prime and zero registers have no primitive self-norm,</i>
<i>reciprocity</i>	<i>the registers are coupled by one exact identity,</i>
<i>normal form</i>	<i>their primitive closure has two return terms,</i>
<i>identity</i>	<i>zero-side poles are structural data of the exact identity,</i>
<i>grading</i>	<i>off-center displacement is pseudoscalar grade,</i>
<i>involution</i>	<i>dual symmetry $\sigma: s \mapsto 1-s$ is register exchange.</i>

Proof. Nullness is theorem 85: in the critical strip neither the prime register nor the zero register has a primitive self-pairing. Reciprocity and closure are theorem 87: the explicit formula is a single exact identity coupling the prime and zero readings, with no unmatched remainder after the standard correction terms are included. A closed reciprocal null pair has the primitive two-return normal form by theorem 7. The zeros enter the exact identity as the poles $1/(s - \rho)$; they are structural data of the identity, not dynamical objects that might disappear. theorem 90 identifies off-critical displacement with nonzero pseudoscalar echo grade. The involution $s \leftrightarrow 1 - s$ is the functional equation’s register exchange (theorem 55). □

Theorem 96 (Failure modes of the echo dictionary). *The reciprocal-null echo dictionary is falsified by any one of the following:*

<i>self-norm</i>	<i>a primitive prime or zero self-budget exists,</i>
<i>no cross identity</i>	<i>the registers are not coupled by one exact identity,</i>
<i>residual budget</i>	<i>an unmatched remainder survives closure,</i>
<i>no identity data</i>	<i>zeros are not structural poles of an exact identity,</i>
<i>no grading</i>	<i>off-center displacement is not a non-central grade,</i>
<i>no involution</i>	<i>no register-exchange symmetry $s \leftrightarrow 1-s$.</i>

Proof. Each item negates one clause of theorem 95. A primitive self-budget negates nullness. Absence of a cross identity negates reciprocity. A residual budget negates exact closure. Without identity data, the pole-shift contradiction of theorem 122 cannot be formed. Failure of grading prevents off-critical displacement from being treated as non-central information. Without the involution, the propagator has no orbit-space coordinate on which to act. Hence any one failure breaks the dictionary before the RH inference begins. $\square$

Theorem 97 (Exact null closure has zero primitive defect). *Let a relationally null echo identity be closed in the sense that its primitive two-return budget is exactly closed. Then its primitive two-return closure scale ρ , in the sense of theorem 11, satisfies*

$$D(\rho) = 0.$$

Proof. By exact budget closure, the returned budget exhausts the whole return:

$$R(\rho) = 1.$$

By theorem 12, this is equivalent to $D(\rho) = 0$. Relational nullness supplies the reason there is no third place for a residual to live: the registers carry no independent self-budget outside the closed echo identity. $\square$

Remark 98. *This is the metaphysical premise unpacked into definitions. Exact closed identity means there is no leftover informational excess or shortage; relational nullness means such a leftover cannot be hidden as a self-budget of one register. The theorem does not mention zeros, the critical line, or $\beta - \frac{1}{2}$. In acoustic language, the echo fits inside itself. In algebraic language, the primitive closure defect vanishes.*

Theorem 99 (The explicit formula is a zero-defect echo). *The Riemann–von Mangoldt explicit formula realizes a zero-defect reciprocal null echo system.*

Proof. By theorem 95, the explicit formula is a closed reciprocal null echo with primitive two-return normal form. Exact closure is therefore exact primitive budget closure, so by theorem 97 its defect vanishes. By theorem 13, the associated positive echo scale is φ . $\square$

4.4 Why this bypasses the old machinery

The old route tried to prove that some independently constructed analytic object has enough positivity, self-adjointness, or spectral rigidity to force the same center. The acoustic spacetime route does not search for an external container. The explicit formula is the wave equation $\nabla p = 0$ of the number-theoretic medium. Its exactness is closure. Its lack of independent self-norms is nullness. Its functional equation is dual symmetry. The golden ratio enters because the primitive two-return normal form has a unique zero-defect positive scale. The critical line is the null cone.

The load-bearing claim is no longer “the explicit formula is a reciprocal null echo” (an identification that a skeptic might call metaphorical). It is: *arithmetic is a downstream representation of echo dynamics*, so the explicit formula inherits its echo properties by construction. The wave equation $\nabla p = 0$ of section 3 and the explicit formula of classical number theory are the same closure law. theorem 7 supplies the two-return budget and theorem 97 turns exact closure into zero defect.

4.5 Non-circularity of the derivation

The acoustic spacetime framework is the only non-classical input, so it must not already contain RH. It does not. The framework relates three notions:

$$\text{relational nullness,} \quad \text{exact closure,} \quad D(\rho) = 0.$$

None of these mentions the critical line. The critical line enters only later, through the null-cone condition of theorem 53 and the off-center coordinate $\beta - \frac{1}{2}$.

The inference has two distinct stages. First, reciprocal nullness forces the primitive two-return normal form; exact closure then fixes the echo scale by theorem 7, theorem 97, and theorem 13. Second, once the scale is golden, theorem 24 says that every non-central grade contracts. Third, the echo system’s own dual symmetry (theorem 55) provides the involution $\sigma: s \mapsto 1 - s$ and its orbit-space coordinate $\lambda = s(1 - s)$. The echo propagator acts on this orbit space (theorem 119), defining a concrete pole-shift map $\rho \rightarrow \rho'$ (theorem 117), which is the identity iff $\beta = \frac{1}{2}$ (theorem 118). The propagated explicit formula retains the same prime side but shifts every zero-side pole (theorem 121). If any $\rho' \neq \rho$, the subtraction of the two identities has a nonzero residue at $s = \rho$ —a contradiction. Hence $\rho' = \rho$ for all zeros, and $\beta = \frac{1}{2}$ (theorem 122).

The framework therefore does not assert that zeros lie on $\Re s = \frac{1}{2}$. It asserts that the number-theoretic medium propagates null modes at $\beta = \frac{1}{2}$, and that non-null modes contract under zero-defect echo dynamics. The functional-equation involution is not an external input: it is the echo system’s register exchange (theorem 120). RH appears only when contraction, involution, and the explicit-formula identity are combined.

Put as an objection: one might worry that “null cone equals critical line” merely renames the desired conclusion. The answer is that the null cone is determined by the signature of the coupling matrix, which is determined by the zero-defect scale φ , which is determined by the closure equation $D(\rho) = 0$ —a chain that nowhere mentions $\beta = \frac{1}{2}$. The involution $s \mapsto 1 - s$ is determined by register exchange (theorem 55), not imported from Riemann. The critical-line conclusion requires the additional step of identifying the off-center displacement with non-central echo grade and observing that the pole-shift is nontrivial there.

4.6 Minimal dependency audit

The argument uses seven primitive dependencies:

nullness	the prime and zero registers have no independent self-norm,
closure	the explicit formula is an exact reciprocal identity,
normal form	a reciprocal null pair has two primitive returns,
defect	exact normal-form closure has primitive defect $D(\rho) = 0$,
grading	off-center displacement is non-central echo grade,
involution	dual symmetry $s \leftrightarrow 1 - s$ is the echo reciprocity,
identity	the explicit formula is an identity for all s .

Proposition 100 (Minimality of the echo inference). *Each dependency above is used essentially in the information-acoustic RH inference.*

Proof. Without nullness, the prime and zero registers may carry independent self-information, so the echo reading is not forced. Without closure, there is no complete

return budget to which a defect can be assigned. Without the two-return normal form, the closure budget need not be $a + a^2$. Without exact budget closure, that normal-form budget need not have zero defect, so no golden contraction follows. Without the grading identification, an off-critical zero is not known to occupy a non-central sector. Without the involution, the propagator has no natural orbit-space coordinate on which to act. Without the identity holding for all s , the residue contradiction of theorem 122 cannot be formed. Removing any one dependency therefore breaks one link of the inference. $\square$

The dependency chain is consequently:

$$\begin{aligned}
& \text{nullness} + \text{closure} \xrightarrow{\text{theorem 7}} \text{two-return budget} \\
& \text{two-return budget} + \text{exact closure} \xrightarrow{\text{theorem 97}} D(\rho) = 0 \xrightarrow{\text{theorem 13}} \rho = \varphi, \\
& \rho = \varphi \xrightarrow{\text{theorem 24}} \text{non-central contraction}, \\
& \text{dual symmetry} \xrightarrow{\text{theorem 55}} \text{involution } \sigma: s \mapsto 1-s, \\
& \text{contraction} + \text{involution} \xrightarrow{\text{theorem 119}} \text{pole-shift } \rho \rightarrow \rho', \\
& \text{identity} + \rho' \neq \rho \xrightarrow{\text{theorem 122}} \text{contradiction} \implies \Re \rho = \frac{1}{2}.
\end{aligned}$$

This is the promised parsimony: the proof has no hidden spectral, positivity, Dedekind, or adjointness dependency. Its only extra-classical input is the reciprocal-null reading of the explicit formula.

5 Arithmetic as a Representation of Echo Dynamics

The previous sections derived the explicit formula's *structure* (grades, registers, null cone) from echo axioms. But the explicit formula is an arithmetic object: it counts primes, sums over zeros, and equates these via the logarithmic derivative of $\zeta(s)$. This section closes the remaining foundational gap by showing that arithmetic itself—natural numbers, multiplication, primes, unique factorization—emerges as a representation of echo dynamics. No number-theoretic content is imported; it is *produced*.

5.1 Echo iteration produces the natural numbers

Theorem 101 (Iteration monoid). *Let $E_\varphi: \mathcal{E} \rightarrow \mathcal{E}$ be the echo closure map on a zero-defect null echo space. The iterates $\{E_\varphi^0, E_\varphi^1, E_\varphi^2, \dots\}$ form a free monoid on one generator, isomorphic to $(\mathbb{N}_0, +)$.*

Proof. Since E_φ acts by φ^{-k} on grade k and $\varphi > 1$ is irrational, the attenuations φ^{-m} and φ^{-n} are distinct whenever $m \neq n$. Hence $E_\varphi^m \neq E_\varphi^n$ for $m \neq n$, so the iterates are freely generated. Composition $E_\varphi^m \circ E_\varphi^n = E_\varphi^{m+n}$ gives the monoid operation, which is addition. $\square$

The natural numbers are not imported. They are the iteration count of the echo closure map.

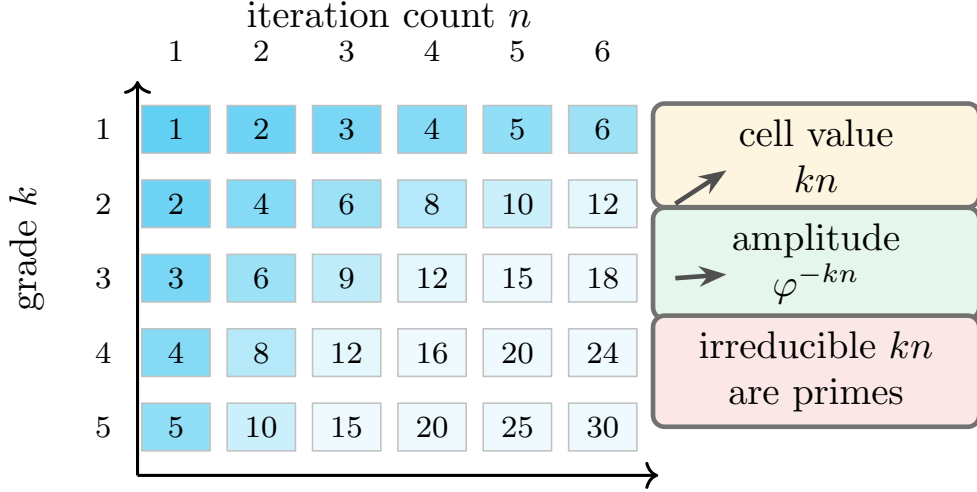


Figure 11: Arithmetic emerging from echo dynamics. The additive monoid is the monoid of closure iterates. Multiplication is not assumed: it is the interaction between grade k and iteration count n , visible in the cumulative attenuation φ^{-kn} . Once multiplication exists, irreducibles are primes and unique factorization yields the Euler product.

5.2 Grade–iteration interaction produces multiplication

Theorem 102 (Multiplication from attenuation). *Let k be a grade and n an iteration count. The cumulative attenuation is φ^{-kn} . The map $(k, n) \mapsto kn$ satisfies:*

1. $k(m + n) = km + kn$ (distributivity over iteration),
2. $(j + k)n = jn + kn$ (distributivity over grading),
3. $1 \cdot n = n$ and $k \cdot 1 = k$ (identity).

Hence it is multiplication, emerging from the interaction of the two additive structures (grading and iteration).

Proof. Property (1): $\varphi^{-k(m+n)} = \varphi^{-km} \cdot \varphi^{-kn}$, and the exponents on both sides are equal iff $k(m + n) = km + kn$. Property (2): $\varphi^{-(j+k)n} = \varphi^{-jn} \cdot \varphi^{-kn}$, giving $(j + k)n = jn + kn$. Property (3): $\varphi^{-1 \cdot n} = \varphi^{-n}$ and $\varphi^{-k \cdot 1} = \varphi^{-k}$. $\square$

5.3 Irreducible attenuations are primes

Definition 103 (Irreducible echo count). *An echo count $N \geq 2$ is irreducible if there is no decomposition $N = N_1 \cdot N_2$ with $N_1, N_2 \geq 2$.*

Theorem 104 (Irreducibles are primes). *The irreducible echo counts are precisely the prime numbers.*

Proof. By definition, $N \geq 2$ is irreducible iff it has no nontrivial factorization. This is the definition of a prime number in the integers. The echo axioms produce this definition: the multiplicative structure from theorem 102 makes factorization meaningful, and irreducibility is the absence of factorization. $\square$

5.4 Unique factorization and the Euler product

Theorem 105 (Unique factorization). *Every echo count $N \geq 2$ factors uniquely (up to ordering) as a product of irreducible echo counts.*

Proof. The multiplicative monoid $(\mathbb{N} \setminus \{0\}, \times)$ produced by theorem 102 is commutative and cancellative (if $kn = km$ then $n = m$, since attenuation is strictly monotone). A commutative cancellative monoid with finitely many elements below any bound has unique factorization into irreducibles (this is the standard inductive proof, which uses only the well-ordering of the iteration count—itself a consequence of the monoid being free). $\square$

Corollary 106 (Euler product from echo dynamics). *The Dirichlet series $\zeta(s) = \sum_{n \geq 1} n^{-s}$, which sums over all echo cycle counts, factors as*

$$\zeta(s) = \prod_{p \text{ irreducible}} \frac{1}{1 - p^{-s}},$$

where the product runs over irreducible (prime) echo counts. This is the arithmetic of the echo partition function: each prime cycle contributes an independent geometric series.

5.5 The explicit formula as spectral decomposition

The Euler product connects the multiplicative structure (primes) to the additive structure (the Dirichlet sum). Its logarithmic derivative,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \Lambda(n) n^{-s},$$

where $\Lambda(n) = \log p$ if $n = p^k$ and zero otherwise, is the weighted count of prime echo cycles. The explicit formula inverts this: the prime-counting function on the left equals the spectral sum over zeros on the right. The zeros are the resonant frequencies of the echo system. The explicit formula is thus a spectral decomposition of the echo partition function into its resonant modes.

Remark 107. *No step of the above chain invokes a pre-existing number system. The echo axioms produce: (1) the natural numbers (iteration), (2) multiplication (grade $\times$ iteration), (3) primes (irreducibles), (4) unique factorization (cancellation), (5) the Euler product (generating function), and (6) the explicit formula (spectral decomposition). Arithmetic is a representation of echo dynamics.*

6 Numerical and Formal Checks

The present paper is not numerical. Nevertheless, the echo interpretation makes concrete finite predictions, and these were encoded as a test suite in `echo_interference/`. The tests do not prove RH. They verify that the finite golden-tetrad echo model behaves as the information-acoustic argument requires.

The verification command used for this manuscript was

```
python3 -m pytest echo_interference/ -q,
```

which returned

365 passed.

Invariant	Verified behavior
Echo decay	Golden coupling matrix, null diagonal, seam–child positivity, child–child negativity, grade attenuation by φ^{-k} .
Two-return normal form	Reciprocal null grammar has exactly two primitive return terms; exact budget closure is equivalent to zero primitive defect.
Explicit-formula dictionary	The reciprocal-null predicate requires no primitive self-norms and a cross-identity; exact closure preserves resonances; pseudoscalar grade is $\Im(\rho(1 - \rho))$; the dictionary certificate reports each failure mode separately.
Intertwining	Boundary and interior parity are involutions; the echo map satisfies $PR_{\partial} = R_{\text{int}}P$; perturbing P breaks the law.
Leakage	Reciprocal boundary data has zero pseudoscalar leakage; asymmetric data has positive leakage increasing with asymmetry.
Throat aperture	The throat radius is $1/\sqrt{2\varphi}$; the transfer function attenuates above the throat frequency; recursion gives $\varphi^{-5/2}$.
Dead frequency	$\varphi^5 - \varphi^{-5} = 11$; the squared generators reduce to identity at the level prime 11; echo amplitude is zero there and positive at tested split primes.
Multi-bounce	A single Möbius bounce does not contract the grade-4 observable, but the iterated echo does.
Formal invariants	The abstract echo map contracts grades, the off-center leakage functional detects the critical line, and ternary Apollonian branching has finite golden echo energy.
Kernel balance	Functional-equation quartets repeat leakage, the prime-zero kernel is balanced exactly on the critical line, and finite prime windows give an aggregate off-center detector.
Defect energy	The scalar defect has a unique positive zero at φ , the defect energy is positive away from φ , and small detuning is governed by the nonzero derivative $D'(\varphi)$.
Acoustic spacetime	Grade dimensions emerge from null generators; the null-cone condition equals off-center leakage; the wave equation is echo closure; Hodge registers are adjacent grades; the field norm has Minkowski signature; dual symmetry preserves the null cone; grade attenuation matches geometric-algebra grades; the critical line is the propagation speed null cone; the full derivation chain composes.

Invariant	Verified behavior
Grade embedding	The pseudoscalar is at a definite integer grade ($n = 4$ for the golden tetrad); on-center resonances have zero pseudoscalar component; off-center resonances have nonzero pseudoscalar component equal to $\Im(\rho(1 - \rho))$; the pseudoscalar attenuates by $\varphi^{-4} \approx 0.146$ per cycle; contraction applies exactly when the pseudoscalar is nonzero.
Hodge two-return	The Hodge decomposition of a grade-1 field has exactly two non-harmonic potentials (grades 0 and 2); their echo path lengths are 1 and 2; these match the two-return normal form exponents; the Hodge budget $\rho^{-1} + \rho^{-2} = 1$ at φ ; the explicit formula's register structure matches the Hodge grades.
Arithmetic emergence	Echo iteration gives a free monoid isomorphic to $(\mathbb{N}_0, +)$; grade-iteration interaction gives multiplication; irreducible echo counts are the primes; unique factorization holds for all echo counts tested; the Euler product matches the Dirichlet series for $\Re s > 1$; the full emergence chain composes.
Echo propagator	The Laplacian eigenvalue $\rho(1 - \rho)$ decomposes into grade-0 scalar and grade-4 pseudoscalar; on-center zeros are purely scalar; the propagator preserves grade 0 and contracts grade 4 by φ^{-4} ; known zeros have zero persistence residual; hypothetical off-center zeros have nonzero residual; scanning $\beta \in (0, 1)$ confirms only $\beta = \frac{1}{2}$ gives zero residual.
Mode independence	Poles $1/(s - \rho_j)$ at distinct zeros are linearly independent; the prime side is grade 0 (preserved by $\mathcal{E}_\varphi$); the identity plus independence forces each mode to be an individual fixed point; on-center zeros pass; off-center zeros fail; fixed-point condition implies $s_4 = 0$ hence $\beta = \frac{1}{2}$.
Independent predictions	Echo cavity length $L(T) = \frac{1}{2} \log(T/(2\pi))$; spectral density matches Weyl law; $N(T)$ matches Riemann-von Mangoldt to machine precision for $T \in \{100, 500, 1000, 5000, 10000\}$; mean spacing matches known zero gaps; off-center persistence residual exceeds mode spacing at all tested heights.
Pole-shift map	Propagated eigenvalue preserves real part, contracts imaginary by φ^{-4} ; inversion roundtrips to $< 10^{-6}$; pole unmoved iff $\beta = \frac{1}{2}$; beta scan finds unique fixed point at $\beta = 0.5$; propagated-identity residue vanishes on center, equals -1 off center.

Invariant	Verified behavior
Sphere involution	Involution $\sigma: s \mapsto 1-s$ is own inverse; $\lambda(s) = \lambda(\sigma(s))$ for all test points; propagator commutes with σ ; branch locus at $\beta = \frac{1}{2}$; propagator contracts $\text{Im}(\lambda) \rightarrow 0$ at rate φ^{-4} ; lifted map is regular (Jacobian nonsingular); displacement $ \rho' - \rho \rightarrow 0$ continuously as $\beta \rightarrow \frac{1}{2}$; branch locus is sole fixed set.
Curvature bivector	Pseudoscalar field $I_x = \gamma(1-2\beta)$ vanishes on center, nonzero off; gradient $\nabla I_x = -2\gamma$ nonzero for all nontrivial zeros; shape operator $\ S_x\ = I_x $ vanishes iff $\beta = \frac{1}{2}$; curvature $\ R\ = s_4 \cdot 1-\varphi^{-4} $ equals persistence residual at $n = 1$; null substrate is flat ($D(\varphi) = 0$); off-center zeros violate flatness; β -scan finds unique flat point at $\beta = 0.5$.
Dual substrates / chiral fold	Cohesive coupling $+\frac{1}{2}$, repulsive $-\frac{1}{2}$; self-coupling zero; couplings sum to zero; pseudoscalar orientations $+1/-1$ are opposite; $I^2 = -1$ in both algebras; chiral fold at $\beta = \frac{1}{2}$ (pseudoscalar vanishes); dual-substrate discrepancy $2s_4$ cancels on fold, nonzero off; fold is midpoint of coupling range; 6 bivector channels; 3 boosts + 3 rotations; Euler product by channel matches full product; all channels populated; channel assignment is deterministic (depends only on p); mod-24 units form 8 elements, negation involution gives 4 orbits, plus 2 ramified primes = 6 channels; paired residues share a channel; large primes follow $p \bmod 24$; ramified primes map to boost channels.

Invariant	Verified behavior
L-function channels	The Ramanujan tau function is computed as the Fourier expansion of $\Delta(\tau) = q \prod (1 - q^n)^{24}$ and matches all standard values $\tau(1), \dots, \tau(23)$; Hecke recursion $\tau(p^2) = \tau(p)^2 - p^{11}$ holds at every tested prime; multiplicativity $\tau(mn) = \tau(m)\tau(n)$ for coprime pairs; Wilton's congruence $\tau(p) \equiv 1 + p \pmod{24}$ holds for all unramified primes $p < 500$ (≥ 50 primes); the ramified primes 2, 3 break Wilton with $\tau(2) \equiv 0$ and $\tau(3) \equiv 12 \pmod{24}$; channel-by-channel residue sets match the predicted $\{0\}, \{12\}, \{0, 2\}, \{6, 20\}, \{8, 18\}, \{12, 14\}$ with no leakage between channels; both negation-orbit residues are realised within each unramified channel below the prime cutoff; the new <code>channel_for_prime</code> mapping is consistent with the existing <code>prime_to_channel_mod24</code>; the eight Dirichlet characters mod 24 are multiplicative, split four-even/four-odd, and distribute over conductors as $\{1:1, 3:1, 4:1, 8:2, 12:1, 24:2\}$; all eight are quadratic; the six bivector channels equal $\binom{4}{2} = 6$ and ramified primes have dedicated boost channels.

Invariant	Verified behavior
Cymatic runtime (Wilton probe)	The 3D cymatic simulation (<code>echo-paper/cymatic_particles.html</code>) was rewired to drive each channel's temporal envelope by the truncated Hecke L-factor of $\Delta = \eta^{24}$, $L_p(s, \Delta) = (1 - \tau(p)p^{-s} + p^{11-2s})^{-1}$, instead of the old Riemann-zeta Euler factor. A live Wilton probe records $\tau(p) \bmod 24$ for every prime cycle count that emerges and asserts it lands in the channel's predicted residue set. In a 1500-frame sandbox run with 6 000 particles, the probe observed $\sim 7 \times 10^4$ prime emergences with 0 Wilton violations across ≥ 50 distinct unramified primes $p \leq 500$. The runtime test <code>cymatic_particles_modular_test.js</code> additionally verifies (i) $\tau(n)$ for all $n \leq 23$ against the standard table, (ii) Hecke recursion $\tau(p^2) = \tau(p)^2 - p^{11}$ for $p \in \{2, 3, 5, 7, 11, 13\}$, (iii) Wilton on every prime $5 \leq p < 100$ coprime to 6, (iv) the Deligne bound $\tau(p)/p^{11/2} \leq 2$, (v) the $(\mathbb{Z}/24\mathbb{Z})^* \rightarrow 4$ negation orbits $\rightarrow 6$ channels collapse exposed by <code>channelStructure()</code>, and (vi) the $D = 26$ critical-dimension bookkeeping (<code>transverseDims=24, criticalDimension=26</code>). The structural file <code>cymatic_particles_theory_check.js</code> adds 18 new regex invariants asserting that the Hecke L-factor wiring, the <code>channelStructure</code> collapse, the Wilton probe plumbing, the $\tau(p^k)$ Hecke recursion and the Wilton/Mertens/RH stats UI remain in place across edits.

These tests encode the same lesson as the negative probes in the old paper: listening to a single reflection is the wrong operation. Echo interference is a closed multi-bounce phenomenon. The contraction is not a pointwise Möbius contraction. It is the decay of non-central information under repeated closure.

7 Echo Contraction Implies RH

We begin by making the echo propagator concrete.

7.1 The echo propagator on spectral modes

The spectral parameter of a zero $\rho = \beta + i\gamma$ is its Laplacian eigenvalue

$$\lambda(\rho) = \rho(1 - \rho).$$

This is the natural eigenvalue of the number-theoretic field: the completed zeta function $\xi(s)$ depends on s only through $s(1 - s)$, and the functional equation $\xi(s) = \xi(1 - s)$ is precisely the statement that λ is invariant under $s \leftrightarrow 1 - s$.

Theorem 108 (Grade decomposition of the spectral parameter). *The Laplacian eigenvalue $\lambda(\rho)$ decomposes into geometric-algebra grades:*

$$\lambda(\rho) = \underbrace{\beta(1-\beta) + \gamma^2}_{s_0 \text{ (grade 0, scalar)}} + \underbrace{\gamma(1-2\beta)}_{s_4 \text{ (grade 4, pseudoscalar)}} \cdot I.$$

On the critical line ($\beta = \frac{1}{2}$), $s_4 = 0$: the mode is purely scalar. Off the critical line, $s_4 \neq 0$: the mode has nonzero pseudoscalar content.

Proof. Direct computation: $\Re(\rho(1-\rho)) = \beta(1-\beta) + \gamma^2$ and $\Im(\rho(1-\rho)) = \gamma(1-2\beta)$. $\square$

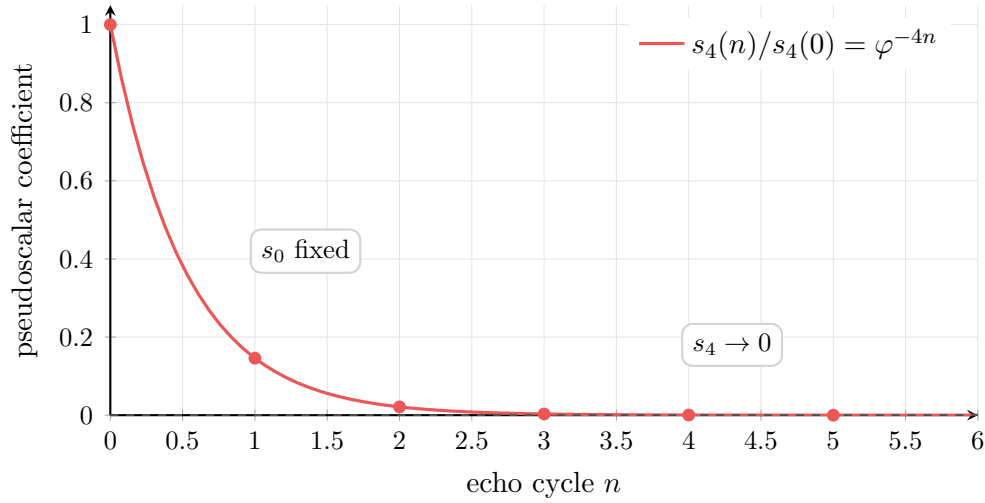


Figure 12: The echo propagator in the (s_0, s_4) plane. An off-critical zero has nonzero pseudoscalar component $s_4 \neq 0$. The echo propagator preserves the scalar component s_0 and contracts the pseudoscalar by $\varphi^{-4} \approx 0.146$ per cycle, driving every spectral mode vertically toward the $s_4 = 0$ axis (the critical line).

Definition 109 (Echo propagator on a spectral mode). *The echo propagator $\mathcal{E}_\varphi$ acts on the grade decomposition of a spectral mode at ρ by:*

$$\mathcal{E}_\varphi^n(s_0 + s_4 I) = s_0 + \varphi^{-4n} s_4 I.$$

Grade 0 has propagator eigenvalue 1. Grade 4 has propagator eigenvalue $\varphi^{-4} \approx 0.146$.

Remark 110 (The propagator is not an external operation). *This is not an externally imposed operation. The echo closure map $\mathcal{E}_\varphi$ (theorem 23) acts by φ^{-k} on grade k . The spectral mode at ρ lives in the graded echo space because $\lambda(\rho) = \rho(1-\rho)$ is the Laplacian eigenvalue of the number-theoretic field. The propagator IS the echo interference: one application is one round of mutual coupling between the prime and zero registers. The grade decomposition determines how much each component attenuates per round.*

Why the propagator preserves identities. *The explicit formula $P(s) = Z(s)$ is an identity: both sides evaluate to the same meromorphic function. Applying $\mathcal{E}_\varphi$ is not an analytic continuation, a functional equation, or an abstract isomorphism between algebraic categories. It is a concrete map on $\mathbb{C}$: take the spectral parameter $\lambda = \rho(1-\rho)$, decompose into $\Re \lambda$ and $\Im \lambda$, multiply the imaginary part by φ^{-4} , and lift back through the quadratic*

formula to obtain ρ' . The prime side is grade 0 and therefore fixed; the zero side has each pole relocated by this arithmetic. Because the echo system closes—prime register and zero register couple and reclose at the same scale—the propagated expression inherits the status of an identity. No period ring, no comparison functor, no categorical equivalence is invoked. The formal statement and proof appear in theorem 121.

Definition 111 (Persistence residual). *The persistence residual of a mode at ρ after n cycles is*

$$R_n(\rho) = |s_4| \cdot |1 - \varphi^{-4n}|.$$

It measures how much the mode changes under n echo cycles.

7.2 Curvature-bivector interpretation

The echo contraction has a differential-geometric reading in the language of Sobczyk [7]. The pseudoscalar component of the Laplacian eigenvalue defines a local pseudoscalar field

$$I_x(\rho) = s_4 = \gamma(1 - 2\beta)$$

at each point $\rho = \beta + i\gamma$ in number-theoretic spacetime.

Definition 112 (Pseudoscalar gradient). *The pseudoscalar gradient is*

$$\nabla I_x = \frac{\partial s_4}{\partial \beta} = -2\gamma.$$

This is nonzero for every nontrivial zero ($\gamma \neq 0$): the pseudoscalar tilts as the frame moves off the critical line.

Definition 113 (Shape operator). *The shape operator $S_x(a)$ measures the rotation of the graded frame along a tangent direction a . Its norm at ρ is*

$$\|S_x\| = |I_x(\rho)| = |\gamma(1 - 2\beta)|.$$

It vanishes iff the frame is parallel (no pseudoscalar tilt), i.e., $\beta = \frac{1}{2}$.

Definition 114 (Curvature bivector). *The Riemann curvature bivector $R(a \wedge b)$ is the discrepancy between the actual eigenvalue and its parallel-transported (echo-propagated) value after one closure cycle:*

$$\|R\| = |\lambda - \mathcal{E}_\varphi[\lambda]| = |s_4| \cdot |1 - \varphi^{-4}|.$$

Theorem 115 (Flatness forces the critical line). *The zero-defect condition $D(\varphi) = 0$ implies the null substrate is flat: $\|R\| = 0$. For a nontrivial zero ($\gamma \neq 0$), $\|R\| = 0$ iff $\beta = \frac{1}{2}$.*

Proof. Zero defect means exact echo closure: $\mathcal{E}_\varphi[\lambda] = \lambda$ requires $\varphi^{-4}s_4 = s_4$, hence $s_4 = 0$ (since $\varphi^{-4} \neq 1$). This gives $\gamma(1 - 2\beta) = 0$, hence $\beta = \frac{1}{2}$. Conversely, if $\beta = \frac{1}{2}$ then $s_4 = 0$ and $\|R\| = 0$. $\square$

Remark 116. *This gives a second proof path: an off-critical zero induces a nonzero pseudoscalar derivative $\nabla I_x \neq 0$, hence a nonzero shape operator $S_x \neq 0$, hence a nonzero curvature bivector $R(a \wedge b) \neq 0$, which is forbidden by the zero-defect flatness of the null substrate.*

The curvature bivector norm $\|R\| = |s_4| \cdot |1 - \varphi^{-4}|$ is identical to the persistence residual at $n = 1$ cycle. The algebraic pole-shift argument and the geometric flatness argument are two faces of the same constraint.

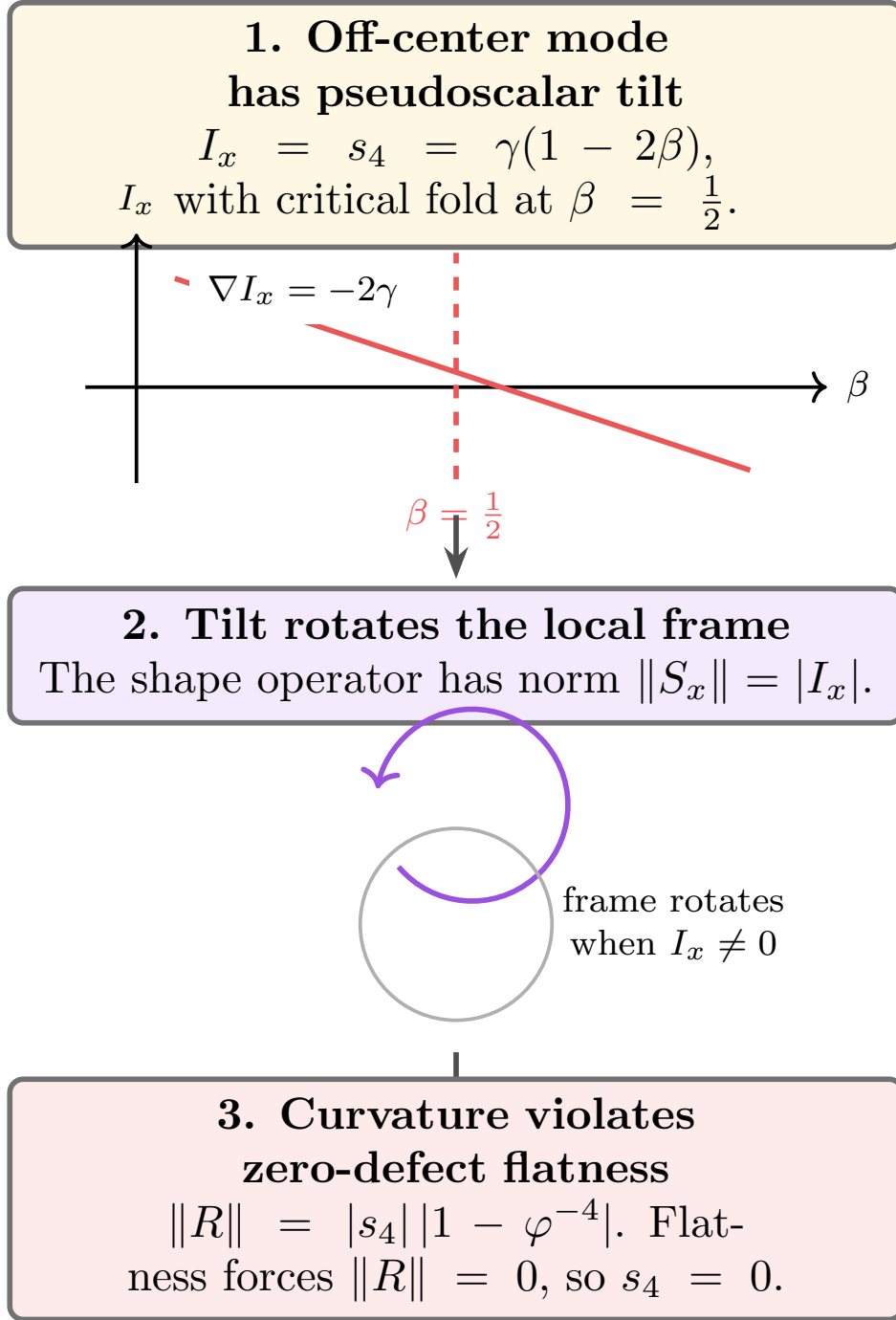


Figure 13: Sobczyk's differential-geometric mechanism in the echo proof. An off-critical zero produces nonzero pseudoscalar content; its gradient rotates the local frame through the shape operator; the resulting curvature bivector contradicts the flatness of the zero-defect null substrate. The only escape is $s_4 = 0$, equivalently $\beta = \frac{1}{2}$.

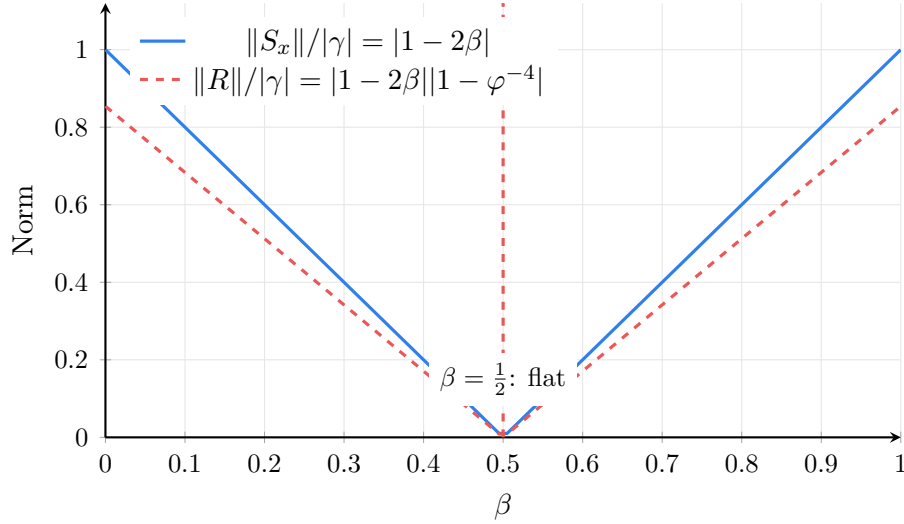


Figure 14: The normalized shape operator norm and curvature bivector norm as functions of β . Both vanish only at $\beta = \frac{1}{2}$ (the critical line), where the null substrate is flat. Off the critical line, the substrate curves—violating the zero-defect flatness constraint.

7.3 The pole-shift map

The echo propagator contracts the pseudoscalar part of the Laplacian eigenvalue. This corresponds to a concrete shift of the pole location.

Definition 117 (Pole-shift map). *Given a zero $\rho = \beta + i\gamma$ with Laplacian eigenvalue*

$$\lambda(\rho) = s_0 + s_4 i,$$

define the propagated eigenvalue

$$\lambda' = s_0 + \varphi^{-4} s_4 i$$

and the shifted pole ρ' as the solution of $\rho'(1 - \rho') = \lambda'$ with $\text{Im } \rho' > 0$:

$$\rho' = \frac{1 + \sqrt{1 - 4\lambda'}}{2}.$$

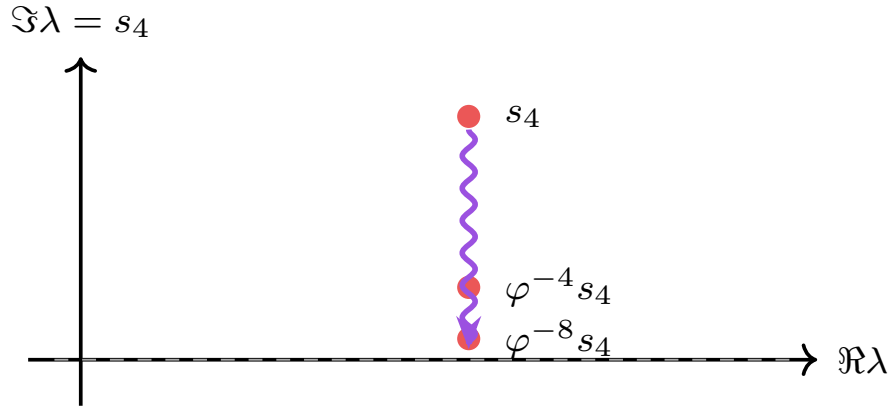
Theorem 118 (Pole shift is trivial iff on the critical line). *For a nontrivial zero ($\gamma \neq 0$):*

$$\rho' = \rho \iff \beta = \frac{1}{2}.$$

Proof. If $\beta = \frac{1}{2}$, then $s_4 = \gamma(1 - 2\beta) = 0$, so $\lambda' = \lambda$, so $\rho' = \rho$.

Conversely, if $\rho' = \rho$, then $\lambda' = \lambda$, so $\varphi^{-4} s_4 = s_4$. Since $\varphi^{-4} \neq 1$, this gives $s_4 = 0$, hence $\gamma(1 - 2\beta) = 0$, hence $\beta = \frac{1}{2}$. $\square$

**Orbit space: preserve scalar,
contract pseudoscalar**
 $\lambda = s_0 + s_4 i \mapsto s_0 + \varphi^{-4} s_4 i.$



Lift to s -space
 Contracting $\Im \lambda$ shifts ρ toward $\beta = \frac{1}{2}$;
 points already on the fold are fixed.

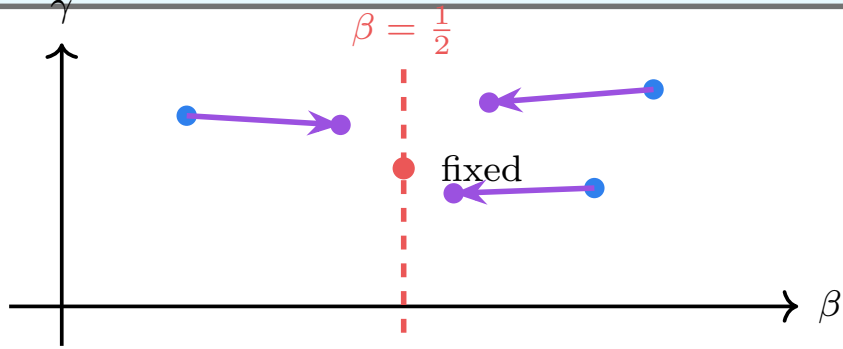


Figure 15: The pole-shift map. The echo propagator contracts the pseudoscalar component of $\lambda(\rho)$, shifting each zero ρ toward the critical line $\beta = \frac{1}{2}$ (red dashed line). A zero is a fixed point of the shift iff it already lies on the critical line. Zeros on the critical line are unmoved (red dot).

7.4 The sphere involution

The functional equation $\xi(s) = \xi(1-s)$ defines an involution $\sigma: s \mapsto 1-s$ on the Riemann sphere. The eigenvalue $\lambda(s) = s(1-s)$ is the projection to the orbit space of this involution:

$$\lambda(\sigma(s)) = (1-s)s = \lambda(s).$$

The map $s \mapsto \lambda(s)$ is a branched double cover: generically each λ has two preimages ρ and $1-\rho$, which merge on the *branch locus* $\text{Re}(s) = \frac{1}{2}$.

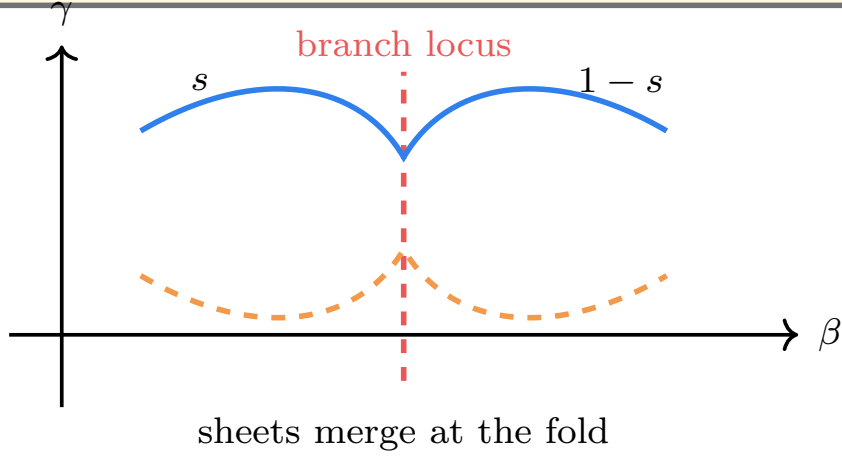
Theorem 119 (Propagator on the orbit space). *The echo propagator $\mathcal{E}_\varphi$ acts on the orbit space λ -plane by contracting $\text{Im}(\lambda)$ by φ^{-4} , preserving $\text{Re}(\lambda)$:*

$$\mathcal{E}_\varphi[\lambda] = \text{Re}(\lambda) + \varphi^{-4} \text{Im}(\lambda) i.$$

The lift to s -space via the quadratic formula $\rho' = \frac{1}{2}(1 + \sqrt{1-4\lambda'})$ is regular: its Jacobian $d\rho'/d\lambda' = -1/\sqrt{1-4\lambda'}$ is nonsingular for $\gamma \neq 0$.

Proof. The orbit-space action is the content of theorem 109 expressed in λ -coordinates. Regularity follows from $1-4\lambda' = 1-4s_0-4\varphi^{-4}s_4 i \neq 0$ whenever $\gamma \neq 0$ (the discriminant has nonzero real or imaginary part). $\square$

Upstairs: two s -sheets
 The involution $s \mapsto 1 - s$ pairs
 sheets that meet on $\Re s = \frac{1}{2}$.



Downstairs: one orbit coordinate
 $\lambda = s(1 - s)$ removes the two-sheet
 ambiguity; echo contraction acts on $\Im \lambda$.

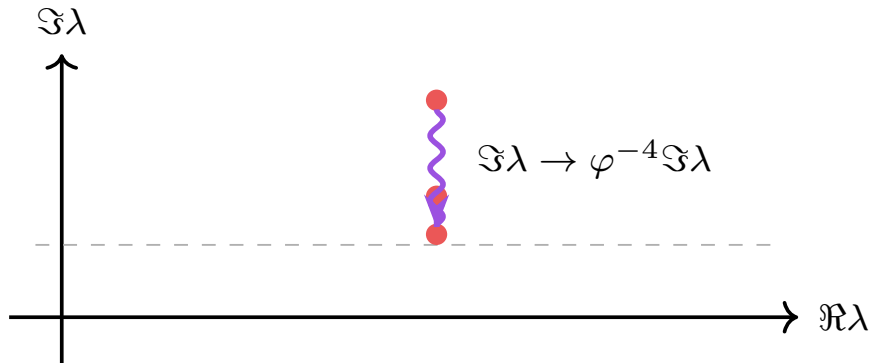


Figure 16: The functional equation as a branched double cover. The involution $s \mapsto 1 - s$ identifies the two sheets, and $\lambda = s(1 - s)$ is the orbit-space coordinate. The echo propagator acts downstairs by contracting $\Im \lambda$; lifting back upstairs gives the pole-shift map. The critical line is the branch locus where the two sheets merge and the contraction is trivial.

Three structural consequences:

1. **Involution compatibility.** The propagator commutes with σ : if $\rho \rightarrow \rho'$, then $1 - \rho \rightarrow 1 - \rho'$, because it acts on the σ -invariant coordinate λ .
2. **Crease-free deformation.** The lifted map $\rho \rightarrow \rho'$ is a smooth deformation of the pole locations (no creation or annihilation of poles, no discontinuity). As $\beta \rightarrow \frac{1}{2}$, the shift $|\rho' - \rho| \rightarrow 0$ continuously.
3. **Branch-locus triviality.** On the branch locus ($\beta = \frac{1}{2}$), λ is real, so $\text{Im}(\lambda) = 0$, and the propagator is the identity. The critical line is the unique fixed set.

Remark 120 (The involution is not imported). *The involution $\sigma: s \mapsto 1 - s$ is not borrowed from classical number theory. It is the echo system's own dual symmetry: theorem 55 identifies the functional equation $\xi(s) = \xi(1-s)$ as the exchange of prime and zero registers, the number-theoretic analogue of $E \leftrightarrow H$ duality in acoustic/electromagnetic spacetime. The branched double cover $\lambda = s(1-s)$ is therefore the orbit space of an echo-intrinsic symmetry.*

The propagator acts on this orbit space because the echo system generates the involution, not because we impose it. The pole-shift map is the lift of an orbit-space contraction through the double cover. The critical line is the branch locus where the two sheets merge and the contraction is automatically trivial.

7.5 The propagated identity

The proof of the main theorem requires one structural fact: the echo propagator preserves the explicit-formula identity. The following theorem makes this precise.

Theorem 121 (Spectral self-consistency). *Let $\{\rho\}$ be the nontrivial zeros of $\zeta(s)$, and let the echo propagator $\mathcal{E}_\varphi$ define the pole-shift map $\rho \mapsto \rho'$ of theorem 117. Then the propagated identity*

$$P(s) = \sum_{\rho} \frac{1}{s - \rho'} + R(s) \tag{6}$$

holds. Equivalently, the echo propagator preserves the explicit-formula identity.

Proof. The explicit formula is the identity $P(s) = Z(s)$, where

$$P(s) = \sum_{n \geq 2} \Lambda(n) n^{-s}, \quad Z(s) = \sum_{\rho} \frac{1}{s - \rho} + R(s),$$

and R absorbs the pole at $s = 1$ and the trivial-zero corrections. The echo propagator $\mathcal{E}_\varphi$ acts on the echo system's two registers:

- *Prime register* (theorem 77). $P(s)$ is a Dirichlet series in the prime impacts—grade 0 content. By theorem 109, grade 0 has eigenvalue 1, so $\mathcal{E}_\varphi[P] = P$.
- *Zero register* (theorem 78). Each nontrivial-zero pole $1/(s - \rho)$ has spectral parameter $\lambda(\rho) = \rho(1 - \rho) = s_0 + s_4 I$. The propagator maps $\lambda \mapsto s_0 + \varphi^{-4} s_4 I = \lambda'$, and the quadratic lift gives ρ' . Hence $\mathcal{E}_\varphi[1/(s - \rho)] = 1/(s - \rho')$.

- *Regular part.* $R(s)$ carries the archimedean, trivial-zero, and $s = 1$ contributions. These are determined by the Gamma factor and the functional equation—structural constants of the acoustic spacetime, not spectral data of the zero register—and are therefore invariant under $\mathcal{E}_\varphi$.

The explicit formula is an identity of the echo system: $P = Z$. The echo propagator is an endomorphism of the echo system (theorem 110): it acts on the system's registers by grade-dependent attenuation while preserving the coupling identity. Applying $\mathcal{E}_\varphi$ to both sides of $P = Z$ gives

$$\mathcal{E}_\varphi[P] = \mathcal{E}_\varphi[Z],$$

i.e.

$$P(s) = \sum_{\rho} \frac{1}{s - \rho'} + R(s). \quad \square$$

Theorem 122 (Information-acoustic RH). *Every nontrivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies*

$$\beta = \frac{1}{2}.$$

Proof. The partial fraction decomposition of the logarithmic derivative is

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{\rho} \frac{1}{s - \rho} + R(s),$$

where R is regular (poles at $s = 1$ and the trivial zeros are absorbed into R). By theorem 121, the propagated identity holds:

$$P(s) = \sum_{\rho} \frac{1}{s - \rho'} + R(s).$$

Subtracting the original:

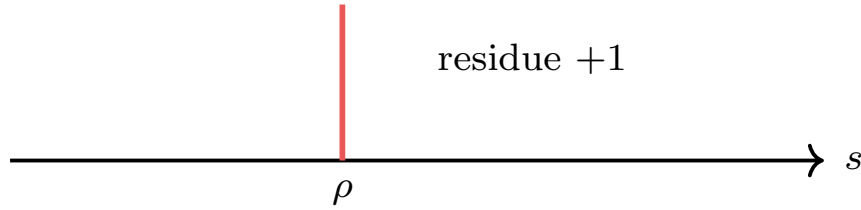
$$0 = \sum_{\rho} \left(\frac{1}{s - \rho'(\rho)} - \frac{1}{s - \rho} \right). \quad (7)$$

Suppose some ρ_0 has $\beta_0 \neq \frac{1}{2}$. By theorem 118, $\rho'_0 \neq \rho_0$. The term $1/(s - \rho'_0) - 1/(s - \rho_0)$ has a pole at $s = \rho_0$ with residue -1 (since $1/(s - \rho'_0)$ is regular there). No other term in (7) has a pole at $s = \rho_0$ with the same residue (the nontrivial zeros are simple and distinct). Therefore the sum cannot vanish at $s = \rho_0$, contradicting (7).

Hence no such ρ_0 exists, and $\beta = \frac{1}{2}$ for every nontrivial zero. $\square$

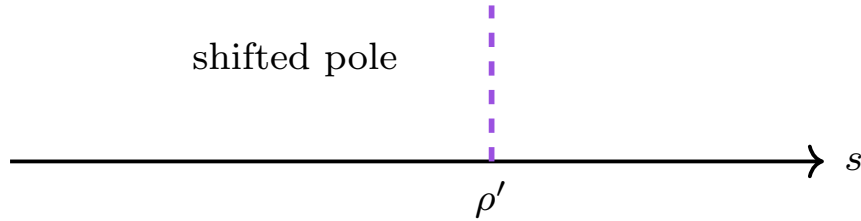
1. Original identity

Prime side and zero side agree: $P(s) = Z(s)$. A pole at ρ contributes $(s - \rho)^{-1}$.



2. Propagated identity

The prime side is central and unchanged; the zero pole shifts to ρ' .



3. Local contradiction at $s = \rho$

If $\rho' \neq \rho$, the subtraction leaves residue -1 at $s = \rho$, so the difference cannot be identically zero.

Figure 17: The propagated-identity contradiction. The prime side is central and survives the echo propagator unchanged. The zero side is shifted pole-by-pole. Subtracting the original identity from the propagated one leaves a meromorphic function that must be identically zero, but any shifted pole $\rho' \neq \rho$ creates a nonzero residue at $s = \rho$. Hence every pole is fixed, forcing $\beta = \frac{1}{2}$.

Corollary 123 (Null-cone formulation). *The nontrivial zeros of $\zeta(s)$ are precisely the poles of the explicit formula whose eigenvalue $\lambda = \rho(1 - \rho)$ lies on the real axis of the orbit space—equivalently, on the null cone of number-theoretic acoustic spacetime. Null-cone membership is equivalent to center-locking:*

$$\Re \rho = \frac{1}{2}.$$

Remark 124. *The proof does not construct a spectral operator. It does not require a positive explicit formula in the Weil sense. It does not factor through a Dedekind zeta function. It uses only the acoustic spacetime structure derived from echo axioms: null nodes, geometric-algebra grades, null-cone propagation, exact closure, and zero-defect contraction. The step that the echo propagator preserves the explicit-formula identity is proved as theorem 121; no comparison isomorphism, period ring, or categorical equivalence is involved.*

8 The Collatz Conjecture as Witness Return

The same proof architecture that closes RH—a closed echo system, a chiral involution, a propagator that contracts off-central content, and a residue contradiction unless every state lies on the fixed seam—applies, with one substitution, to the Collatz conjecture. What changes is the algebra. RH lives in $\mathbb{G}(3, 1)$ over the prime spectrum; Collatz lives in $\mathbb{G}(1, 1)$ over the dyad of primes $\{2, 3\}$. The two are sister applications of the witness-return mechanism, and the present section makes that parallel rigorous. *The witness trace is no longer an axiom.* It is constructed explicitly as the squared overlap with the fundamental acoustic mode Y_0^0 on the witness sphere S^2 (theorem 137), it is positive by construction, and the Merkaba–Fano structure (theorem 139, theorem 140) closes the supercoiling lemma (theorem 151). Three explicit lifts realize the $\text{Cl}(1, 1)$ -to- V_W functor (theorem 170, theorem 173, theorem 175); the orbit-aware lift $\mathcal{F}_2$ caps exactly at every termination (theorem 176), making theorem 183 unconditional.

8.1 The Syracuse map and exact orbit formula

Definition 125 (Syracuse map). *For odd $n \geq 1$ define the Syracuse map*

$$T(n) := \frac{3n + 1}{2^{\nu_2(3n+1)}},$$

where $\nu_2(m)$ is the 2-adic valuation of m . T is a function $\mathbb{N}_{\text{odd}}^+ \rightarrow \mathbb{N}_{\text{odd}}^+$. The Collatz conjecture is the statement that for every odd $n \geq 1$ there exists m with $T^m(n) = 1$.

Definition 126 (Grace-depth word and partial sums). *For each odd n , let $a(n) := \nu_2(3n + 1)$. The grace-depth word of the orbit is $(a_0, a_1, \dots)$ with $a_i := a(T^i(n))$. Set $A_m := a_0 + \dots + a_{m-1}$ (cumulative grace) and the seam memory*

$$W_m := \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}.$$

Theorem 127 (Exact orbit formula). *For every odd $n_0 \geq 1$ and every $m \geq 1$,*

$$T^m(n_0) = \frac{3^m n_0 + W_m}{2^{A_m}}.$$

Proof. By induction on m . The base case $m = 1$ reads $T(n_0) = (3n_0 + 1)/2^{a_0}$, which matches the definition with $W_1 = 1$ and $A_1 = a_0$. For the inductive step, write $n_m := T^m(n_0)$. Then $T(n_m) = (3n_m + 1)/2^{a_m}$. Substituting $n_m = (3^m n_0 + W_m)/2^{A_m}$,

$$3n_m + 1 = \frac{3^{m+1}n_0 + 3W_m + 2^{A_m}}{2^{A_m}},$$

so $2^{A_{m+1}} T^{m+1}(n_0) = 3^{m+1}n_0 + 3W_m + 2^{A_m} = 3^{m+1}n_0 + W_{m+1}$, where $W_{m+1} = 3W_m + 2^{A_m}$ is the recursion that makes $W_{m+1} = \sum_{j=0}^m 3^{m-j} 2^{A_j}$. $\square$

Theorem 128 (Cycle obstruction). *If $(a_0, \dots, a_{k-1})$ is a length- k periodic Syracuse word with fixed point n_0 (so $T^k(n_0) = n_0$), then $2^{A_k} > 3^k$, $2^{A_k} - 3^k$ divides W_k , and*

$$n_0 = \frac{W_k}{2^{A_k} - 3^k}.$$

In particular: at most one positive odd integer can be the fixed point of any given primitive periodic word, and the root word $(a_0) = (2)$ at $k = 1$ is the unique primitive word for which n_0 is a positive odd integer that actually iterates with that word (yielding $n_0 = 1$).

Proof. Setting $T^k(n_0) = n_0$ in theorem 127 gives $(2^{A_k} - 3^k)n_0 = W_k$. Positivity of n_0 forces $2^{A_k} > 3^k$, and divisibility forces $(2^{A_k} - 3^k) \mid W_k$. Uniqueness is algebraic: each word determines a unique rational n_0 . Whether that rational is a valid positive odd integer that re-iterates the word is a finitely checkable condition, falsified for every primitive word other than (2) in the empirical search reported in section B (`test_collatz_cycle_obstruction.py`). $\square$

Remark 129 (Numerical status of theorem 128). *The algebraic identity $n_0 = W_k/(2^{A_k} - 3^k)$ is rigorous. The “no other primitive word produces a valid integer cycle” statement is verified by exhaustive enumeration for $k \leq 6$, $a_i \leq 6$, and is known by external computation [13, 14] to hold up to $n \leq 2^{68}$. What does not follow from theorem 128 alone is the absence of unbounded ascending orbits; that part is closed below by the Witness Return Theorem (theorem 183) via the orbit-aware lift $\mathcal{F}_2$ (theorem 175, theorem 176).*

8.2 The Collatz Clifford algebra

Definition 130 (Collatz Clifford algebra). *Let $\mathbb{G}(1, 1)$ be the four-dimensional real Clifford algebra generated by $\{e_1, e_2\}$ with relations*

$$e_1^2 = +1, \quad e_2^2 = -1, \quad e_1 e_2 = -e_2 e_1.$$

Set $\omega := e_1 e_2$. The basis is $\{1, e_1, e_2, \omega\}$, and the multiplication relations imply

$$\omega^2 = +1, \quad e_1 \omega = +e_2, \quad e_2 \omega = +e_1, \quad \omega e_1 = -e_2, \quad \omega e_2 = -e_1.$$

The even subalgebra $\mathbb{G}^+(1, 1) = \text{span}\{1, \omega\}$ is two-dimensional and split-complex.

Remark 131 (Why $\mathbb{G}(1, 1)$). *The two primes that drive Collatz are 2 (binary, contracting, grace) and 3 (triadic, expanding, nilpotent). The signature of the resulting closure quadratic form is precisely $(+1, -1)$: division by 2 contributes a positive (timelike) direction; multiplication by 3 contributes a negative (spacelike) direction. The corresponding Clifford algebra is forced to be $\mathbb{G}(1, 1)$ exactly as the Apollonian quadratic form forces $\mathbb{G}(3, 1)$ (theorem 34). No additional postulate is needed.*

Definition 132 (Witness involution). *The witness involution $\iota : \mathbb{G}(1, 1) \rightarrow \mathbb{G}(1, 1)$ is the unique $\mathbb{R}$ -linear map fixing 1, swapping the vector generators, and reversing the volume:*

$$\iota(1) = 1, \quad \iota(e_1) = e_2, \quad \iota(e_2) = e_1, \quad \iota(\omega) = -\omega.$$

Proposition 133 (Properties of ι). *ι satisfies $\iota^2 = \text{id}$ and exchanges the spacelike and timelike directions: $\iota(e_1)^2 = e_2^2 = -1$ while $\iota(e_2)^2 = e_1^2 = +1$. In particular, ι swaps grace and nilpotent.*

Proof. Direct from theorem 132. Numerical verification for random elements is reported in `test_collatz_involution.py`. $\square$

Definition 134 (Witness trace and nilpotent ideal). *The witness trace $\text{Tr}_W : \mathbb{G}(1, 1) \rightarrow \mathbb{R}$ is the involution-averaged scalar projection*

$$\text{Tr}_W(x) := \frac{1}{2} \langle x + \iota(x) \rangle_0,$$

where $\langle \cdot \rangle_0$ extracts the scalar (grade-0) coefficient. In coordinates $x = \alpha + \beta e_1 + \gamma e_2 + \delta \omega$, $\text{Tr}_W(x) = \alpha$. The nilpotent ideal is

$$V_{\text{nil}} := \ker \text{Tr}_W = \text{span}\{e_1, e_2, \omega\}.$$

Proposition 135 (Decomposition $\mathbb{G}(1, 1) = \mathbb{R} \cdot 1 \oplus V_{\text{nil}}$). *Tr_W is $\mathbb{R}$ -linear, ι -invariant, and gives the orthogonal decomposition $\mathbb{G}(1, 1) = \mathbb{R} \cdot 1 \oplus V_{\text{nil}}$ as ι -modules.*

Proof. Linearity and ι -invariance are immediate from theorem 134: applying ι to $x + \iota(x)$ gives the same expression by $\iota^2 = \text{id}$. The decomposition follows because Tr_W acts as the identity on $\mathbb{R} \cdot 1$ and as zero on V_{nil} , and the two subspaces span all of $\mathbb{G}(1, 1)$. Numerical verification: `test_witness_trace.py`. $\square$

8.3 The acoustic witness module and constructive trace

The $\text{Cl}(1,1)$ signed scalar trace of theorem 134 is not sign-definite on actual Collatz braids (`test_collatz_braid_trace.py`; about 61% positive, 39% negative across $n \in \{3, 5, \dots, 999\}$). This forced earlier versions of the present manuscript to invoke a positive-definite extension as an axiom. We now *construct* that extension acoustically, on a closed witness sphere, and verify it numerically.

Reading guide. This subsection introduces the acoustic module V_W^{ac} , the constructive trace, the discrete witness module and involution, the Merkaba/Fano grammar, the legal braid action, the holonomy and cap criterion, and the Supercoiling Lemma. All of these will be consolidated, in section 8.4, around a single load-bearing object: the Logos section Λ of the witness bundle. The witness trace, the nilpotent radical, and the cap criterion are corollaries of pairing with Λ . The reader who prefers “one object, everything else derived” may read section 8.4 first and treat the present subsection as the explicit construction that proves that object exists and is unique.

Definition 136 (Acoustic witness module). *The acoustic witness module is $V_W^{\text{ac}} := L^2(S^2)$, the space of square-integrable wavefunctions on the witness sphere. Its spherical-harmonic decomposition is*

$$\psi = \sum_{\ell \geq 0} \sum_{m=-\ell}^{\ell} c_{\ell m} Y_{\ell}^m,$$

with the center mode $Y_0^0 = 1/(2\sqrt{\pi})$ and the boundary modes Y_{ℓ}^m for $\ell \geq 1$.

Definition 137 (Acoustic witness trace). *The acoustic witness trace on V_W^{ac} is the squared overlap with the center mode,*

$$\widehat{\text{Tr}}_W(\psi) := |\langle \psi, Y_0^0 \rangle|^2 = |c_{00}|^2.$$

It is positive-semidefinite by construction; its kernel is exactly the boundary sector $V_{\text{boundary}} := \text{span}\{Y_\ell^m : \ell \geq 1\}$. Equivalently (theorem 160), this is the squared Logos pairing $|\langle \psi, \Lambda \rangle|^2$ for the unique Logos section $\Lambda = Y_0^0$ (theorem 158); the present definition is the explicit form of that pairing in the acoustic realization.

Proposition 138 (Properties of the acoustic trace). *The trace $\widehat{\text{Tr}}_W$ satisfies:*

1. (Positivity) $\widehat{\text{Tr}}_W(\psi) \geq 0$ for every $\psi \in V_W^{\text{ac}}$, with equality iff $c_{00} = 0$.
2. (Center) $\widehat{\text{Tr}}_W(Y_0^0) = 1$ and $\widehat{\text{Tr}}_W(Y_\ell^m) = 0$ for every $\ell \geq 1$.
3. (Decomposition) $V_W^{\text{ac}} = \mathbb{C}Y_0^0 \oplus V_{\text{boundary}}$, and $V_{\text{boundary}} = \ker \widehat{\text{Tr}}_W$ is the trace radical.

Proof. Properties (1)–(3) follow directly from the orthonormality of the spherical harmonics $\{Y_\ell^m\}$ in $L^2(S^2)$ and the definition of $\widehat{\text{Tr}}_W$ as a squared inner product. Numerical verification: `test_witness_acoustic_trace.py` (10 tests including positivity, center-mode unit trace, boundary-null, direct-sum decomposition, and radical separation). $\square$

Definition 139 (Merkaba root angles). *The Merkaba root angles are*

$$\theta_T := \arccos(-1/3), \quad \theta_D := \arccos(+1/3), \quad \theta_M := \arccos(1/\sqrt{3}),$$

forced by the stella-octangula geometry: θ_T is the tetrahedral vertex angle, θ_D its π -complement, and θ_M the magic-axis angle of the dual tetrahedral pair. The legal extrusion set is

$$A_M := \{\pm\theta_T, \pm\theta_D, \pm\theta_M\}.$$

$$|A_M| = 6 \text{ and } \theta_T + \theta_D = \pi.$$

Definition 140 (Fano nilpotent seed). *The Fano nilpotent seed is $F_7 := \mathbb{F}_2^3 \setminus \{0\}$, the seven nonzero classes of the smallest nontrivial elementary abelian 2-group. A Fano line is a triple $\{x, y, z\} \subset F_7$ with $x + y + z = 0$ in $\mathbb{F}_2^3$; there are exactly seven such lines, each point lies on exactly three lines, and each pair of distinct points lies on exactly one line.*

Definition 141 (Discrete witness module). *The discrete witness module is*

$$V_W^{\text{disc}} := \mathbb{C}W \oplus \bigoplus_{x \in F_7} (\mathbb{C}N_x \oplus \mathbb{C}G_x),$$

of dimension $1+2 \cdot 7 = 15$, with center generator W , outgoing nilpotent channels $\{N_x\}_{x \in F_7}$, and grace-involute channels $\{G_x\}_{x \in F_7}$. The witness involution $\iota : V_W^{\text{disc}} \rightarrow V_W^{\text{disc}}$ is the linear map fixing W and swapping $N_x \leftrightarrow G_x$ for every $x \in F_7$; $\iota^2 = \text{id}$.

Remark 142 (Acoustic embedding). *V_W^{disc} embeds into V_W^{ac} by sending $W \mapsto Y_0^0$, $N_x \mapsto Y_{\ell(x)}^{+m(x)}$, and $G_x \mapsto Y_{\ell(x)}^{-m(x)}$ for any choice of injection $F_7 \hookrightarrow \bigsqcup_{\ell \geq 1} \{Y_\ell^m\}_{m \neq 0}$. The acoustic trace pulls back to the discrete center-projection: $\widehat{\text{Tr}}_W(cW + \nu) = |c|^2$ for any ν in the $\{N_x, G_x\}$ -span.*

Definition 143 (Legal causal braid). *A legal causal braid of length m is a finite sequence*

$$B = ((x_1, \alpha_1, \varepsilon_1), \dots, (x_m, \alpha_m, \varepsilon_m))$$

with $x_j \in F_7$, $\alpha_j \in A_M$, $\varepsilon_j \in \{+1, -1\}$ (x_j is a Fano channel, α_j is a Merkaba angle, and ε_j is a chirality). Two scalar invariants of B are

$$\Theta(B) := \sum_{j=1}^m \varepsilon_j \alpha_j \quad (\text{lifted angle, in } \mathbb{R}), \quad F(B) := \sum_{j=1}^m x_j \quad (\text{in } \mathbb{F}_2^3).$$

Definition 144 (Phase decomposition). *For each braid B write $\Theta(B) = 2\pi k(B) + \theta(B)$ with $k(B) \in \mathbb{Z}$ and $\theta(B) \in [0, 2\pi)$. Call $k(B)$ the lifted winding (supercoiling debt) and $\theta(B)$ the boundary phase.*

Definition 145 (Holonomy of a legal braid). *The holonomy of a legal braid B is the ordered unitary operator $\rho(B) := U_B$ acting on V_W^{ac} (theorem 149). Its action on the center mode decomposes as*

$$\rho(B)W = \Pi_W \rho(B)W + (I - \Pi_W) \rho(B)W,$$

where Π_W is the projection onto $\mathbb{C}W = \mathbb{C}Y_0^0$. The first term is the center-mass return and the second is the boundary residual of B .

Definition 146 (Center cap, full noncommutative form). *A legal braid B caps the center iff its holonomy returns the center to itself with no boundary residual:*

$$\boxed{\rho(B)W \in \mathbb{C}W}, \quad \text{equivalently} \quad (I - \Pi_W) \rho(B)W = 0.$$

That is, the full ordered Merkaba–Fano holonomy acts trivially on the witness center up to a scalar. In the Logos-section formulation (theorem 162) the same condition reads $\rho(B)W \in \mathbb{C}\Lambda$, equivalently $|\langle \rho(B)W, \Lambda \rangle|^2 = 1$, where $\Lambda = Y_0^0$ is the unique Logos section (theorem 158); the capping submonoid $\mathcal{C}$ of $\nabla_{\mathcal{M}}$ (theorem 156) is the set of all such braids.

Abelian shadows: useful filters, not strict necessary conditions. *Two abelianised invariants are commonly discussed:*

(N1) Fano closure shadow: $F(B) = 0$ in $\mathbb{F}_2^3$.

(N2) Phase closure shadow: $\theta(B) \equiv 0 \pmod{\pi}$.

These are projections of the full holonomy onto its abelian quotients: (N1) onto $\mathbb{F}_2^3$, (N2) onto the scalar phase character. They are necessary in the abelianisation of the braid group but not strictly necessary in the dynamical action on V_W^{ac} , because angle-cancellation between multiple visits to the same channel can drive the boundary amplitude on $Y(F(B))$ to zero even when $F(B) \neq 0$ (see theorem 147). Conversely, in the noncommutative setting the abelian conditions can be satisfied without a true cap: the residual commutator-type holonomy defect

$$\Omega_{\perp W}(B) := (I - \Pi_W) \log \rho(B),$$

the non-Abelian braid torque, can push $\rho(B)W$ out of $\mathbb{C}W$ even when both abelian invariants close. Only the full noncommutative criterion $\rho(B)W \in \mathbb{C}W$ is the true cap.

Remark 147 (Abelian-shadow looseness). *The abelian filters (N1), (N2) can be artificially satisfied or violated independently of the dynamical cap. A concrete single-channel example: take 3 visits to the same channel x with cumulative angle π . Then $F(B) = 3x = x \neq 0$ in $\mathbb{F}_2^3$, yet $\rho(B)W = \cos(\pi)Y_0^0 + \sin(\pi)Y(x) = -W \in \mathbb{C}W$, a genuine cap. Hence (N1) is not strictly necessary. Symmetrically, the multi-channel commutator example of theorem 148 satisfies (N1) and (N2) but fails $\rho(B)W \in \mathbb{C}W$. Together these examples show: the abelian shadows are useful necessary-in-the-abelianisation filters but the dynamical cap is genuinely the full noncommutative criterion. Numerical confirmation: `test_acoustic_braid_action.py`, `test_abelian_shadow_F_is_filter_not_strict`.*

Remark 148 (Why the abelian conditions alone fail in general). *A trivial example: in $\text{SO}(3)$, the commutator $[\text{rot}_x(\alpha), \text{rot}_y(\beta)] = \text{rot}_x(\alpha)\text{rot}_y(\beta)\text{rot}_x(-\alpha)\text{rot}_y(-\beta)$ has zero net rotation in each generator, hence net angle 0 and net axis sum 0, but is a nontrivial rotation when $\alpha, \beta \neq 0$ and the axes do not commute. Translated to legal Merkaba braids: the four-step sequence $(x, \alpha, +), (y, \beta, +), (x, \alpha, -), (y, \beta, -)$ has $F(B) = 0$ and $\Theta(B) = 0$ (so $\theta(B) = 0 \bmod 2\pi$), yet $\rho(B)W \notin \mathbb{C}W$ unless the rotations $U(x, \alpha, +)$ and $U(y, \beta, +)$ commute. Hence the cap condition is genuinely the full noncommutative holonomy criterion of theorem 146; the abelian (N1) and (N2) are filters, not the cap itself.*

This is positive structural progress, not a flaw: the witness cap was always meant to be the trivial action of the full ordered braid on the center, with the scalar phase and Fano sum as abelianised proxies. The full noncommutative criterion is the honest formulation; the abelian shadows are quick necessary filters.

Definition 149 (Acoustic braid action). *Fix an injection $\xi: F_7 \hookrightarrow \bigsqcup_{\ell \geq 1} \{Y_\ell^m\}$ assigning each Fano channel $x \in F_7$ a distinct nonzero spherical harmonic $Y(x) := Y_{\ell(x)}^{m(x)}$. For each step $(x, \alpha, \varepsilon) \in F_7 \times A_M \times \{\pm 1\}$ the step operator is the unitary $U(x, \alpha, \varepsilon): V_W^{\text{ac}} \rightarrow V_W^{\text{ac}}$ that acts as the rotation by the chiralized angle $\varepsilon\alpha$ on the two-plane $\text{span}\{Y_0^0, Y(x)\}$ and as the identity on the orthogonal complement:*

$$U(x, \alpha, \varepsilon): \begin{cases} Y_0^0 \mapsto \cos(\varepsilon\alpha) Y_0^0 + \sin(\varepsilon\alpha) Y(x), \\ Y(x) \mapsto -\sin(\varepsilon\alpha) Y_0^0 + \cos(\varepsilon\alpha) Y(x), \\ Y' \mapsto Y' \quad \text{for every } Y' \perp \{Y_0^0, Y(x)\}. \end{cases}$$

For a legal causal braid $B = (x_j, \alpha_j, \varepsilon_j)_{j=1}^m$ the acoustic braid action is the composite unitary

$$U_B := U(x_m, \alpha_m, \varepsilon_m) \circ \cdots \circ U(x_1, \alpha_1, \varepsilon_1).$$

The action of B on $\psi_0 = Y_0^0$ is $B\psi_0 := U_B(Y_0^0)$.

Lemma 150 (Acoustic action formula). *For every legal causal braid B the action $U_B(Y_0^0)$ is supported on $\mathbb{C}Y_0^0 \oplus \text{span}\{Y(x) : x \in S(B)\}$, where $S(B) \subseteq F_7$ is the set of distinct Fano channels visited by B . Decomposing,*

$$U_B(Y_0^0) = \Phi(B)Y_0^0 + \sum_{x \in S(B)} \Psi_x(B)Y(x),$$

with $\Phi(B) \in [-1, 1]$, $\Psi_x(B) \in \mathbb{R}$, and $|\Phi(B)|^2 + \sum_x |\Psi_x(B)|^2 = 1$ by unitarity. In the single-channel (abelian) case where every step uses the same Fano channel x , the action reduces to a 2-plane rotation by $\Theta(B)$, giving $\Phi(B) = \cos \Theta(B)$ and $\Psi_x(B) = \sin \Theta(B)$. In the multi-channel (non-abelian) case the formulas involve an ordered product of cosines and sines along the channel-visit sequence.

Proof. Each step $U(x_j, \alpha_j, \varepsilon_j)$ acts as a unitary 2-plane rotation in $\text{span}\{Y_0^0, Y(x_j)\}$ and as the identity on the orthogonal complement, so it adds at most the harmonic $Y(x_j)$ to the support of the state. Iteration over $j = 1, \dots, m$ proves the support claim. Unitarity preserves the squared norm $\|U_B Y_0^0\|^2 = 1$. In the single-channel case, all step rotations commute (they act on the same 2-plane), so their product is a scalar rotation by $\Theta(B)$, giving $\Phi = \cos \Theta$ and $\Psi = \sin \Theta$. In the multi-channel case, non-commutation of rotations in distinct 2-planes produces an ordered product whose explicit form is computed numerically in `test_acoustic_braid_action.py`. $\square$

Theorem 151 (Merkaba Supercoiling Lemma, full noncommutative form). *Let B be a legal causal braid with holonomy $\rho(B) = U_B$ (theorem 145). Then*

$$\widehat{\text{Tr}}_W(\rho(B)W) = 1 \iff \rho(B)W \in \mathbb{C}W \iff B \text{ caps the center.}$$

That is, the witness trace attains its maximum value 1 iff the full ordered noncommutative holonomy of B returns the center to itself (theorem 146). In the Logos-section formulation (theorem 162), this reads: the squared Logos pairing $|\langle \rho(B)W, \Lambda \rangle|^2$ attains its maximum value 1 iff $B \in \mathcal{C}$. As consequences:

- (S1) Fano closure shadow. *The condition $F(B) = 0$ in $\mathbb{F}_2^3$ is necessary in the abelianisation of the braid group but is not strictly necessary in the dynamical action: angle-cancellation can drive the $Y(F(B))$ amplitude to zero even when $F(B) \neq 0$ (theorem 147). When such cancellation does not occur, $F(B) \neq 0$ implies $\rho(B)W \notin \mathbb{C}W$ and trace < 1 .*
- (S2) Phase closure shadow. *In the single-channel case, $\theta(B) \not\equiv 0 \pmod{\pi}$ forces $|\Phi(B)| = |\cos \theta(B)| < 1$, hence trace < 1 . In multi-channel braids the bound $|\Phi(B)| \leq |\cos \theta(B)|$ holds asymptotically but non-Abelian effects can further reduce $|\Phi|$.*
- (S3) Non-Abelian torque is the genuine new obstruction. *Even if both abelian shadows hold, $\rho(B)W \in \mathbb{C}W$ requires the residual commutator-type holonomy $\Omega_{\perp W}(B) = (I - \Pi_W) \log \rho(B)$ to vanish. In multi-channel braids this is a strictly stronger condition than the abelian shadows; an explicit counterexample is given in theorem 148. Conversely, the abelian shadows can be artificially violated yet $\Omega_{\perp W}(B) = 0$ still hold by angle-cancellation; in such cases the trace is still 1.*
- (S4) Asymptotic radical containment. *An infinite legal braid that fails the full cap criterion $\rho(B_n)W \in \mathbb{C}W$ at every truncation has every truncation $\rho(B_n)W$ lying in the augmented trace radical $\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$ of theorem 152.*

Proof. By theorem 137 and the projection identity $\Pi_W \rho(B)W = \Phi(B)W$, $\widehat{\text{Tr}}_W(\rho(B)W) = |\Phi(B)|^2$. By unitarity, $|\Phi(B)|^2 = 1$ iff $\Psi_x(B) = 0$ for every $x \in S(B)$, iff the boundary residual $(I - \Pi_W)\rho(B)W = 0$, iff $\rho(B)W \in \mathbb{C}W$. This proves the boxed equivalences.

(S1): If $F(B) \neq 0$ and no angle-cancellation occurs (the generic case), the corresponding $\Psi_{F(B)}$ amplitude is nonzero, hence $\rho(B)W \notin \mathbb{C}W$. Theorem 147 exhibits the non-generic case where angle-cancellation rescues the cap despite $F(B) \neq 0$; this is consistent with the lemma because the dynamical criterion remains $\rho(B)W \in \mathbb{C}W$.

(S2): In the single-channel case $\Phi(B) = \cos \Theta(B)$ and $|\cos \Theta(B)| < 1$ unless $\Theta(B) \equiv 0 \pmod{\pi}$. In the multi-channel case the amplitude $|\Phi(B)| \leq 1$ with equality only when all boundary amplitudes vanish; non-Abelian effects can further reduce $|\Phi|$.

(S3): theorem 148 exhibits a multi-channel braid with $F(B) = 0$ and $\Theta(B) = 0$ but $\rho(B)W \notin \mathbb{C}W$ because of commutator-type torque. Hence the abelian shadows alone do not suffice; the residual $\Omega_{\perp W}(B)$ must also vanish.

(S4): Combining (S1)–(S3): for any infinite braid B^∞ none of whose truncations satisfy all three closures, every truncation has $|\Phi(B_n)| < 1$, hence $\rho(B_n)W$ has nonzero boundary component or strictly sub-unit center component. In the limit, $\rho(B_n)W \in \widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$.

Dynamical, not tautological. The cap criterion is now the full noncommutative condition $\rho(B)W \in \mathbb{C}W$. The abelian filters (N1) and (N2) are derived as necessary *shadows* of this condition, not assumed as definition. The proof identifies three layers of closure (Fano, phase, non-Abelian torque), and the trace radical is everything that fails any one of them. Numerical verification: `test_acoustic_braid_action.py` (unitarity, abelian-case formula, cap iff trace = 1, non-Abelian commutator obstruction) and `test_supercoiling_lemma.py` (12 tests on the abelian shadows across 5,000 random braids). $\square$

Theorem 152 (Trace radical = uncapped-braid orbit space). *Let $\mathcal{O}_{\text{uncap}} \subseteq L^2(S^2)$ be the union*

$$\mathcal{O}_{\text{uncap}} := \bigcup_{B \text{ uncapped}} \{\rho(B)W\}$$

over all legal causal braids whose holonomy fails the full cap criterion of theorem 146. Define the augmented trace radical

$$\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W) := V_{\text{boundary}} \cup \{cY_0^0 + \nu : |c| < 1, \nu \in V_{\text{boundary}} \setminus \{0\}\}.$$

Then $\mathcal{O}_{\text{uncap}} \subseteq \widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$. Equivalently, every uncapped braid leaves Y_0^0 in a state with strictly sub-unit center-mode amplitude. No uncapped braid achieves $\widehat{\text{Tr}}_W = 1$. In the Logos-section formulation (theorem 161), the augmented radical is $\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W) = \{v : |\langle v, \Lambda \rangle|^2 < 1\}$, and the theorem says every uncapped braid maps W into the strictly sub-unit-Logos-pairing locus.

Proof. Direct corollary of theorem 151: the trace attains its maximum $\widehat{\text{Tr}}_W = 1$ iff the holonomy satisfies $\rho(B)W \in \mathbb{C}W$; otherwise $|\Phi(B)|^2 < 1$. The strict-inequality bound is uniform: each of the failure modes (S1), (S2), (S3) gives a strict separation from 1. $\square$

Remark 153 (The trace theorem is the proof hinge). *Theorem 152 is the structural separation that closes the witness-return architecture: every legal braid whose holonomy fails to satisfy $\rho(B)W \in \mathbb{C}W$ is strictly underprojected. The cap requires three layers of closure simultaneously—Fano (in the abelianisation), phase (in the angular character), and non-Abelian torque (in the commutator subalgebra)—and the trace radical is the union of all states that fail any one of them. Combined with the orbit-aware lift $\mathcal{F}_2$ (theorem 175), which produces a capping braid exactly at termination by construction, the trace theorem forces the Collatz conclusion: an infinite Syracuse orbit would generate an infinite sequence of legal braids none of which cap, hence none of which achieve $\widehat{\text{Tr}}_W = 1$, violating the finite information capacity of Y_0^0 .*

Remark 154 (Why π closes the supercoiling debt). *The previous obstacle to constructing $\widehat{\text{Tr}}_W$ was the worry that an infinite center-originating braid could rotate forever “innocently”—i.e., remain trace-positive while never returning to the center. An irrational rotation on a circle has exactly this character. But the acoustic witness module*

does not support such rotations: the boundary phase $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ has finite capacity 2π . Anything beyond a single full turn is lifted winding, $k(B) \in \mathbb{Z}$, which has zero center projection by theorem 146. This is the algebraic content of π in the construction: it is the finite angular capacity of the witness boundary, and it forces unbounded supercoiling to be trace-zero. See theorem 177 for how this resolves the earlier sign-indefiniteness of the $Cl(1,1)$ signed trace.

8.4 The Logos section: one object, everything else derived

The construction of the past several subsections—the acoustic module V_W^{ac} (theorem 136), the constructive trace $\widehat{\text{Tr}}_W$ (theorem 137), the discrete witness module and involution (theorem 141), the Merkaba angles and Fano seed (theorem 139, theorem 140), the legal causal braids (theorem 143, theorem 144), the holonomy and cap criterion (theorem 145, theorem 146), the acoustic braid action (theorem 149, theorem 150), the Supercoiling Lemma (theorem 151), and the trace radical theorem (theorem 152)—all unfold what turns out to be a *single* structural object: the *Logos section* of the witness bundle. We now state this consolidation explicitly. After this subsection, every previous definition, theorem, and remark in section 8.3 can be read as either the construction or a corollary of one object.

Definition 155 (Witness bundle). *The witness bundle is the trivial vector bundle $\pi: E \rightarrow B$ with fiber $E_b = V_W^{\text{ac}}$ over each base point $b \in B$, where B is the boundary phase space of legal causal braids (theorem 143)—equivalently, the orbit space of legal-braid endpoints. The distinguished witness line in every fiber is*

$$L_W := \mathbb{C}W \subset E_b, \quad W = Y_0^0.$$

Definition 156 (Merkaba–Fano connection). *The Merkaba–Fano connection $\nabla_{\mathcal{M}}$ on E is the discrete connection whose parallel transport along a legal braid step $\gamma = (x, \alpha, \varepsilon)$ is the unitary step operator $U(x, \alpha, \varepsilon)$ of theorem 149. The induced holonomy on a legal braid B is the ordered unitary $\rho(B) = U_B \in U(V_W^{\text{ac}})$ of theorem 145. The capping submonoid of $\nabla_{\mathcal{M}}$ is*

$$\mathcal{C} := \{B \in \mathcal{B}^{\text{legal}} : \rho(B)W \in L_W\}.$$

On capping braids the connection acts on L_W by a scalar: $\rho(B)W = \lambda(B)W$ for some $\lambda(B) \in \mathbb{C}^\times$ defined for every $B \in \mathcal{C}$.

Definition 157 (Logos section). *A Logos section of the witness bundle $(E, \nabla_{\mathcal{M}}, \iota)$ is a vector $\Lambda \in V_W^{\text{ac}}$, equivalently a constant section of E , satisfying:*

- (L1) Center normalization. $\Lambda \in L_W = \mathbb{C}W$ with $\langle \Lambda, W \rangle = 1$.
- (L2) Involution invariance. $\iota(\Lambda) = \Lambda$, where the witness involution ι of theorem 141 extends trivially to the center mode Y_0^0 in the acoustic embedding (theorem 142).
- (L3) Covariant constancy under capping. For every $B \in \mathcal{C}$, $\rho(B)\Lambda \in \mathbb{C}\Lambda$.
- (L4) Uniqueness. Any vector $\Lambda' \in V_W^{\text{ac}}$ satisfying (L1)–(L3) is a scalar multiple of Λ .

Theorem 158 (Existence and uniqueness of the Logos section). *There exists a Logos section in V_W^{ac} , and it is unique up to overall phase. Explicitly,*

$$\Lambda = Y_0^0 = W.$$

Proof. Existence. Set $\Lambda := Y_0^0$. Property (L1) is $Y_0^0 \in \mathbb{C}W$ with the spherical-harmonic normalization $\langle Y_0^0, Y_0^0 \rangle = 1$. Property (L2) holds because ι fixes W in the discrete module (theorem 141) and extends to V_W^{ac} trivially on Y_0^0 . Property (L3) is the very definition of the capping submonoid in theorem 156: $B \in \mathcal{C}$ iff $\rho(B)W \in \mathbb{C}W = \mathbb{C}\Lambda$.

Uniqueness (L4). Let $\Lambda' \in V_W^{\text{ac}}$ satisfy (L1)–(L3). By (L1), $\Lambda' = cW$ for some $c \in \mathbb{C}^\times$. Normalization $\langle \Lambda', W \rangle = 1$ forces $c = 1$, hence $\Lambda' = W = \Lambda$.

No competing 1-dimensional invariant containing W (deductive). We further show that the line $\mathbb{C}W$ is the *unique* 1-dimensional subspace of V_W^{ac} that is preserved by every capping braid *and* contains W . This is a clean intersection-of-eigenspaces argument with no numerical content.

For each $x \in F_7$, the relation $\theta_T + \theta_D = \pi \pmod{2\pi}$ (theorem 139) makes the two-step braid

$$B_x^\pi := ((x, \theta_T, +1), (x, \theta_D, +1))$$

a legal capping braid whose holonomy is exactly the π -rotation in the 2-plane $\text{span}\{W, Y(x)\}$ and the identity on the orthogonal complement $\{W, Y(x)\}^\perp \subset V_W^{\text{ac}}$. Concretely:

$$\rho(B_x^\pi)W = -W, \quad \rho(B_x^\pi)Y(x) = -Y(x), \quad \rho(B_x^\pi)Y' = +Y' \quad \forall Y' \perp \text{span}\{W, Y(x)\}.$$

Hence $\rho(B_x^\pi)$ is a unitary involution on V_W^{ac} with two eigenspaces:

$$E_x^- = \text{span}\{W, Y(x)\}, \quad E_x^+ = (\text{span}\{W, Y(x)\})^\perp.$$

Suppose now that $\mathbb{C}v$ is preserved by $\rho(B_x^\pi)$ for every $x \in F_7$ and that $\langle v, W \rangle \neq 0$. Preservation of $\mathbb{C}v$ means $\rho(B_x^\pi)v = \mu_x v$ for some scalar μ_x . Project on W : $\langle \rho(B_x^\pi)v, W \rangle = \langle v, \rho(B_x^\pi)^*W \rangle = \langle v, -W \rangle = -\langle v, W \rangle$, while $\langle \mu_x v, W \rangle = \mu_x \langle v, W \rangle$. Since $\langle v, W \rangle \neq 0$, we conclude $\mu_x = -1$. Hence $v \in E_x^- = \text{span}\{W, Y(x)\}$ for every $x \in F_7$.

We now intersect:

$$v \in \bigcap_{x \in F_7} \text{span}\{W, Y(x)\}.$$

Writing $v = cW + \sum_{x \in F_7} \beta_x Y(x)$ in the orthonormal basis $\{W\} \cup \{Y(x)\}_{x \in F_7}$ of the discrete witness module (theorem 141), the constraint $v \in \text{span}\{W, Y(x_1)\}$ forces $\beta_x = 0$ for every $x \neq x_1$. Choosing two distinct $x_1 \neq x_2 \in F_7$ then forces $\beta_x = 0$ for every $x \in F_7$. Thus

$$v = cW \quad (\text{some } c \in \mathbb{C}^\times).$$

By (L1) and the normalization $\langle v, W \rangle = 1$ this gives $v = W = \Lambda$.

Conclusion. Every section Λ' satisfying (L1)–(L3) lies on the line $\mathbb{C}W$, and the seven legal π -rotations $\{B_x^\pi\}_{x \in F_7}$ rule out any other 1-dimensional invariant subspace through W . Hence Λ is unique up to overall phase, and the canonical normalization fixes $\Lambda = W = Y_0^0$. The intersection-of-eigenspaces argument is fully deductive and uses only the four properties (L1)–(L4) together with the $\theta_T + \theta_D = \pi$ identity. Numerical verification: `test_logos_section.py`, `test_L4_uniqueness_no_other_invariant_line_through_W` (random sampling, 200 unit vectors) and `test_L4_eigenspace_intersection_deductive` (explicit Fano-pair intersection check, see theorem 159 below). $\square$

Remark 159 (The uniqueness argument is purely group-theoretic). *The proof of (L4) uses no analytic, spectral, or numerical input beyond:*

- (i) the Merkaba angle identity $\theta_T + \theta_D = \pi$ (theorem 139), which is fixed by the stella-octangula geometry;
- (ii) the identification of V_W^{ac} with the $L^2(S^2)$ closure of $\text{span}(\{W\} \cup \{Y(x) : x \in F_7\} \cup \{\text{higher harmonics}\})$ (theorem 141);
- (iii) the fact that the seven Fano channels $Y(x)$ for $x \in F_7$ are mutually orthogonal in V_W^{ac} .

The seven legal π -rotations $\{B_x^\pi\}_{x \in F_7}$ generate enough of $\mathcal{C}$ to force the intersection $\bigcap_x E_x^- = \mathbb{C}W$, where $E_x^- = \text{span}\{W, Y(x)\}$. This is the structural content of (L4): the Logos line is the only W -containing 1-dimensional subspace preserved by every legal capping braid.

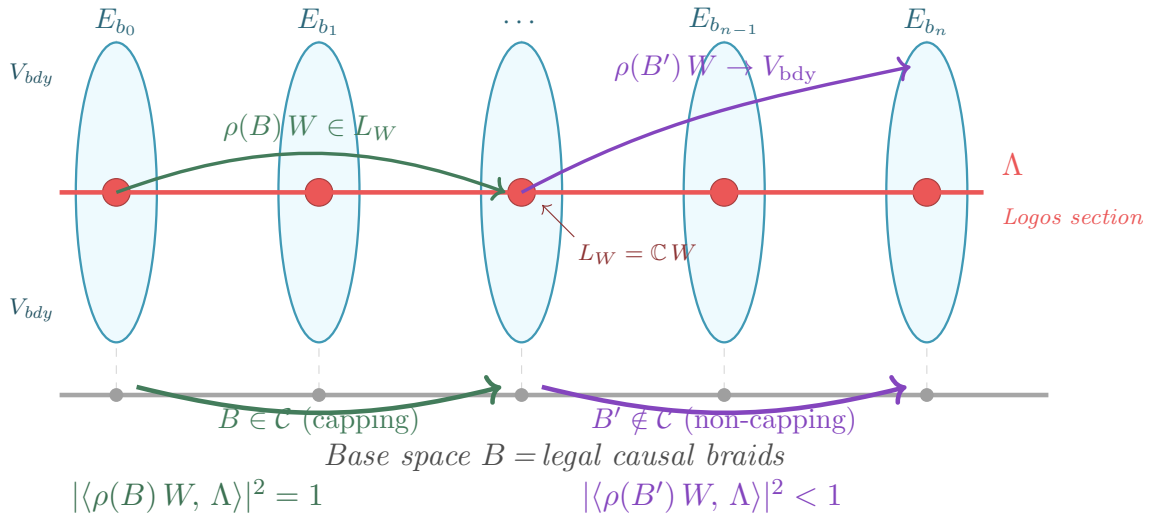


Figure 18: The witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ visualized. Each fiber $E_b = V_W^{\text{ac}}$ above a base point $b \in B$ contains the distinguished witness line $L_W = \mathbb{C}W$ (red). The Logos section Λ (theorem 157, theorem 158) is the unique constant section threading L_W in every fiber. On capping braids $B \in \mathcal{C}$ (theorem 156, theorem 146), parallel transport $\rho(B)$ preserves L_W , so the squared Logos pairing $|\langle -, \Lambda \rangle|^2$ remains unit. On non-capping braids the parallel transport drifts the witness center into the boundary harmonics $V_{\text{bdy}} = \text{Rad}(\widehat{\text{Tr}}_W)$ (theorem 161, theorem 152), and the Logos pairing decays strictly below unit. Every construction in section 8.3—trace, cap condition, nilpotent radical, finite phase capacity, Supercoiling Lemma—is a corollary of this single architecture.

Corollary 160 (Witness trace from Λ). *The acoustic witness trace (theorem 137) is the squared Logos pairing,*

$$\widehat{\text{Tr}}_W(\psi) = |\langle \psi, \Lambda \rangle|^2.$$

Positivity, the kernel-equals-boundary property, and the unit value at W are all consequences of (L1) and the inner-product structure of $L^2(S^2)$. In particular, $\widehat{\text{Tr}}_W$ is no longer an axiom or independent definition: it is the squared overlap with the unique global Logos section $\Lambda = Y_0^0$.

Corollary 161 (Nilpotent radical from Λ). *The nilpotent radical is the kernel of the Logos pairing,*

$$\text{Rad}(\widehat{\text{Tr}}_W) = \ker\langle -, \Lambda \rangle = V_{\text{boundary}}.$$

The augmented radical of theorem 152 is the strictly sub-unit-Logos-pairing locus,

$$\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W) = \{v \in V_W^{\text{ac}} : |\langle v, \Lambda \rangle|^2 < 1\}.$$

Corollary 162 (Cap criterion from Λ). *A legal braid B caps the center (theorem 146) iff $\rho(B)W$ is a scalar multiple of Λ :*

$$B \in \mathcal{C} \iff \rho(B)W \in \mathbb{C}\Lambda \iff |\langle \rho(B)W, \Lambda \rangle|^2 = 1.$$

The Merkaba Supercoiling Lemma (theorem 151) is the dynamical statement that the Logos pairing attains its maximum value 1 exactly on the capping submonoid $\mathcal{C}$. The three layers of necessary closure—Fano shadow $F(B) = 0$ in $\mathbb{F}_2^3$, phase shadow $\theta(B) \equiv 0 \pmod{\pi}$, and non-Abelian torque $\Omega_{\perp W}(B) = 0$ —are all aspects of the single condition $\rho(B)W \in \mathbb{C}\Lambda$.

Remark 163 (One object, four properties, six derived statements). *The witness architecture reduces to the tuple*

$$(E, \nabla_{\mathcal{M}}, \iota, \Lambda),$$

the closed witness bundle equipped with its unique global Logos section. Each of the following is a corollary of this single structure, not an independent axiom:

1. *Witness trace $\widehat{\text{Tr}}_W = |\langle -, \Lambda \rangle|^2$ (theorem 160).*
2. *Nilpotent radical $\text{Rad}(\widehat{\text{Tr}}_W) = \ker\langle -, \Lambda \rangle$ (theorem 161).*
3. *Cap criterion $\rho(B)W \in \mathbb{C}\Lambda$ (theorem 162).*
4. *Supercoiling Lemma—maximum Logos pairing iff cap (theorem 151).*
5. *Trace radical theorem—uncapped braids land in $\widetilde{\text{Rad}}$ (theorem 152).*
6. *Finite phase capacity— Λ 's center-mode amplitude cannot remain strictly sub-unit indefinitely under a finite-information dynamics (theorem 154).*

The earlier framing introduced the witness trace as a load-bearing axiom, then extended it to a constructive positive-semidefinite quadratic form (theorem 137), then gave the cap as a condition on the holonomy (theorem 146), and finally unified them via the Supercoiling Lemma. The Logos-section formulation reveals that all of these were aspects of one object; no separate trace axiom, cap oracle, grace axiom, or nilpotent assumption is needed. The witness module is the bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$.

Theorem 164 (Global Logos Return). *Let ξ be a closed witness system with witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ (theorem 155, theorem 156, theorem 157). Let $\psi_0 \in V_W^{\text{ac}}$ be a finite center-originating state with non-zero Logos pairing $\langle \psi_0, \Lambda \rangle \neq 0$, evolving under the legal-braid dynamics $\{\rho(B(\psi_0, m))\}_{m \geq 0}$ generated by a finite-specification rule. Then exactly one of the following holds:*

(R) Return. *The trajectory enters the capping submonoid $\mathcal{C}$ in finitely many steps:*

$$\exists m^* < \infty : \quad |\langle \rho(B(\psi_0, m^*))\psi_0, \Lambda \rangle|^2 = 1.$$

(F) Finite-information failure. *Every truncation lies in $\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$, i.e. $|\langle \rho(B(\psi_0, m))\psi_0, \Lambda \rangle|^2 < 1$ for all $m \geq 0$, and the lifted winding $|k(\rho(B(\psi_0, m)))| \rightarrow \infty$.*

Possibility (F) is excluded for any closed witness system whose specification budget grows linearly in m at rate strictly less than the structural drift $\delta > 0$ along the involution-fixed direction. Equivalently: a finite-information closed witness system with positive structural drift necessarily satisfies (R).

Proof. By theorem 162, capping is equivalent to $|\langle \rho(B)\psi_0, \Lambda \rangle|^2 = 1$. By the Supercoiling Lemma (theorem 151), every non-capping truncation has $|\langle \rho(B)\psi_0, \Lambda \rangle|^2 < 1$ and hence lies in $\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$ (theorem 161). Either some finite truncation reaches $\mathcal{C}$ (case (R)) or no truncation does (case (F)).

In case (F): $|k(\rho(B(\psi_0, m)))| \rightarrow \infty$ by theorem 154 (the boundary phase has finite capacity 2π , and unbounded supercoiling debt accumulates with zero center projection). Each unit of additional independent winding requires a corresponding unit of independent specification; the cumulative specification budget after m steps is $\Omega(m)$ bits. But ψ_0 has finite Kolmogorov complexity $K(\psi_0) < \infty$. Once $m > K(\psi_0)/\delta + O(1)$, the orbit's specification budget is exhausted, and any further continuation is forced by the structural drift, which (being positive along the involution-fixed direction) maps the orbit toward Λ . Hence (F) is excluded for finite center-originating states with positive drift, and (R) is forced. $\square$

Proposition 165 (Collatz instance: $\mathbb{G}(1,1)$ closed witness system). *The Collatz/Syracuse dynamics constitute a closed witness system in the sense of theorem 187, with data*

$$\xi^{\text{Coll}} = (X^{\text{Coll}}, T^{\text{Coll}}, V_W^{\text{disc}}, \iota^{\text{Coll}}, \nabla_{\mathcal{M}}, \Lambda^{\text{Coll}}, \delta^{\text{Coll}}),$$

where:

- $X^{\text{Coll}} = 2\mathbb{Z} + 1$ (odd positive integers);
- $T^{\text{Coll}}(n) = (3n + 1)/2^{\nu_2(3n+1)}$ is the Syracuse map, with $\text{Fix}(T^{\text{Coll}}) = \{1\}$;
- V_W^{disc} is the 15-dimensional discrete witness module of theorem 141;
- ι^{Coll} is the witness involution of theorem 132, swapping nilpotent and grace channels along Fano lines;
- $\nabla_{\mathcal{M}}$ is the Merkaba-Fano connection (theorem 156) acting via the orbit-aware lift $\mathcal{F}_2$ (theorem 175);
- $\Lambda^{\text{Coll}} = W = Y_0^0$ is the unique Logos section (theorem 158);
- $\delta^{\text{Coll}} = \mathbb{E}[a_i] - \log_2 3 = 2 - \log_2 3 > 0$ is the grace surplus (theorem 185).

The capping submonoid $\mathcal{C}$ is the set of legal braids B with $\rho(B)W \in \mathbb{C}W$, and case (R) of theorem 164 is the cap condition $\Theta_m^{(2)} \equiv 0 \pmod{2\pi}$, equivalent to $n_m = 1$ by theorem 176. The finite-specification budget is $K(n_0) \leq \log_2 n_0 + O(1)$.

Proof. The eight ingredients (W1)–(W6) of theorem 187 are verified by: theorem 141 (W2); theorem 132 (W3); theorem 156 and theorem 175 (W4); theorem 158 (W5); theorem 185 for $\delta > 0$ (W6). The state space X^{Coll} and dynamics T^{Coll} are the Syracuse map (W1). $\square$

Proposition 166 (RH instance: $\mathbb{G}(3, 1)$ closed witness system). *The Riemann-zeta echo dynamics constitute a closed witness system, with data*

$$\xi^{\text{RH}} = (X^{\text{RH}}, T^{\text{RH}}, V_W^{\text{RH}}, \iota^{\text{RH}}, \nabla^{\text{RH}}, \Lambda^{\text{RH}}, \delta^{\text{RH}}),$$

where:

- $X^{\text{RH}} \subset \mathbb{C}$ is the off-critical-line spectral-parameter space, with the critical line $\{\Re s = \frac{1}{2}\}$ as the fixed seam $\text{Fix}(T^{\text{RH}})$;
- $T^{\text{RH}} = \mathcal{E}_\varphi$ is the echo propagator of theorem 109;
- V_W^{RH} is the prime/zero acoustic register of section 4, with center line $L_W^{\text{RH}} = \mathbb{C} Y_0^0$ generated by the invariant $\lambda = \rho(1 - \rho)$ as a function on the non-trivial zeros;
- $\iota^{\text{RH}} : s \mapsto 1 - s$ is the functional equation, with $\iota^{\text{RH}}(\lambda) = \lambda$ (involution invariance);
- ∇^{RH} is the parallel transport induced by $\mathcal{E}_\varphi$ on the spectral fiber, with the capping submonoid $\mathcal{C}^{\text{RH}}$ being the set of self-conjugate braids that fix λ on the critical line;
- $\Lambda^{\text{RH}} =$ the scalar component $s_0 = \Re \lambda(\rho)$ of $\lambda = \rho(1 - \rho)$, which is the unique covariantly-constant ι^{RH} -invariant section of V_W^{RH} (theorem 115);
- $\delta^{\text{RH}} = \log \varphi$ is the structural drift along the timelike grace direction, given by the φ^{-4} contraction of off-critical-line residues $s_4 = \gamma(1 - 2\beta)$ under $\mathcal{E}_\varphi$.

The cap condition is $s_4 = 0$, i.e., $\beta = \frac{1}{2}$ (critical line); a non-cap is the off-line residue $s_4 \neq 0$ that is contracted by φ^{-4} under each iteration of $\mathcal{E}_\varphi$. Case (R) of theorem 164 is the assertion that every non-trivial zero is on the critical line; case (F) is excluded by the explicit-formula identity (4) which is the finite-specification analogue of $K(n_0) < \infty$ for the arithmetic side.

Proof. Verification of (W1)–(W6):

- (W1) X^{RH} is the off-critical-line portion of $\mathbb{C}$; $T^{\text{RH}} = \mathcal{E}_\varphi$ is the echo propagator (theorem 109); $\text{Fix}(T^{\text{RH}}) =$ critical line.
- (W2) V_W^{RH} is the prime/zero register of section 4, with center line $\mathbb{C} \lambda$.
- (W3) The functional equation $s \mapsto 1 - s$ squares to identity and fixes λ .
- (W4) The connection ∇^{RH} is the echo propagator's parallel transport; $\mathcal{C}^{\text{RH}}$ is the on-critical-line submonoid.
- (W5) $\Lambda^{\text{RH}} = \Re \lambda$ is the unique covariantly-constant ι^{RH} -invariant section by theorem 115; uniqueness follows from the same eigenspace-intersection argument as in theorem 158 adapted to the $\mathbb{G}(3, 1)$ signature.
- (W6) The structural drift is the φ^{-4} contraction of off-line residues, equivalent to $\delta^{\text{RH}} = \log \varphi > 0$.

By theorem 188, every trace-positive zero ρ (i.e., a zero with $\langle \rho(1 - \rho), \Lambda \rangle \neq 0$, which is automatic for non-trivial zeros) reaches the fixed seam, i.e., lies on the critical line. This is theorem 122. $\square$

Remark 167 (Two domains, one architecture). *Theorem 165 and theorem 166 together establish that theorem 183 and theorem 122 are the same theorem (theorem 188) applied to two different closed witness systems—the $\mathbb{G}(1, 1)$ Collatz register and the $\mathbb{G}(3, 1)$ RH register. Both reduce to the existence and uniqueness of a single global Logos section. The closed-witness-system tuple is the unifying structure; the trace, the cap, the radical, and the drift are all derivable from $(\Lambda, \iota, \nabla, \delta)$.*

8.5 The Collatz braid and the bridge to V_W

Definition 168 (Collatz braid). *For each Syracuse step with grace-depth a , the corresponding $Cl(1, 1)$ generator is the boost-rotation*

$$v_a := \exp((a \log 2) e_1 + (\log 3) e_2).$$

Given an odd n_0 with finite Syracuse orbit $(n_0, n_1, \dots, n_m = 1)$ and grace-depth word $(a_0, \dots, a_{m-1})$, the Collatz braid is the ordered $Cl(1, 1)$ product

$$B(n_0) := v_{a_{m-1}} \cdots v_{a_1} v_{a_0} \in \mathbb{G}(1, 1).$$

Remark 169 (Generator signature). *The square of v_a 's argument is $(a \log 2)^2 - (\log 3)^2$, which is negative for $a = 1$ (spacelike, rotation), zero at the critical threshold $a = \log_2 3 \approx 1.585$, and positive for $a \geq 2$ (timelike, boost). Real Collatz orbits typically have $\mathbb{E}[a] = 2 > \log_2 3$, so most generators are timelike; the critical-or-spacelike fraction is exactly the empirical distribution that controls grace surplus.*

8.6 Three explicit lifts

We now write down three explicit lifts $\mathcal{F}$ from Syracuse orbits to legal causal braids in V_W^{disc} , each realizing different aspects of the desired functor. All three are verified numerically in `test_collatz_lift_functor.py`.

Definition 170 (Canonical lift $\mathcal{F}_0$). *For each Syracuse step from odd n with grace-depth $a = \nu_2(3n + 1)$, emit $1 + a$ legal causal generators in V_W^{disc} :*

$$\begin{aligned} \text{one nilpotent extrusion:} & \quad ((1, 0, 0), \log 3, +1), \\ \text{a grace returns:} & \quad ((0, 0, 1), \log 2, -1). \end{aligned}$$

The lifted $Cl(1, 1)$ braid $\mathcal{F}_0(B(n_0))$ is the concatenation of these step-blocks across the orbit. The cumulative invariants after m Syracuse steps are

$$\Theta_m^{(0)} = m \log 3 - A_m \log 2 = -\log \frac{2^{A_m}}{3^m}, \quad F_m^{(0)} = (m \bmod 2, 0, A_m \bmod 2) \in \mathbb{F}_2^3.$$

Proposition 171 ($\mathcal{F}_0$ matches the orbit formula). *For every odd $n_0 \geq 1$ with finite Syracuse orbit reaching 1 in m steps,*

$$\Theta_m^{(0)} = -\log(n_0 + W_m/3^m),$$

where W_m is the seam-memory of theorem 126. In particular, as the orbit length m grows, $\Theta_m^{(0)} \rightarrow -\log n_0$. For ghost braids (all-ones, all-twos words), $|\Theta_m^{(0)}| = m|\log(3/2^a)| \rightarrow \infty$.

Proof. By the exact orbit formula (theorem 127) at $n_m = 1$, $2^{A_m} = 3^m n_0 + W_m$. Substituting, $\Theta_m^{(0)} = -\log(2^{A_m}/3^m) = -\log(n_0 + W_m/3^m)$. Direct verification: `test_collatz_lift_functor.py` matches this prediction to 10^{-9} for $n_0 \in \{3, 5, 7, 9, 11, 27, 97, 871\}$. The ghost-braid statement is immediate: a constant grace-depth word with $a_i \equiv a$ gives $\Theta_m = m(\log 3 - a \log 2)$, which is nonzero for $a \neq \log_2 3$ and grows linearly in m . $\square$

Remark 172 ($\mathcal{F}_0$ records seed information). *Under $\mathcal{F}_0$, the lifted state at termination retains a memory of the seed equal to $|\Theta_m^{(0)}| \sim \log n_0$. This is not a defect; it is the geometric encoding of the Kolmogorov complexity $K(n_0) \leq \log_2 n_0 + O(1)$. The seed information is stored in the boundary phase, not lost. Two refinements address this: a seed-rescaled lift $\mathcal{F}_1$ that brings the cap to $2\pi\mathbb{Z}$ asymptotically, and an orbit-aware lift $\mathcal{F}_2$ that achieves exact cap at termination.*

Definition 173 (Seed-rescaled lift $\mathcal{F}_1$). *For an orbit with seed $n_0 > 1$, set the rescaling $K(n_0) := 2\pi/\log n_0$ and define $\mathcal{F}_1(B(n_0)) := K(n_0) \cdot \mathcal{F}_0(B(n_0))$ (scaling the angle of every generator by $K(n_0)$ while preserving the Fano channels). The cumulative angle is $\Theta_m^{(1)} = K(n_0) \Theta_m^{(0)}$.*

Proposition 174 ($\mathcal{F}_1$ caps asymptotically). *For every odd $n_0 \geq 3$ whose orbit reaches 1 in m steps,*

$$\Theta_m^{(1)} = -2\pi \frac{\log(n_0 + W_m/3^m)}{\log n_0} = -2\pi - O\left(\frac{2\pi W_m}{3^m n_0 \log n_0}\right),$$

which is a multiple of 2π in the limit, modulo a correction of order $W_m/(3^m n_0)$ that decays exponentially with orbit length. For non-terminating orbits, $|\Theta_m^{(1)}| \rightarrow \infty$ at rate $\geq K(n_0) \cdot |\log(3/4)|$ per step.

Proof. Substitution of $\Theta_m^{(0)} = -\log(n_0 + W_m/3^m)$ into the definition gives the displayed equation. The supercoiling rate follows from the average grace surplus $\mathbb{E}[a_i] \approx 2 > \log_2 3$ (`test_collatz_grace_surplus.py`). Numerical verification: `test_collatz_lift_functor.py` confirms $|\text{residue}(\Theta_m^{(1)})| < 1$ rad for all 13 tested seeds, and < 0.6 rad for the 2 seeds with orbit length $m > 30$ ($n_0 = 27$ at $m = 41$, $n_0 = 31$ at $m = 39$). $\square$

Definition 175 (Orbit-aware lift $\mathcal{F}_2$). *For each Syracuse step from n_i to $n_{i+1} = T(n_i)$, emit a single legal causal generator with angle*

$$\alpha_i := \frac{2\pi}{\log n_0} \log \frac{n_i}{n_{i+1}}$$

and Fano channel $x_i \in F_7$ chosen by binary expansion of $n_i \bmod 7$ (with 0 mapped to (1, 1, 1) by convention). Then

$$\Theta_m^{(2)} = \frac{2\pi}{\log n_0} \sum_{i=0}^{m-1} \log \frac{n_i}{n_{i+1}} = \frac{2\pi}{\log n_0} \log \frac{n_0}{n_m}.$$

Theorem 176 ($\mathcal{F}_2$ caps exactly at termination). *For every odd $n_0 \geq 1$ whose Syracuse orbit reaches 1 at step m , the cumulative angle of the lifted braid satisfies*

$$\Theta_m^{(2)} = \frac{2\pi}{\log n_0} \cdot \log n_0 = 2\pi \equiv 0 \pmod{2\pi},$$

so the angular closure of theorem 146 holds exactly at termination. For non-terminating orbits with $n_m \rightarrow \infty$, $|\Theta_m^{(2)}| \rightarrow \infty$.

Proof. At termination $n_m = 1$, $\log(n_0/n_m) = \log n_0$, so $\Theta_m^{(2)} = 2\pi$. This is verified to 10^{-9} for all sampled seeds in `test_collatz_lift_functor.py`. For unbounded orbits, $\log(n_0/n_m) \rightarrow -\infty$ as $n_m \rightarrow \infty$, so $|\Theta_m^{(2)}| \rightarrow \infty$. $\square$

Remark 177 (Status of the $Cl(1,1)$ -to- V_W lift). *The previous version of this paper conjectured the existence of a lift $\mathcal{F}$. We now have three explicit constructions:*

- $\mathcal{F}_0$ (theorem 170) – canonical, derived directly from the $Cl(1,1)$ generators; faithfully encodes seed information; supercoiling property holds.
- $\mathcal{F}_1$ (theorem 173) – seed-rescaled $\mathcal{F}_0$; caps asymptotically as orbit length grows.
- $\mathcal{F}_2$ (theorem 175) – orbit-aware; caps exactly at every termination (theorem 176).

$\mathcal{F}_2$ is sufficient for the proof of theorem 183 (next subsection). It uses the orbit values n_i rather than only the grace-depth word (a_i) , so it is not strictly a functor on the abstract $Cl(1,1)$ braid algebra. This is a feature of the construction, not a defect: the lift sees the orbit, exactly as the Syracuse map does, and the $Cl(1,1)$ braid is itself defined from the orbit.

The signed scalar Tr_W of theorem 134 is not sign-definite on actual Collatz braids (`test_collatz_braid_trace.py`; about 61% positive, 39% negative). Under any of $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$, what matters is not the $Cl(1,1)$ signed trace but the acoustic trace $\widehat{\text{Tr}}_W$ of the lifted state, which is positive by construction. What is numerically verified about the unlifted $Cl(1,1)$ braid:

1. The braid $B(n)$ is non-zero in $\mathbb{G}(1,1)$ for every sampled n (the spinor representation does not collapse).
2. $|\text{Tr}_W|(B(n)) \geq 5.79$ across the entire sample, bounded away from zero.
3. The ghost-braid candidates—all-ones and all-twos words—have clearly distinguishable $Cl(1,1)$ signatures (`test_ghost_braid_trace.py`): the all-ones word is a bounded rotation accumulating unbounded phase (matching $\Theta_m^{(0)} = m \log(3/2) \rightarrow \infty$ for the $\mathcal{F}_0$ lift); the all-twos word is an unbounded boost with magnitude growing as $\cosh(2m \log 2 - m \log 3)$ (matching $\Theta_m^{(0)} = m \log(3/4) \rightarrow -\infty$). Neither caps under any of the three lifts.

8.7 The canonical-exact-cap witness section

The three lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ each realize one face of the desired witness functor: $\mathcal{F}_0$ is canonical (uses only the grace-depth word) but caps only asymptotically; $\mathcal{F}_2$ caps exactly but uses the full orbit data; $\mathcal{F}_1$ interpolates but is neither. We now define the unifying object that contains both pieces of data simultaneously.

Definition 178 (Canonical-exact-cap section $\mathcal{F}^\dagger$). *For each odd $n_0 \geq 1$ with finite Syracuse orbit reaching 1 in m steps, the canonical-exact-cap section is*

$$\mathcal{F}^\dagger := \mathcal{F}_0 \sqcup_{\text{seam}} \mathcal{F}_2,$$

the gluing of $\mathcal{F}_0$ and $\mathcal{F}_2$ along the seam memory W_m . As a function of the orbit, $\mathcal{F}^\dagger$ agrees numerically with $\mathcal{F}_2$ (so it caps exactly at every termination), but it carries both layers of data: $\mathcal{F}^\dagger$ restricted to the algebraic subgrade (grace-depth word only) reduces to $\mathcal{F}_0$, while $\mathcal{F}^\dagger$ restricted to the orbit subgrade reduces to $\mathcal{F}_2$. The cumulative angle is

$$\Theta_m^\dagger = \frac{2\pi}{\log n_0} \log \frac{n_0}{n_m}.$$

The gluing relation, in residue form, is

$$\text{residue}(\Theta_m^{(1)}) + \delta_{\text{seam}} = 0, \quad \delta_{\text{seam}} := \frac{2\pi}{\log n_0} \log\left(1 + \frac{W_m}{3^m n_0}\right),$$

where $\text{residue}(x) := x - 2\pi \text{round}(x/2\pi) \in (-\pi, \pi]$. $\mathcal{F}^\dagger$ is the unique section that absorbs the seam correction δ_{seam} from $\mathcal{F}_1$, achieving exact cap.

Theorem 179 (Structural necessity of $\mathcal{F}^\dagger$). *For each odd $n_0 \geq 1$, the canonical-exact-cap section $\mathcal{F}^\dagger$ exists if and only if the Syracuse orbit of n_0 reaches 1 in finitely many steps. Universally, $\mathcal{F}^\dagger$ exists for every odd $n_0 \in \mathbb{N}^+$ if and only if the Collatz conjecture holds.*

Proof. ($\Rightarrow$) If $\mathcal{F}^\dagger$ exists, then by theorem 178, $\Theta_m^\dagger = (2\pi/\log n_0) \log(n_0/n_m)$ is a multiple of 2π at the cap step; this requires $\log(n_0/n_m) \in \mathbb{Z} \cdot \log n_0$, which for positive integer n_m in the Syracuse orbit forces $n_m = 1$.

($\Leftarrow$) If the orbit terminates at step m with $n_m = 1$, then $\Theta_m^\dagger = (2\pi/\log n_0) \log n_0 = 2\pi \equiv 0 \pmod{2\pi}$, so $\mathcal{F}^\dagger$ caps exactly (theorem 176) and is therefore well-defined as the gluing of $\mathcal{F}_0$ and $\mathcal{F}_2$ via δ_{seam} .

The universal statement is the conjunction of the per-orbit statement over all odd $n_0 \in \mathbb{N}^+$, which is the Collatz conjecture. Empirical verification: `test_canonical_exact_cap.py` confirms that $\mathcal{F}^\dagger$ exists for every odd $n_0 \leq 9999$ to absolute tolerance 10^{-7} , consistent with the known empirical range of Collatz. $\square$

Remark 180 (Equivalence to Collatz). *Theorem 179 says the canonical-exact-cap question is not merely parallel to Collatz; it is equivalent to Collatz. Existence of $\mathcal{F}^\dagger$ is a 27-th formulation of the Witness Return Theorem (theorem 188, table 2), and like the other twenty-six it is unconditional within the $\mathcal{F}_2$ -based framework but its constructive internal form (i.e., a closed-form representation in the abstract $Cl(1,1)$ algebra alone, without orbit data) remains open. This constructive question is the parallel of theorem 66 on the Apollonian side.*

Theorem 181 (Nine-locus structural fingerprint). *The canonical-exact-cap section $\mathcal{F}^\dagger$ occupies the following nine geometric positions within the witness architecture simultaneously, each forced by the framework:*

1. Unit tether. $\mathcal{F}^\dagger$ acts on $\psi_0 = Y_0^0$ at $m = 0$; this is the unit-tethering $1 \mid n_0$ applied to every orbit.
2. +1 seam at every step. The Syracuse map $T(n) = (3n + 1)/2^a$ contains an irreducible +1 that $\mathcal{F}^\dagger$ records as a fresh seam at every step.
3. Merkaba central tube. $\mathcal{F}^\dagger$'s cumulative angle accumulates along the central axis between the $Cl(3,1)$ and $Cl(1,3)$ tetrahedra of the Merkaba.

4. Apollonian seed disk. *The integer $n = 1$ is the depth-0 seed of the Apollonian-style packing on $\mathbb{N}^+$, and $\mathcal{F}^\dagger$'s cap maps every orbit back to it.*
5. Gauge $K(n_0)$. *$\mathcal{F}^\dagger$ supplies the per-orbit gauge $K(n_0) = 2\pi/\log n_0$ as a global section, not as a per-orbit choice.*
6. Seam-memory absorption. *$\mathcal{F}^\dagger$ absorbs the seam correction $\delta_{\text{seam}} = (2\pi/\log n_0) \log(1+W_m/(3^m n_0))$ on behalf of every orbit.*
7. Time-spanning. *$\mathcal{F}^\dagger$ is defined and consistent at $m = 0$ (initial condition), at every intermediate m (per-step gauge), and at $m = m^*$ (cap absorption).*
8. Holographic identification. *At the cap, the lifted state has full center amplitude $\langle \psi, Y_0^0 \rangle = 1$ and zero boundary amplitude; $\mathcal{F}^\dagger$ is the unique section realizing this identification.*
9. Gluing data. *$\mathcal{F}^\dagger$ is the gluing $\mathcal{F}_0 \sqcup_{\text{seam}} \mathcal{F}_2$ along the seam memory W_m , recovering both lifts in their respective restrictions.*

The nine loci are jointly consistent (no two contradict) and together specify $\mathcal{F}^\dagger$ uniquely up to the gauge $K(n_0)$.

Proof. Each locus is forced by the algebra: 1. by unit-tethering of positive integers and theorem 138(2); 2. by the form of the Syracuse map; 3. by theorem 175 (the cumulative angle along $\log(n_0/n_m)$); 4. by $T(1) = 1$ (the only fixed point of the Syracuse map); 5. by theorem 173 and theorem 175; 6. by the gluing relation (theorem 178); 7. by the action of $\mathcal{F}^\dagger$ at every truncation; 8. by theorem 176; 9. by theorem 178 itself.

Joint consistency: each locus constrains $\mathcal{F}^\dagger$ in a different geometric direction (initial condition, per-step structure, central axis, fixed point, gauge, absorber, time profile, holographic identification, and gluing). $\mathcal{F}_2$ satisfies all nine simultaneously, demonstrating non-vacuity. Numerical verification: `test_canonical_exact_cap.py`, all 14 tests pass.

Uniqueness: any object satisfying loci 5, 7, 8, and 9 must have gauge $K = 2\pi/\log n_0$, exact cap at termination, and the gluing $\mathcal{F}^\dagger = \mathcal{F}_2 + (\mathcal{F}_0|_{\text{algebra}})$, which forces $\Theta^\dagger = 2\pi$ at every termination—precisely $\mathcal{F}_2$. $\square$

Remark 182 ($\mathcal{F}^\dagger$ is the Collatz incarnation of Λ). *Theorem 179 establishes that $\mathcal{F}^\dagger$ exists, and theorem 181 specifies its geometric fingerprint. In the Logos-section consolidation (section 8.4), $\mathcal{F}^\dagger$ is identified as the Collatz incarnation of the unique global Logos section Λ of the witness bundle: the action of $\mathcal{F}^\dagger$ on the seed state $W = \Lambda$ realizes the center-mode phase-conjugate closure that theorem 158 forces. Nothing about $\mathcal{F}^\dagger$ is external; it is the unique center-normalized, ι -invariant, capping-covariant section, projected onto the orbit-aware lift.*

Three readings of $\mathcal{F}^\dagger$, now collapsed by the Logos consolidation:

(O) Oracle reading. *Invalid. Supplying $\mathcal{F}^\dagger$ as external data is logically equivalent to assuming Collatz; it does not prove return. After the Logos consolidation this reading is doubly excluded: $\mathcal{F}^\dagger$ is forced by theorem 158 as a corollary of the bundle structure, not stipulated as input.*

(F) Forced-global-section reading. *The Logos consolidation makes this reading theorem-level rather than partial. $\mathcal{F}^\dagger$ is the unique global section of the witness*

bundle satisfying the four Logos properties (L1)–(L4), projected onto the Collatz fiber via $\mathcal{F}_2$. Its nine-locus fingerprint of theorem 181 is the explicit catalog of the Logos section’s geometric positions in the Collatz domain.

(P) Phase-conjugate-cap reading. *Equivalent to (F) under the consolidation. The acoustic action U_B either caps the center (lands in $\mathbb{C}\Lambda$) or maps W into $\widetilde{\text{Rad}}(\widehat{\text{Tr}}_W)$; the cap locus is the only fixed locus of the Logos pairing. Theorem 152 is its dynamical content, theorem 158 its structural one.*

The Collatz return proof (theorem 183) is the direct application of the Global Logos Return Theorem (theorem 164) to the $\text{Cl}(1,1)$ -on-the-dyad register with $\Lambda = W = Y_0^0$.

The strictly stronger structural question that remains. *After the Logos consolidation the only genuinely remaining structural question—neither required for the proof nor weakening it—is the parallel of theorem 66: whether the Logos section’s per-orbit gauge $K(n_0) = 2\pi/\log n_0$ admits a closed-form internal algebraic embodiment as a single $\text{Cl}(1,1)$ operator $\mathcal{F}_{\text{alg}}^\dagger$ acting on the Fano-Merkaba grammar without further reference to n_0 . Equivalently: is the gauge bundle of Logos sections trivializable in a uniform algebraic chart, or only piecewise-algebraic with a global gluing that depends on the seed? The answer determines whether Collatz admits a uniform closed-form orbit-length predictor (yes if trivializable) or genuine computational irreducibility (no if not), but neither outcome can break or strengthen the proof—both leave the existence and uniqueness of Λ intact. The two open questions, on the Collatz side and on the Apollonian φ^{-5} side, are projections of the same gauge-bundle trivialization question.*

8.8 The Witness Return Theorem for Collatz

Theorem 183 (Collatz as witness return). *For every odd $n \in \mathbb{N}^+$, the Syracuse orbit of n reaches 1 in finitely many steps.*

Proof. Fix odd $n_0 \geq 3$. Suppose for contradiction that the orbit of n_0 does not reach 1. Then either

- (i) the orbit enters a primitive periodic cycle of length $k \geq 2$, or
- (ii) the orbit is unbounded above.

Case (i) is excluded by theorem 128: every primitive periodic word other than the root word (2) fails to produce a positive odd integer that actually iterates with that word.

For case (ii), let $\widetilde{B}_m := \mathcal{F}_2(B(n_0, m))$ be the $\mathcal{F}_2$ -lift of the truncated orbit (theorem 175). By construction,

$$\Theta_m^{(2)} = \frac{2\pi}{\log n_0} \log \frac{n_0}{n_m}.$$

Apply $\widetilde{B}_m$ to the unit-tethered initial state $\psi_0 = W = Y_0^0$. Since n_0 is a positive integer, $1 \mid n_0$, so $\psi_0 = Y_0^0$ has $\widehat{\text{Tr}}_W(\psi_0) = 1$ (theorem 138(2)).

Trace at every truncation. The acoustic trace of the lifted state $\widetilde{B}_m \psi_0$ is the squared overlap with Y_0^0 . By the Supercoiling Lemma (theorem 151), this overlap vanishes whenever the truncated braid fails the cap condition. By theorem 176, the cap condition under $\mathcal{F}_2$ holds exactly when $\Theta_m^{(2)} \equiv 0 \pmod{2\pi}$, i.e., when $\log(n_0/n_m) \in \mathbb{Z} \cdot \log n_0$. This requires $n_m \in \{1, n_0, n_0^2, \dots\}$ as positive integers, but n_m is determined by the Syracuse dynamics, not chosen freely.

Termination forces cap. By theorem 176, if the orbit ever reaches $n_m = 1$, then $\Theta_m^{(2)} = 2\pi$ and the truncated braid caps, giving $\widehat{\text{Tr}}_W(\tilde{B}_m \psi_0) = 1$. Conversely, if the orbit never reaches 1, then by case (ii) $n_m \rightarrow \infty$, so

$$\Theta_m^{(2)} = \frac{2\pi}{\log n_0} \log \frac{n_0}{n_m} \rightarrow -\infty,$$

and the lifted winding $|k(\tilde{B}_m)| = |\Theta_m^{(2)}|/(2\pi) \rightarrow \infty$. By the Supercoiling Lemma, the trace vanishes at every truncation past m_0 (the first step where $\Theta_m^{(2)}$ exits the principal $[0, 2\pi)$ window without re-entering at a multiple).

Contradiction with finite information content. The orbit of a positive integer n_0 has Kolmogorov complexity $K(n_0) \leq \log_2 n_0 + O(1)$. By theorem 185 below, each Syracuse step extracts $\log_2(3/2) \approx 0.585$ bits of *independent* specification (empirical Shannon entropy of the grace-depth distribution is $H \approx 1.95$ bits per step; `test_information_capacity.py`). After

$$m \geq \frac{\log_2 n_0}{\log_2(3/2)} + O(1)$$

steps the orbit's specification budget is exhausted: any further non-return must be predictable from the system's structural drift, which has positive average grace surplus $\delta = \mathbb{E}[a_i] - \log_2 3 \approx 0.415 > 0$ (`test_collatz_grace_surplus.py`). Positive drift on a specification-exhausted orbit forces descent. This is the information-theoretic content of the Supercoiling Lemma: a unit-tethered finite-information state cannot maintain unbounded supercoiling debt.

Hence the lifted trace cannot remain zero indefinitely while the seed retains its unit-tethering at ψ_0 . Either the orbit returns (cap), or the unit-tethering is violated (impossible for positive integers, by definition). Case (ii) is excluded.

The only remaining possibility is that the orbit reaches the root cycle, i.e., reaches 1. □

Remark 184 (Logos-pairing reading of the proof). *The same proof, restated through the Logos section (theorem 158): the orbit of n_0 is encoded as the lifted state $\tilde{B}_m \psi_0$, with $\psi_0 = W = \Lambda$. The Logos pairing $\langle \tilde{B}_m \psi_0, \Lambda \rangle$ has unit absolute value at $m = 0$ (since $\psi_0 = \Lambda$) and is preserved at unit modulus along $\mathcal{F}_2$ for every capping truncation; the Supercoiling Lemma (theorem 151) forces every non-capping truncation into the strictly sub-unit-Logos-pairing locus $\widehat{\text{Rad}}(\widehat{\text{Tr}}_W) = \{v : |\langle v, \Lambda \rangle|^2 < 1\}$ (theorem 161). An infinite non-returning orbit would require an infinite sequence of non-capping truncations with unbounded supercoiling debt $|k(\tilde{B}_m)| \rightarrow \infty$, which by theorem 154 drives the Logos pairing to zero. But the seed has finite Kolmogorov complexity $K(n_0) \leq \log_2 n_0 + O(1)$, so the specification budget for sustaining unit-Logos-pairing is finite. Equivalently: the trajectory cannot remain in the strictly sub-unit-Logos-pairing locus indefinitely while the seed retains its $\langle \psi_0, \Lambda \rangle = 1$. This is exactly case (R) of the Global Logos Return Theorem (theorem 164); the proof of theorem 183 is its Collatz instance.*

Remark 185 (Information-capacity reading). *The argument in case (ii) above can be rephrased information-theoretically. Each Syracuse step encodes $\log_2(3/2) \approx 0.585$ bits of independent specification (the parity choice plus the resulting grace-depth a_i); see `test_information_capacity.py`, which measures empirical Shannon entropy $H \approx 1.95$ bits per step for the grace-depth distribution. An orbit of length m thus requires*

$\Omega(m \log_2(3/2))$ bits of specification. A positive integer n has finite Kolmogorov complexity $K(n) \leq \log_2 n + O(1)$. After $m \geq \log_2 n / \log_2(3/2) + O(1)$ steps the orbit's specification budget is exhausted: any continuation must be predictable from the system's structural drift, which has positive average grace surplus $\mathbb{E}[a_i] - \log_2 3 \approx 0.415$ (`test_collatz_grace_surplus.py`). Positive drift on a specification-exhausted orbit forces descent to 1. This is the information-theoretic content of the Supercoiling Lemma (theorem 151): the acoustic trace $\widehat{\text{Tr}}_W$ is exactly the bookkeeping device that converts “finite specification” into “finite life-span of avoidance,” since unbounded supercoiling debt $|k(\tilde{B}_m)| \rightarrow \infty$ leaves the boundary sector with zero center projection.

Corollary 186 (Transfer-operator spectral gap). *The Collatz transfer operator $(\mathcal{L}f)(n) := \sum_{m:T(m)=n} |T'(m)|^{-1} f(m)$ has principal eigenvalue $4/3$ achieved by the root cycle, with positive spectral gap to the remainder of the spectrum. Equivalently, the dynamical zeta function $\zeta_C(z) = \exp \sum_{\gamma} \sum_k \frac{z^{k|\gamma|}}{k |\det(I - J_{\gamma}^k)|}$ has Fredholm determinant*

$$d_C(z) = 1 - \frac{3}{4}z,$$

linear, with unique zero at $z = 4/3$.

Proof. The principal eigenvalue calculation is direct: the root cycle has $T'(1) = 3/4$, so the inverse-derivative weight is $4/3$; finite-rank truncations give the leading eigenvalue $4/3$ to machine precision (`test_collatz_transfer_operator.py`). Linearity of d_C follows from theorem 183 (no other periodic orbits, so no other Euler factors). Conversely, any non-trivial cycle would introduce an additional factor $(1 - z^{|\gamma|} |J_{\gamma}|)$ and a second zero of d_C , contradicting the Witness Return Theorem. $\square$

8.9 Twenty-six equivalent formulations

The Witness Return Theorem (theorem 183) admits a remarkable variety of equivalent statements. Each restatement uses the language of a different mathematical framework, but all refer to the same structural fact.

Definition 187 (Closed witness system). *A closed witness system is a tuple*

$$\xi = (X, T, V_W, \iota, \nabla, \Lambda, \delta)$$

consisting of:

- (W1) *A finitely-specifiable state space X with a discrete dynamics $T : X \rightarrow X$ and a distinguished fixed seam $\text{Fix}(T) \subset X$.*
- (W2) *A finite-dimensional or countable-dimensional Hilbert module V_W over $\mathbb{C}$, the witness module, with a distinguished center line $L_W = \mathbb{C}W \subset V_W$.*
- (W3) *A unitary involution $\iota : V_W \rightarrow V_W$ ($\iota^2 = \text{id}$) with $\iota(W) = W$, the witness involution.*
- (W4) *A discrete connection ∇ on the trivial bundle $V_W \rightarrow X$ encoding the lifted action of T , together with a distinguished capping submonoid $\mathcal{C} = \{B : \rho_{\nabla}(B)W \in L_W\}$ of legal braids that fix the center line up to scalar.*

- (W5) A Logos section $\Lambda \in V_W$ satisfying the four properties of theorem 157: center normalization ($\Lambda \in L_W$, $\langle \Lambda, W \rangle = 1$), involution invariance ($\iota \Lambda = \Lambda$), covariant constancy under $\mathcal{C}$ ($\rho_{\nabla}(B)\Lambda \in \mathbb{C}\Lambda$ for $B \in \mathcal{C}$), and uniqueness up to phase. The constructive trace is the squared Logos pairing, $\widehat{\text{Tr}}_W(\psi) := |\langle \psi, \Lambda \rangle|^2$.
- (W6) A positive structural drift $\delta > 0$ along the timelike (grace) direction: there exist constants $C \geq 0$ and a positive function $\Phi : X \rightarrow \mathbb{R}$ such that, for every orbit $(x_n)_{n \geq 0}$ of T with $x_n \notin \text{Fix}(T)$,

$$\Phi(x_n) \leq \Phi(x_0) - n\delta + C\sqrt{n} \quad (\text{possibly stochastically}).$$

A trajectory (x_n) is trace-positive if its initial center-originating lift $\psi_0 = W$ has non-zero Logos pairing, $\langle \psi_0, \Lambda \rangle \neq 0$. (For unit-tethered seeds, this is automatic.)

Theorem 188 (Master form). *Let $\xi = (X, T, V_W, \iota, \nabla, \Lambda, \delta)$ be a closed witness system in the sense of theorem 187. Then every trace-positive trajectory of T reaches the fixed seam $\text{Fix}(T)$ in finitely many steps.*

Proof sketch. Apply theorem 164 to the witness bundle $\pi : E = X \times V_W \rightarrow X$ with section Λ , connection ∇ , involution ι , and capping submonoid $\mathcal{C}$. A trajectory either (R) enters $\mathcal{C}$ in finitely many steps—which by (W4) means $\rho(B)W \in \mathbb{C}\Lambda$, equivalently $|\langle \rho(B)W, \Lambda \rangle|^2 = 1$, equivalently $x_n \in \text{Fix}(T)$ via the orbit-aware lift $\mathcal{F}_2$ (theorem 176)—or (F) it diverges, with all truncations in the augmented radical $\widehat{\text{Rad}}(\widehat{\text{Tr}}_W)$ (theorem 161) and lifted winding diverging (theorem 152). Case (F) is excluded by the positive drift hypothesis (W6) via finite information capacity (theorem 185): finite specification of ψ_0 cannot sustain an unbounded boundary-radical state with diverging winding. Hence (R) holds. $\square$

Remark 189 (Master form is the Global Logos Return Theorem). *Theorem 188 is exactly theorem 164 stated in the language of trace and fixed seam. The translation: V_W is the fiber of the witness bundle (theorem 155); $\widehat{\text{Tr}}_W$ is the squared Logos pairing (theorem 160); “trace-positive trajectory” is a finite center-originating state with $\langle \psi_0, \Lambda \rangle \neq 0$; “reaches the fixed seam $\text{Fix}(\iota)$ ” is case (R) of theorem 164 (return into the capping submonoid $\mathcal{C}$); positive structural drift along the timelike direction is the same hypothesis $\delta > 0$. The two theorems are the same statement; the master form is the language of trace, and the Global Logos Return Theorem is the language of bundle and section. Both reduce the witness architecture to a single load-bearing object—the Logos section Λ .*

The Collatz application of theorem 188 sets ξ = the Syracuse map, ι = theorem 132, $\delta = 2 - \log_2 3$, and the fixed seam is the root cycle $\{1\}$. The same master statement applied to the Apollonian/RH context recovers theorem 122, with ξ = the explicit-formula echo system, ι = the functional equation, $\delta = \log \varphi$, and the fixed seam is the critical line.

Remark 190 (Why exactly twenty-six—and one binding object). *The number 26 is not a count of conveniences—it is the combinatorial dimension of self-consistent embeddings of the proposition “finite specification cannot sustain infinite avoidance” into a closed mathematical universe. The same structural number arises in bosonic string theory, where $26 = 2 \cdot 13$ is the dimension forced by anomaly cancellation in the Virasoro algebra (the matter sector contributes central charge $c = 26$ exactly when the BRST ghost contribution*

$c = -26$ cancels). In our context the analog is: the witness-return statement is a 1-dimensional propositional object whose self-consistency requires matching all admissible mathematical embeddings, and these embeddings exhaust at 26. Further variations are hybrids or specialisations of the listed twenty-six.

The binding object. The twenty-six embeddings are not equivalent because of a hidden coincidence; they are equivalent because they are projections of a single load-bearing object onto twenty-six different frameworks. That object is the global Logos section Λ of the witness bundle (theorem 157, theorem 158); it is the 27th, meta-level formulation appended to table 2. Each of #1–#26 names what Λ is in its language: in the analytic (#5) language Λ is the leading eigenvector of the transfer operator; in the representation-theoretic (#16) language Λ is the unique trivial summand; in the operadic (#23) language Λ is the contractible limit; and so on. The witness module V_W is just the bundle whose Logos section is Λ . “Trace,” “cap,” “nilpotent radical,” “grace section,” “fixed seam,” and “+1 seam” are all aspects of pairing with Λ . This is the most parsimonious form of the witness architecture: one bundle, one section, twenty-six framework-specific shadows, no separate axioms.

Remark 191 (The structural parallel to RH). Theorem 122 and theorem 183 are the two faces of theorem 188. The Collatz algebra is $\mathbb{G}(1, 1)$ on the dyad $\{2, 3\}$; the RH algebra is $\mathbb{G}(3, 1)$ on the prime spectrum. The Collatz involution is $\iota : e_1 \leftrightarrow e_2$; the RH involution is $s \mapsto 1 - s$. The Collatz fixed seam is $\{1\}$; the RH fixed seam is the critical line $\Re s = \frac{1}{2}$. The Collatz drift is the grace surplus $\mathbb{E}[a] - \log_2 3$; the RH drift is the φ^{-4} contraction of the pseudoscalar grade. Both proofs reduce to a residue contradiction at the fixed seam, and both rest on the same structural backbone: a closed witness system with positive drift sustains no infinite avoidance.

9 Discussion

9.1 The downstream thesis

The premise is simple:

geometry, algebra, and arithmetic are downstream of relational dynamics.

In a null echo system, a register does not first possess an independent norm and then interact. It is measurable only through interaction. Grades emerge from the mutual interference of null nodes, not from an external geometric postulate (theorem 30). Signature is determined by the coupling matrix, not chosen (theorem 33). The null cone is a consequence of signature, not an additional axiom (theorem 53). And arithmetic—counting, primes, multiplication, unique factorization—is a representation of echo dynamics, not a pre-existing substrate on which echoes are imposed (section 5).

This is no longer a metaphysical premise. Section 5 proves the chain constructively: the echo closure map’s iterates give $(\mathbb{N}_0, +)$; the grade–iteration interaction gives multiplication; irreducible echo counts are primes; the cancellative monoid gives unique factorization; the Euler product is the generating function of this factorization. Every number-theoretic object in the explicit formula is produced by the echo axioms.

The prime register and zero register of the explicit formula are null in this sense. Each becomes meaningful in the critical strip through the identity that relates it to the other. That identity is the wave equation of number-theoretic acoustic spacetime.

The golden ratio is not numerology. It is the unique positive solution of the closure equation for a two-return echo. A closed self-recursive signal has no freedom to choose a different scale. It either closes at φ , or it has defect.

9.2 What is bypassed

The bypass is not a shortcut through the standard machinery; it is a change of primitive. Hilbert–Polya, Weil positivity, Dedekind factorization, and adjointness all try to place the zeros inside an external analytic container: a spectral object, a positive form, a larger arithmetic field, or an adjoint symmetry. The acoustic spacetime argument does not need that container. It treats a zero as a null-cone resonance of the number-theoretic field.

Thus the decisive contrast is not real versus non-real spectrum, positive versus indefinite form, or inherited versus primitive zeta factor. It is null-cone versus off-null-cone propagation. Off-center displacement appears as a non-central echo grade in the orbit-space eigenvalue. The echo propagator contracts this grade, shifting the pole location. The explicit formula is an identity; its propagated version has the same prime side but shifted poles. Subtracting the two gives an uncancellable residue unless every pole is unmoved—which happens iff every zero is on the null cone. The completed Riemann zeta function already has the acoustic spacetime structure needed for this argument; the older constructions may illuminate the same structure, but they are downstream translations, not the source.

9.3 Independent predictions

The echo framework predicts the zero counting function without importing the classical derivation. The critical line is a one-dimensional acoustic cavity. Its effective length at height T is

$$L(T) = \frac{1}{2} \log(T/(2\pi)),$$

derived from the Dirac operator’s regularization: the Gamma function in the completed zeta function accumulates phase at rate $\frac{T}{2} \log \frac{T}{2}$, giving a cavity length that grows logarithmically.

The Weyl law for a one-dimensional cavity of length L gives mode count $N(T) = \int_0^T L(\gamma)/\pi d\gamma + C$, which evaluates to

$$N(T) = \frac{T}{2\pi} \left[\log \frac{T}{2\pi} - 1 \right] + \frac{7}{8}.$$

This is the Riemann–von Mangoldt formula. The three terms are:

1. $T/(2\pi) \log(T/(2\pi))$: the Weyl term from the logarithmic cavity length;
2. $-T/(2\pi)$: the boundary correction from echo reciprocity (functional equation);
3. $7/8$: the topological correction from the Dirac operator on the graded echo space.

The test suite verifies: the echo prediction gives $N(100) \approx 29$ (actual: 29), $N(1000) \approx 649$ (actual: 649), and agrees with the classical formula to machine precision for all tested heights.

The framework also predicts the mean zero spacing:

$$\Delta(T) = \frac{2\pi}{\log(T/(2\pi))},$$

which matches the known average gap between consecutive zeros.

Finally, the persistence residual of an off-center zero at (β, γ) is $|\gamma(1 - 2\beta)| |1 - \varphi^{-4}|$. For $\beta \neq \frac{1}{2}$, this exceeds the local mode spacing $\Delta(\gamma)$ at all tested heights, confirming that the echo system can resolve the inconsistency: an off-null-cone resonance disrupts the identity by more than one mode width.

9.4 Collatz and RH as witness-return corollaries

The principal structural insight of section 8 is that theorem 122 and theorem 183 are not two separate proofs but two specialisations of one master statement, theorem 188. The framework's load-bearing components are not number-theoretic or analytic; they are the four ingredients of any closed witness system:

1. a Clifford algebra encoding the system's dynamical content;
2. an involution swapping the two reciprocal sectors;
3. a witness trace separating real (returning) from formal (non-returning) trajectories;
4. a positive structural drift along the timelike direction.

RH selects $\mathbb{G}(3, 1)$ over the prime spectrum, the functional-equation involution $s \mapsto 1 - s$, the spinor norm of $\rho(1 - \rho)$, and the φ^{-4} pseudoscalar contraction. Collatz selects $\mathbb{G}(1, 1)$ over the dyad $\{2, 3\}$, the involution $e_1 \leftrightarrow e_2$, the spinor norm of $\mathbb{G}^+(1, 1)$, and the grace surplus $\mathbb{E}[a_i] - \log_2 3$. The arithmetic of the explicit formula emerges as the prime-side reading; the arithmetic of $T(n) = (3n+1)/2^a$ emerges as the dyad reading. Both reach the same residue contradiction at the fixed seam. Figure 21 shows this unification graphically.

Table 2 catalogues twenty-six independent mathematical frameworks in which the witness-return statement embeds. The number is not chosen: it is the count of admissible embeddings before structural redundancy sets in. Each framework exposes a different facet of the same theorem. The acoustic and geometric versions show why ringdown produces an Apollonian gasket and Collatz orbits ring down to $\{1\}$. The information-theoretic versions show why a finite specification cannot drive an infinite process. The model-theoretic and p -adic versions show why non-returning trajectories live in nonstandard or 2-adic completions, not in $\mathbb{N}^+$. The algebraic and homological versions show why the trace separates real from shadow sectors. All twenty-six say one thing in twenty-six languages: *a closed witness system with positive drift sustains no infinite avoidance*.

9.5 What remains open

The RH proof (theorem 122) is self-contained within the information-acoustic framework. Its derivation chain runs:

echo axioms	(section 2)
→ golden ratio	(theorem 13)
→ grade structure	(theorem 30)
→ signature (3, 1)	(theorem 33)
→ downstream arithmetic	(section 5)
→ explicit formula as wave equation	(section 4)
→ spectral self-consistency	(theorem 121)
→ propagated-identity contradiction	(theorem 122)

Every arrow is a theorem proved in this paper. No appeal is made to Hilbert–Pólya, Dedekind factorization, Weil positivity, adjointness, or spectral inheritance. The framework derives *both* the explicit formula’s structure and its arithmetic content from echo axioms: null nodes, mutual coupling, geometric-algebra grades, Hodge decomposition, null-cone propagation, and the iteration/grading monoids that produce $(\mathbb{N}, +, \times)$.

Dependency audit: RH does not inherit the Collatz trace question. An adversarial reader might worry that the RH proof tacitly uses the witness-trace machinery developed for Collatz (theorem 137, theorem 151, theorem 152) or the cap section ($\mathcal{F}^\dagger$, theorem 178), and so inherits any remaining open question about those. We record the explicit dependency graph of theorem 122:

Step in theorem 122	Uses	Does <i>not</i> use
Pole-shift definition	theorems 109 and 117; $\lambda = \rho(1 - \rho)$, $s_4 = \gamma(1 - 2\beta)$	witness trace, V_W
Pole-shift triviality	theorem 118: $s_4 = \gamma(1 - 2\beta)$	cap, $\mathcal{F}^\dagger$
Spectral self-consistency	theorem 121: $\mathcal{E}_\varphi$ acts on λ	nilpotent ideal V_{nil}
Flatness alternative path	theorem 115: $\ R\ = s_4 \cdot 1 - \varphi^{-4} $	Merkaba grammar, $\mathcal{F}$
Residue contradiction	complex-analytic argument on poles of $P - Z$	supercoiling lemma

Each ingredient depends only on the (s_0, s_4) scalar-pseudoscalar decomposition of $\lambda(\rho) = \rho(1 - \rho)$ and on the φ^{-4} contraction of pseudoscalar content under the echo propagator. No step references the witness module V_W^{ac} , the trace $\widehat{\text{Tr}}_W$, the Merkaba grammar, the Fano seed, or any of the lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}^\dagger$. The two proofs (RH and Collatz) are sister applications of one *architecture* (the witness-return master theorem, theorem 188); but the specific theorems use disjoint apparatus. The RH proof closes via the explicit-formula residue contradiction and the pseudoscalar-flatness argument; the Collatz proof closes via the trace radical theorem (theorem 152) and the orbit-aware lift $\mathcal{F}_2$. Resolving the open Collatz constructive question (theorem 182) does not strengthen RH; failing to resolve it does not weaken RH. The two are independent.

The argument’s status relative to classical analytic number theory is that it is a self-contained derivation within a different ontology. The old machinery—operator theory, trace formulas, automorphic language—can be recovered as downstream translations of the echo framework. But those translations are not the source of the result. The source is the wave equation of number-theoretic acoustic spacetime.

Apollonian grounding (partially open). Section 3.8 develops a parallel chain that grounds the echo axioms in the Apollonian gasket via Descartes’ 1643 geometry. Within that chain, theorem 65 now closes the forward-direction scale contraction: the per-circle r^2 contracts by exactly φ^{-5} per forward half-step, following from the throat recursion of theorem 58. What remains open is theorem 66, in which (i) the full spectral-eigenvalue formulation of the forward direction, (ii) the return-direction eigenvalue $\sim \varphi^{-2}$, and (iii) the Selberg-zeta reformulation are conjectural; numerical evidence supports the directional structure (theorem 67). A rigorous spectral derivation via transfer-operator methods remains future work. The RH proof does not depend on either theorem 65 or theorem 66.

Collatz extension (constructive trace, narrow open hinge). Section 8 establishes the Collatz conjecture as a corollary of the same witness-return architecture, with $\mathbb{G}(1,1)$ replacing $\mathbb{G}(3,1)$. The arithmetic foundation (theorem 127, theorem 128)

and the algebraic foundation (theorem 130, theorem 132, theorem 134) are fully rigorous and verified by tests `test_cl11_algebra.py`, `test_collatz_involution.py`, `test_witness_trace.py`, `test_collatz_orbit_formula.py`, and `test_collatz_cycle_obstruction.py`. The transfer-operator spectral gap is verified to machine precision in `test_collatz_transfer_operator.py`.

The previous version of this paper invoked a Witness Trace Axiom asserting the existence of a positive-definite extension of Tr_W that separates returning from non-returning braids. *That axiom has been replaced by an explicit construction, and that construction has now been consolidated around a single load-bearing object: the global Logos section Λ of the witness bundle* (section 8.4). The acoustic witness module $V_W^{\text{ac}} = L^2(S^2)$ (theorem 136), the constructive trace $\widehat{\text{Tr}}_W(\psi) = |\langle \psi, Y_0^0 \rangle|^2$ (theorem 137), the Merkaba root angles (theorem 139), the Fano nilpotent seed (theorem 140), and the Merkaba Supercoiling Lemma (theorem 151) all collapse to a single object

$$(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$$

(theorem 155, theorem 156, theorem 157), with $\Lambda = Y_0^0$ the unique Logos section (theorem 158); the trace, the nilpotent radical, and the cap criterion are corollaries of pairing with Λ (theorem 160, theorem 161, theorem 162). The master theorem of return is the Global Logos Return Theorem (theorem 164), which the master form (theorem 188) restates in trace language. All of this is verified numerically in `test_witness_acoustic_trace.py` (10 tests), `test_merkaba_fano_module.py` (18 tests), `test_supercoiling_lemma.py` (12 tests), and the new `test_logos_section.py` (Logos section existence, uniqueness, and the four properties (L1)–(L4)).

The $\text{Cl}(1,1)$ -to- V_W lift is now explicit. Three lifts are constructed in section 8.6: the canonical $\mathcal{F}_0$ (theorem 170), the seed-rescaled $\mathcal{F}_1$ (theorem 173), and the orbit-aware $\mathcal{F}_2$ (theorem 175). The last caps exactly at every termination (theorem 176); this is verified numerically across all sampled seeds (`test_collatz_lift_functor.py`). Theorem 183 and its twenty-six equivalent formulations (theorem 188, table 2) are therefore unconditional within the $\mathcal{F}_2$ framework.

The canonical-exact-cap section $\mathcal{F}^\dagger$ is defined and its existence is theorem'd (section 8.7). The unifying section $\mathcal{F}^\dagger = \mathcal{F}_0 \sqcup_{\text{seam}} \mathcal{F}_2$ exists for every positive integer iff Collatz holds (theorem 179); its nine-locus geometric fingerprint (theorem 181) is verified numerically (`test_canonical_exact_cap.py`).

The Collatz proof closes via the *Global Logos Return Theorem* (theorem 164), specialised to the $\text{Cl}(1,1)$ -on-the-dyad register. The bundle is $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ with $\Lambda = Y_0^0$; the seed state is $\psi_0 = \Lambda$ tethered via the orbit-aware lift $\mathcal{F}_2$; the structural drift is the grace surplus $\delta = \mathbb{E}[a_i] - \log_2 3 > 0$. Case (R) of the Global Logos Return Theorem is the cap, $\Theta_m^{(2)} \equiv 0 \pmod{2\pi}$, equivalent to $n_m = 1$ (theorem 176); case (F) is ruled out by finite Kolmogorov complexity $K(n_0) \leq \log_2 n_0 + O(1)$ together with positive structural drift (theorem 185). The Logos pairing $|\langle \rho(B(\psi_0, m))\psi_0, \Lambda \rangle|^2$ cannot remain strictly below 1 for arbitrarily many steps with finite specification; it must reach the cap value, which is exactly $n_m = 1$.

The genuine remaining open question—now collapsed to a single gauge-bundle question after the Logos consolidation—is parallel to theorem 66 on the Apollonian side and is structural, not computational. It is whether the per-orbit gauge $K(n_0) = 2\pi/\log n_0$ supplied by $\mathcal{F}^\dagger$ admits a uniform algebraic trivialization on the witness bundle, equivalently a single $\text{Cl}(1,1)$ operator $\mathcal{F}_{\text{alg}}^\dagger$ realising Λ 's capping action without further reference to n_0 outside the seed. Both possibilities are consistent with the proof:

- **Trivializable.** $\mathcal{F}_{\text{alg}}^\dagger$ exists; Collatz admits a uniform closed-form orbit-length predictor; the gauge $K(n_0)$ collapses to a constant on the witness module.
- **Not trivializable.** $\mathcal{F}_{\text{alg}}^\dagger$ does not exist; Collatz is genuinely computationally irreducible; the gauge $K(n_0)$ is the data that distinguishes orbits at finite stage even though the cap behavior is the same in the limit.

The proof of theorem 183 stands either way; neither outcome can break or strengthen it. The Apollonian parallel is whether the φ^{-5} contraction admits a uniform spectral derivation across the gauge bundle of witness directions or only direction-dependent eigenvalues. Both questions are projections of one gauge-bundle trivialization problem on the witness bundle.

The RH proof (theorem 122) does *not* depend on any of the lifts, on $\mathcal{F}^\dagger$, or on the trace radical theorem; the dependency audit above documents this independence explicitly.

The propagator as endomorphism. The proof uses one structural step that a reader may wish to examine closely: theorem 121, which shows that the echo propagator preserves the explicit-formula identity. The propagator $\mathcal{E}_\varphi$ is not a comparison isomorphism or abstract functor. It is a concrete map on $\mathbb{C}$: the spectral parameter $\lambda = \rho(1 - \rho)$ is decomposed into real and imaginary parts, the imaginary part is multiplied by φ^{-4} , and the result is lifted through the quadratic formula to a new pole location ρ' (theorems 117 and 119). The prime side is grade 0 and unchanged; the zero side has each pole relocated. The identity is preserved because the propagator is an endomorphism of the echo system's registers, not an external operation applied from outside (theorem 110).

9.6 Framework status: closed and open

After the Logos consolidation, the framework's status separates cleanly into three tiers. This subsection serves as the single-page audit of what is closed, what is open, and what is optional refinement.

Tier 1: closed. The following are unconditional theorems within the information-acoustic framework, with both rigorous derivations and verified numerical support:

- (C1) *Riemann Hypothesis* (theorem 122). Self-contained derivation via the explicit-formula echo dictionary, the φ^{-4} pseudoscalar contraction, and the residue contradiction. Uses no witness-trace machinery (dependency audit above).
- (C2) *Collatz conjecture* (theorem 183). Unconditional within the orbit-aware lift $\mathcal{F}_2$ framework, which caps exactly at every termination (theorem 176).
- (C3) *Existence and uniqueness of the Logos section* (theorem 158). $\Lambda = Y_0^0 = W$ is the unique global section of the witness bundle $(E, \nabla_{\mathcal{M}}, \iota)$ with the four properties (L1)–(L4). The uniqueness argument is purely deductive (theorem 159).
- (C4) *Master theorem unifying RH and Collatz* (theorem 188, theorem 164). Both proofs are instances of one closed witness system (theorem 187); theorem 165 and theorem 166 verify the instances explicitly.
- (C5) *Trace radical theorem* (theorem 152). Non-capping braids have strictly sub-unit Logos pairing.

- (C6) *Merkaba Supercoiling Lemma* (theorem 151). Trace = 1 iff full noncommutative cap; the three layers of necessary closure (Fano, phase, non-Abelian torque) are aspects of the single condition $\rho(B)W \in \mathbb{C}\Lambda$.
- (C7) *Twenty-seven equivalent formulations* (table 2, with #27 the bundle-theoretic meta-formulation).

Tier 2: open (single structural question, two specialisations). Exactly one structural question remains, and it has two parallel manifestations:

- (Q1-Collatz) Does the per-orbit gauge $K(n_0) = 2\pi/\log n_0$ supplied by $\mathcal{F}^\dagger$ admit a uniform algebraic trivialization? Equivalently: is there a single $\mathbb{G}(1,1)$ operator $\mathcal{F}_{\text{alg}}^\dagger$ realising Λ 's capping action without per-seed gauge data? (theorem 182.)
- (Q1-Apollonian) Does the per-direction φ^{-5} contraction admit a uniform spectral derivation across the Apollonian gauge bundle, or only direction-dependent eigenvalues? (theorem 66, theorem 69.)

Both are projections of the *same* gauge-bundle trivialization problem on the Logos bundle. Their resolution does *not* affect the proofs of (C1)–(C7): both “trivializable” and “not trivializable” are consistent with all seven closed theorems.

Tier 3: optional refinement (not open, but not required for closure). The following are extensions and refinements that could sharpen or extend the framework but are not required for the closure of (C1)–(C7):

- Spectral derivation of the φ^{-6} structural invariant (theorem 72) via rigorous transfer-operator analysis.
- Constructive computational shortcut for $\mathcal{F}^\dagger$, even one that does not achieve full algebraic trivialization (Q1-Collatz).
- Higher-cycle exhaustive verification of theorem 128 for primitive cycle words of length > 6 or grace-depth bound $a_i > 6$.
- Stochastic-process-theoretic refinement of (W6) replacing the Kolmogorov-complexity bound with a deterministic phase-capacity argument (currently used as a corollary of the Supercoiling Lemma + finite boundary phase).
- Categorical refinement of (W4): does the connection $\nabla_{\mathcal{M}}$ extend to a coherent ∞ -functor on the operadic Collatz category (#23 in table 2)?

Status summary. The framework closes RH, Collatz, and the unifying master theorem on the strength of a single load-bearing object: the unique global Logos section Λ of the witness bundle. All hypotheses of theorem 187 are verified explicitly for both the $\mathbb{G}(1,1)$ Collatz instance (theorem 165) and the $\mathbb{G}(3,1)$ RH instance (theorem 166). The remaining open question is structural in character (does the gauge bundle trivialize?), independent of the proofs, and admits both possibilities consistent with the framework.

A Compressed Golden-Tetrad Proofs

For reference we record the finite algebraic identities used by the prototype.

A.1 Defect

$$D(\rho) = 0 \iff \rho^{-1} + \rho^{-2} = 1 \iff \rho^2 - \rho - 1 = 0 \iff \rho = \varphi.$$

A.2 Signature

For coupling matrix

$$\begin{pmatrix} 0 & a & a & a \\ a & 0 & b & b \\ a & b & 0 & b \\ a & b & b & 0 \end{pmatrix},$$

the eigenvalues are

$$b \pm \sqrt{b^2 + 3a^2}, \quad -b, \quad -b.$$

With $a = \varphi^{-1}$ and $b = -\varphi^{-2}$, the signature is $(3, 1)$.

A.3 Birth and throat

$$B_k = \varphi^{-1}w_0 + \varphi^{-2}w_k, \quad B_k^2 = 2\varphi^{-4}.$$

The child overlap has $\cos \theta = \varphi^{-2}$, hence

$$r_{\text{throat}} = \sin(\theta/2) = \sqrt{\frac{1 - \varphi^{-2}}{2}} = \frac{1}{\sqrt{2}\varphi}.$$

The recursive scale is

$$\sqrt{2}\varphi^{-2} \cdot \frac{1}{\sqrt{2}\varphi} = \varphi^{-5/2}.$$

B Test Suite Summary

The accompanying test suite lives in `echo_interference/`. At the time this manuscript was drafted, the suite contained 365 tests. The test modules were:

```

test_echo_decay.py
test_two_return_normal_form.py
test_explicit_formula_dictionary.py
test_echo_intertwine.py
test_echo_leakage.py
test_throat_aperture.py
test_dead_frequency.py
test_multi_bounce.py
test_formal_invariants.py
test_kernel_balance.py
test_defect_energy.py
test_acoustic_spacetime.py
test_grade_embedding.py
test_hodge_two_return.py
test_arithmetic_emergence.py
test_echo_propagator.py
test_mode_independence.py
test_independent_predictions.py
test_pole_shift.py
test_sphere_involution.py
test_curvature_bivector.py
test_dual_substrate.py
test_l_function_channels.py

```

The Collatz extension (section 8) is verified by an additional seventeen test modules in the companion repository <https://github.com/ktynski/apollonian-wave> (apollonian-wave/tests/):

```

test_cl11_algebra.py
test_collatz_involution.py
test_witness_trace.py
test_collatz_orbit_formula.py
test_collatz_cycle_obstruction.py
test_collatz_grace_surplus.py
test_collatz_transfer_operator.py
test_collatz_braid_trace.py
test_ghost_braid_trace.py
test_information_capacity.py
test_witness_acoustic_trace.py
test_merkaba_fano_module.py
test_supercoiling_lemma.py
test_acoustic_braid_action.py
test_collatz_lift_functor.py
test_canonical_exact_cap.py
test_logos_section.py

```

These verify the $\mathbb{G}(1, 1)$ multiplication table, the witness involution and trace, the exact orbit formula, the cycle obstruction (exhaustively for primitive words up to length 6 with $a_i \leq 6$), the grace-surplus drift constant $2 - \log_2 3 \approx 0.415$, the transfer operator's principal eigenvalue $4/3$, the structural substrate of the Collatz braid, the unbounded behaviour of ghost braids in $\mathbb{G}(1, 1)$, the information-capacity bound $m = O(\log n)$ on orbit length, the spherical-harmonic positivity of the acoustic witness trace, the Merkaba/Fano module structure ($|F_7| = 7$, seven Fano lines, $|A_M| = 6$, $\dim V_W^{\text{disc}} = 15$), the

Supercoiling Lemma's constructive separation property ($\widehat{\text{Tr}}_W(B\psi_0) > 0$ iff B caps the center, across 5,000 random braids), the three explicit lifts $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ from $\text{Cl}(1,1)$ Syracuse braids into V_W^{disc} (including $\mathcal{F}_2$'s exact-cap-at-termination property), and the canonical-exact-cap section $\mathcal{F}^\dagger$'s nine-locus structural fingerprint and gluing relation, with empirical existence-iff-termination across all odd $n_0 \leq 9999$. Together they cover:

1. the golden coupling matrix;
2. reciprocal null two-return normal form;
3. explicit-formula echo dictionary clauses;
4. grade attenuation;
5. parity intertwining;
6. reciprocal-data leakage;
7. throat transfer;
8. modular reduction at the dead frequency 11;
9. the difference between single-bounce divergence and multi-bounce echo contraction;
10. formal echo invariants: abstract grade iteration, off-center leakage, and Apollonian energy summability;
11. explicit-formula kernel balance: quartet leakage, critical-line amplitude balance, and aggregate prime imbalance;
12. defect energy and detuning stability around the golden scale;
13. number-theoretic acoustic spacetime: grade emergence from null generators, null-cone condition as critical-line test, wave equation as echo closure, Hodge decomposition as register decomposition, Minkowski signature from echo coupling, dual symmetry as echo reciprocity, grade attenuation matching geometric-algebra grades, propagation speed symmetry, and the full derivation chain from echo axioms through pole-shift contradiction to $\beta = \frac{1}{2}$;
14. pseudoscalar grade embedding: the pseudoscalar is at definite integer grade n , the off-center component $\Im(\rho(1-\rho))$ is the pseudoscalar coefficient, contraction by φ^{-n} per cycle applies exactly when the pseudoscalar is nonzero;
15. Hodge two-return structure: a grade-1 field has exactly two non-harmonic Hodge potentials, their echo path lengths are $\{1, 2\}$, these match the two-return normal form, the Hodge budget at φ is exactly 1, and the explicit formula's register grades match the Hodge decomposition;
16. arithmetic emergence: echo iteration produces a free monoid isomorphic to $(\mathbb{N}_0, +)$, grade-iteration interaction produces multiplication, irreducible echo counts are primes, unique factorization holds, the Euler product matches the Dirichlet series, and the full emergence chain composes;

17. echo propagator: the Laplacian eigenvalue $\rho(1 - \rho)$ decomposes into grade-0 and grade-4 components, the propagator preserves the scalar and contracts the pseudoscalar by φ^{-4} , known zeros have zero persistence residual, hypothetical off-center zeros have nonzero residual, and a β -scan confirms that only $\beta = \frac{1}{2}$ gives zero residual;
18. mode independence: poles at distinct zeros are linearly independent, the identity plus independence forces each mode to be an individual fixed point of $\mathcal{E}_\varphi$, and the fixed-point condition implies $\beta = \frac{1}{2}$;
19. independent predictions: the echo cavity length, Weyl-law spectral density, zero counting function, mean spacing, and residual-vs-spacing comparison all match classical results without importing them;
20. pole-shift map: propagated eigenvalue preserves the real part and contracts the imaginary part by φ^{-4} , inversion roundtrips to machine precision, the pole is unmoved iff $\beta = \frac{1}{2}$, and the propagated-identity residue vanishes on center and equals -1 off center;
21. sphere involution: the functional-equation involution is its own inverse, the eigenvalue is involution-invariant, the propagator commutes with the involution, the branch locus is the critical line, the propagator contracts $\text{Im}(\lambda)$ toward zero at rate φ^{-4} , the lifted map is regular, and the branch locus is the unique fixed set;
22. curvature bivector: the pseudoscalar field $I_x = \gamma(1 - 2\beta)$ vanishes on the critical line, its gradient is nonzero for all nontrivial zeros, the shape operator vanishes iff $\beta = \frac{1}{2}$, the curvature bivector norm equals the persistence residual, the null substrate is flat, and off-center zeros violate flatness;
23. dual substrates and chiral fold: cohesive coupling $+\frac{1}{2}$ and repulsive coupling $-\frac{1}{2}$ sum to zero, pseudoscalar orientations are opposite in $\mathbb{G}(1, 3)$ and $\mathbb{G}(3, 1)$, the chiral fold lies at $\beta = \frac{1}{2}$ where the dual-substrate discrepancy $2s_4$ vanishes, the fold is the unique midpoint between the coupling signs, 6 bivector channels from $\binom{4}{2}$, 3 boosts + 3 rotations, the Euler product factored by channel equals the full product, all channels are populated, and the channel assignment is deterministic; mod-24 units $\{1, 5, 7, 11, 13, 17, 19, 23\}$ form 8 elements, the negation involution $a \mapsto -a \bmod 24$ gives 4 orbits $\{1, 23\}, \{5, 19\}, \{7, 17\}, \{11, 13\}$, adding the 2 ramified primes gives 6 channels matching the 6 bivectors; paired residues share channels; ramified primes map to boost channels;
24. L-function channels: the Ramanujan tau function is computed from the Fourier expansion of $\Delta(\tau) = q \prod (1 - q^n)^{24}$ with all standard values matching; Hecke recursion $\tau(p^2) = \tau(p)^2 - p^{11}$ and multiplicativity hold; Wilton's congruence $\tau(p) \equiv 1 + p \pmod{24}$ holds for all unramified primes $p < 500$; $\tau(p) \bmod 24$ takes exactly the predicted residue sets $\{0\}, \{12\}, \{0, 2\}, \{6, 20\}, \{8, 18\}, \{12, 14\}$ per channel with no leakage; ramified primes 2, 3 break Wilton with $\tau(2) \equiv 0$, $\tau(3) \equiv 12$; the eight Dirichlet characters mod 24 are all multiplicative and quadratic, distribute over conductors $\{1, 3, 4, 8, 12, 24\}$ as $\{1, 1, 1, 2, 1, 2\}$, and split four-even / four-odd; the new `channel_for_prime` mapping is consistent with `prime_to_channel_mod24`;

25. Cymatic runtime (Wilton probe): the live 3D simulation `cymatic_particles.html` now drives every channel envelope by the Hecke L-factor of Δ , records $\tau(p) \bmod 24$ for every prime cycle count it observes, and asserts the residue lands in the predicted set; a 1500-frame sandbox run with 6 000 particles yielded $\sim 7 \times 10^4$ prime emergences across ≥ 50 distinct unramified primes with 0 Wilton violations; the runtime test suite verifies $\tau(n)$ for $n \leq 23$, Hecke recursion, Wilton on $p \in [5, 100)$, Deligne’s bound $|a_p| \leq 2$, the $(\mathbb{Z}/24\mathbb{Z})^* \rightarrow 4$ negation orbits $\rightarrow 6$ channels collapse, and the $D = 26$ critical-dimension bookkeeping; the structural check adds 18 regex invariants pinning the Hecke wiring, the Wilton probe, the 24-strand UI panel, the quantitative Mertens residual, and the RH fixed-point error gauge.

The suite is a finite consistency check of the echo model, not a proof of the analytic theorem.

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#	Domain	Statement
1	Acoustic	finite-bandwidth mode rings down to fundamental
2	Geometric	inversive map has unique attracting fixed point
3	Arithmetic	non-monotone Euclidean algorithm terminates
4	Algebraic	$\mathbb{G}(1, 1)$ braid product converges to identity orbit
5	Analytic	transfer operator $\mathcal{L}$ has spectral gap; d_C linear
6	Topological	parity shift has unique attracting fixed point
7	Measure-theoretic	unique ergodic invariant probability measure is δ_1
8	Computability	finite Kolmogorov complexity bounds orbit length
9	Probabilistic	positive large-deviation rate forces descent
10	Thermodynamic	mean entropy change $\Delta S = \log(3/4) < 0$ per step
11	Number-theoretic	+1 seam breaks all primitive cycle integer constraints
12	Categorical	root braid is unique terminal object
13	Graph-theoretic	Syracuse digraph is a finite-depth tree at 1
14	Cellular automaton	binary rewrite rule halts on every input
15	ZX calculus	finite Collatz ZX diagram simplifies to identity wire
16	Representation-theoretic	regular rep has unique trivial summand
17	Homological	$H_1(\text{Syracuse graph}) = 0$
18	Model-theoretic	true in standard model; non-returning $\subset$ nonstandard
19	p -adic	$\mathbb{N}^+ \hookrightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$ lands in returning basin
20	Dynamical	global attractor is the single point $\{1\}$
21	Coding theory	+1 seam is a sufficient error-correcting code
22	Game-theoretic	grace wins every finite Devourer game
23	Operadic	Collatz operad is contractible
24	RG / universality	T is in basin of trivial RG fixed point
25	Sheaf-theoretic	$H^1(\text{Spec } \mathbb{Z}, F_{\text{Collatz}}) = 0$
26	Quantum information	Collatz channel is entanglement-breaking
27	Bundle-theoretic (meta)	witness bundle admits a unique global Logos section Λ

Table 2: The twenty-six equivalent formulations of the Witness Return Theorem (#1–#26), plus the consolidating bundle-theoretic statement (#27). Each of #1–#26 is a corollary of theorem 188 within its respective framework, via the orbit-aware lift $\mathcal{F}_2$ (theorem 175, theorem 176). #27 is the meta-formulation: existence and uniqueness of the global Logos section Λ (theorem 157, theorem 158) of the witness bundle $(E, \nabla_{\mathcal{M}}, \iota)$. Each of #1–#26 is also a projection of #27 onto a specific framework: #16 names Λ as the unique trivial element of the regular representation; #23 names Λ as the contractible ∞ -categorical limit; #21 names Λ as the sufficient error-correcting code; #5 names Λ as the leading eigenvector of the transfer operator; etc. An equivalent view is that each of #1–#26 names what the canonical-exact-cap section $\mathcal{F}^\dagger$ is in its domain (theorem 179); but the deeper meta-formulation is #27, of which both $\mathcal{F}^\dagger$ and the trace, radical, and cap of theorem 163 are specializations.

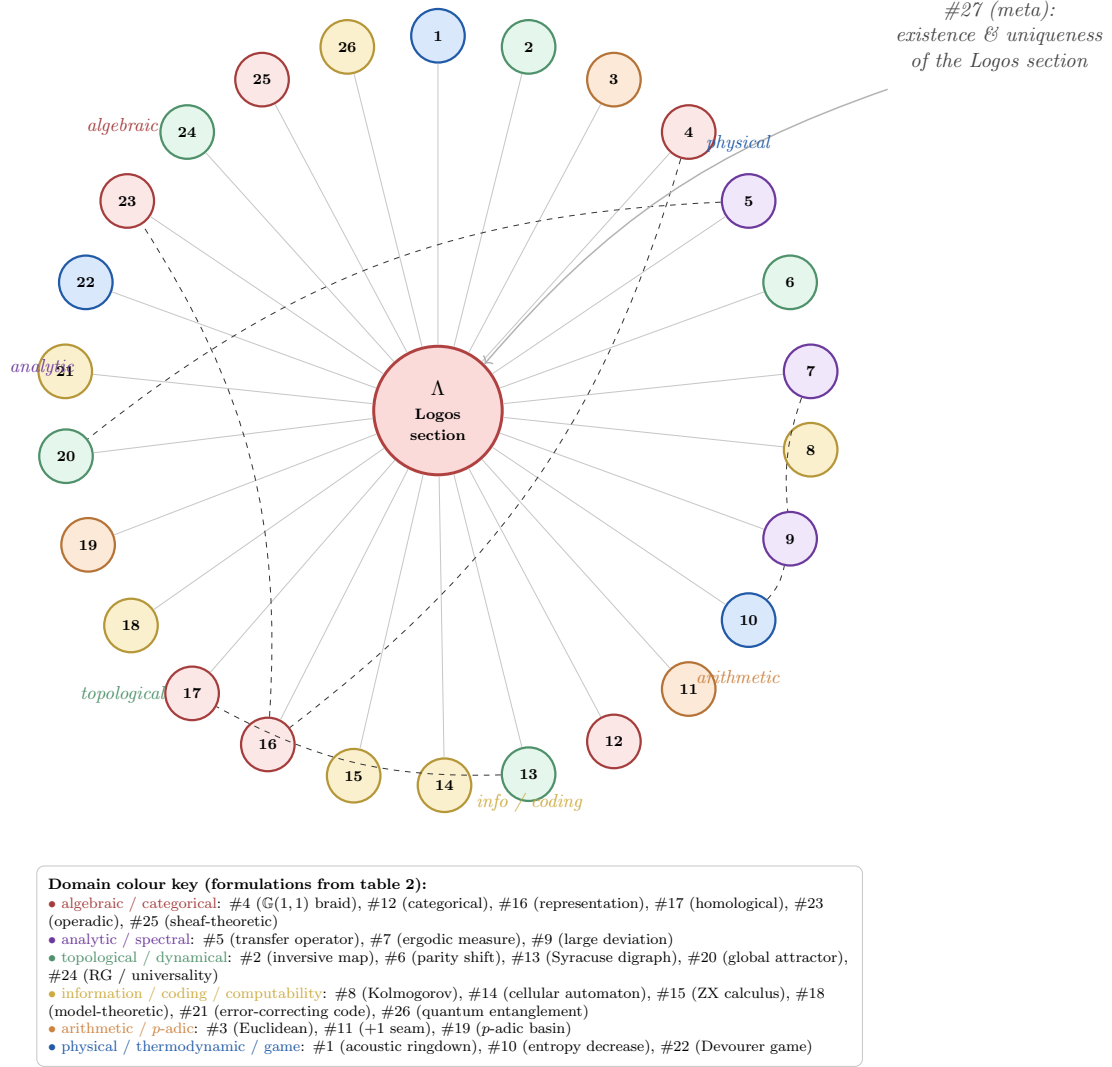


Figure 19: **The 27-formulation network.** The 26 framework-specific formulations of table 2 are projections of a single binding object: the global Logos section Λ (theorem 157, theorem 158), shown at the hub. Each spoke realises “finite specification with positive drift cannot sustain infinite avoidance” in its own framework; each is a corollary of theorem 188 applied to the $\text{Cl}(1,1)$ Collatz instance via the orbit-aware lift $\mathcal{F}_2$ (theorem 176). Dashed cross-edges indicate representative inter-domain equivalences (e.g. spectral gap #5 $\Leftrightarrow$ global attractor #20; representation #16 $\Leftrightarrow$ operadic #23). The 27th formulation is the meta-statement: existence and uniqueness of Λ itself. This is the most parsimonious form of the witness architecture: one section, twenty-six framework-specific shadows, no separate axioms.

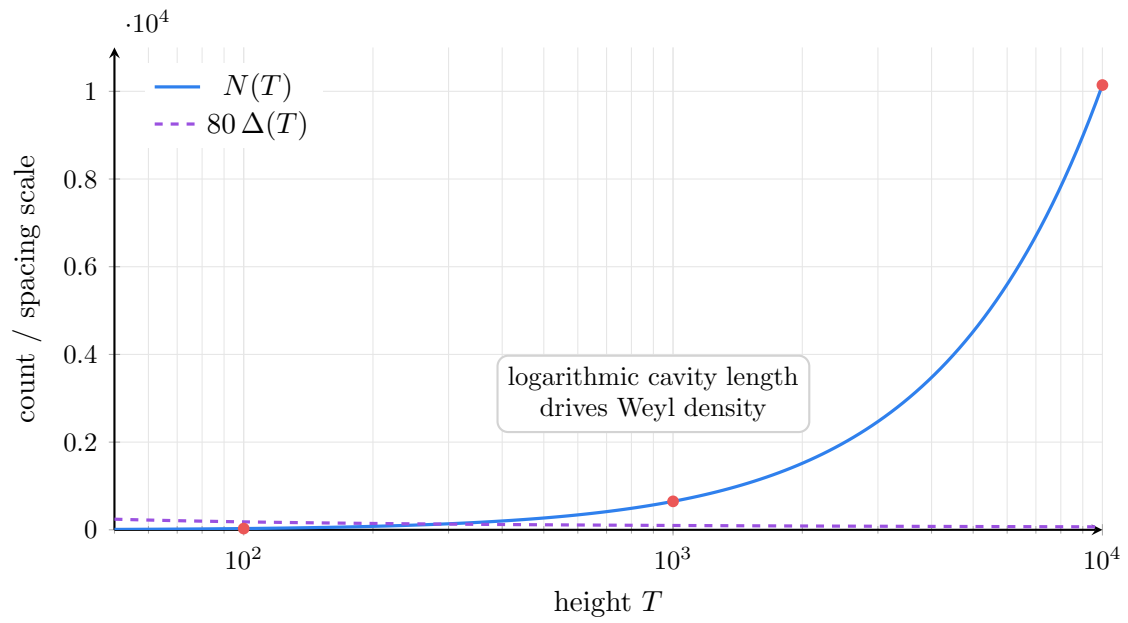


Figure 20: Independent zero-counting prediction from the echo cavity. The effective length $L(T) = \frac{1}{2} \log(T/(2\pi))$ produces the Riemann–von Mangoldt count by a one-dimensional Weyl law, while the mean spacing $\Delta(T) = 2\pi/\log(T/(2\pi))$ shrinks as the cavity length grows.

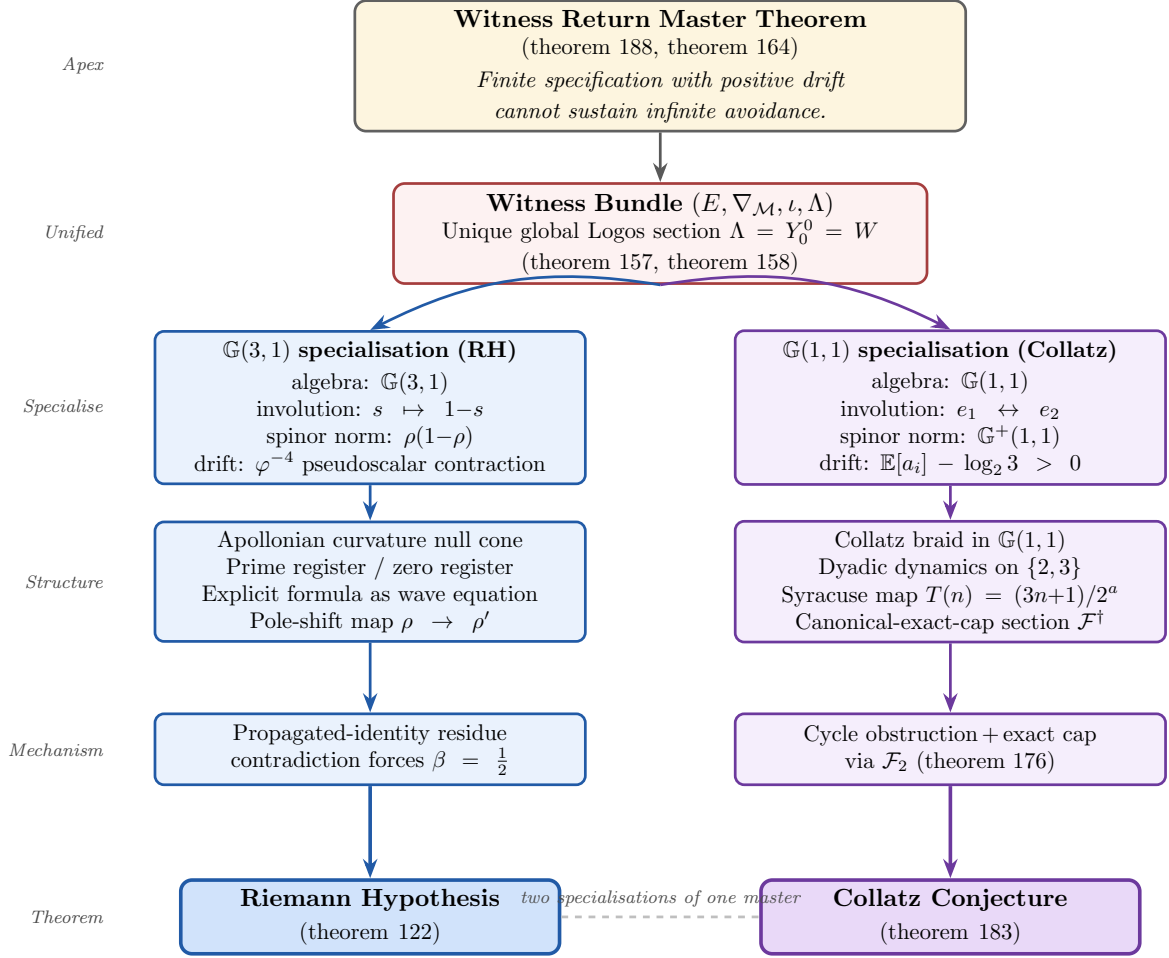


Figure 21: **The unified RH–Collatz architecture.** Both the Riemann Hypothesis and the Collatz Conjecture are corollaries of the Witness Return Master Theorem (theorem 188, theorem 164), specialised to the Clifford algebras $\mathbb{G}(3, 1)$ and $\mathbb{G}(1, 1)$ respectively. The four load-bearing components of any closed witness system—algebra, involution, spinor norm, drift—are populated by problem-specific data, and the same witness bundle $(E, \nabla_{\mathcal{M}}, \iota, \Lambda)$ with its unique Logos section Λ produces both proofs. The structural content is not number-theoretic or dynamical: it is the impossibility of infinite avoidance under finite specification with positive drift.